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1 The ordered sound-change sequence

1.1 Scope and orientation

The sequence begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

1.2 Numbering note

SC numbers remain the established legacy identifiers. The Version 1 book presents the changes in historical chapter order, which differs from the computational cascade order for several rules.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC077 is unused.

2 Chapter 1. From Proto-Germanic to Proto-Northwest Germanic

2.1 Historical interval

This chapter covers developments that took place within the Proto-Germanic period, from the inherited consonant system to the first changes that separate the Northwest Germanic line from the rest of the Germanic family.

The reconstruction labelled Proto-Germanic here is the common ancestor of Gothic, North Germanic, and West Germanic, reconstructed through the classical comparative method from attested descendant languages. The label Proto-Northwest Germanic designates the hypothetical node linking the ancestors of North Germanic (Old Norse and its relatives) and West Germanic (Old English, Old High German, Old Saxon, and Old Frisian among others), to the exclusion of Gothic and the other East Germanic varieties.

2.2 What this chapter contains

Chapter 1 contains one reader-facing sound-change section: the Proto-Germanic loss of a nasal before **x* (`SC103 PGmcNasalLossBeforeX`), with compensatory lengthening and nasalization of the preceding vowel. Its results are shared by every daughter language, Gothic included, so the change belongs to Proto-Germanic itself and precedes every Northwest Germanic and West Germanic development treated in the chapters that follow (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7).

Another historically Proto-Germanic rule, the positional allophony of Proto-Germanic **b* (`SC049 PGmcBAllophony`), executes late in the computational cascade because the alternation interacts with environments shaped by intermediate rule applications; its section therefore appears in Chapter 4, where the divergence between cascade placement and historical stage is noted explicitly.

One other historically Proto-Germanic change, Gm-simplification (`SC002 PGmcGmSimplification`), is documented in the book-entry plan and its literature dossier confirms the source base is narrow (two lexical families: **draugma-* 'dream' and **taugma-* 'team'; (Kroonen 2013, 101, 511)). A reader-facing section for SC002 awaits a stronger explanatory source base and is not yet assembled in the reader-facing sequence.

2.3 Scope and genealogical context

Changes in this chapter are pan-Germanic in scope: they apply to ancestral forms that feed both the North Germanic and West Germanic descendants, or they represent internal Proto-Germanic processes visible across the Germanic family.

The boundary between Proto-Germanic and Proto-Northwest Germanic is not sharp in the textbook literature. Ringe and Taylor treat many of the traditionally "Proto-Germanic" changes as part of a shared innovation package that is diagnostically older than North–West Germanic divergence but not necessarily earlier than the separation of the East Germanic line (Ringe and Taylor 2014, 1–30). For book purposes, the distinction matters primarily because it separates the features inherited uniformly from all Germanic from those shared selectively by North and West Germanic to the exclusion of Gothic.

2.4 A note on the rule names

The CAPR rules implemented in this chapter carry names beginning with `PGmc`. Those names are intended as stable internal identifiers, not as claims about the precise historical stage of every rule so labelled. A rule named `PGmcX` may in some cases be a later development that affects only the West Germanic or Northwest Germanic branch; the chapter assignment in this staging map takes priority over the rule-name prefix for historical organization purposes.

3 Proto-Germanic loss of a nasal before **x*

3.1 Historical discussion

The oldest change treated in this book is common to the whole family. In the group **-nx-* the nasal consonant was lost, the preceding vowel was lengthened in compensation, and that lengthened vowel was nasalized. Its results are shared by every daughter language: Gothic *þeihan* ‘thrive’, *brāhta* ‘brought’, *þūhta* ‘seemed’ stand beside Old High German *dīhan* ‘thrive’, *brāhta, fūht* ‘damp’ and Old English *þēon* ‘thrive’, *brōhte* ‘brought’, *þūhte* ‘seemed’, *fūht* ‘damp’ (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7). Because no daughter keeps the nasal, the change belongs to Proto-Germanic itself, and it precedes every Northwest Germanic, West Germanic, North Sea Germanic and Anglo-Frisian development described in the chapters that follow.

Only **a, *i* and **u* occur in this position. Germanic had already raised **e* to **i* and **o* to **u* before a nasal followed by a consonant, so the mid vowels are absent from the input (Fulk 2018, 55, §4.1).

The vowel that the change creates is long and nasalized, and it is not yet rounded. Fulk emphasizes that the lengthened vowels remained nasalized for a considerable time, well past the close of the Northwest Germanic period, since the nasalized low vowel produced in this way went on to develop to *ō* in Anglo-Frisian and did not fall together with the Old English *ā* that came from **ai* (Fulk 2018, 55, §4.1). The comparative material makes the same point directly: Gothic, Old Norse, Old High German and Old Saxon all reflect the Proto-Germanic nasalized low vowel as unrounded *ā*, and only Old English and Old Frisian show *ō* (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). Ringe describes Proto-Germanic **hanhaną* as **[xą̄xaną]* and observes that its low vowel was rounded, along with the other nasalized low vowels, in the northernmost West Germanic dialects (Ringe 2017, 149–50, §3.2.7). That rounding is a separate and much later change, and it is treated in the chapter on the rounding of the long nasalized low vowel.

3.2 SC103. Proto-Germanic nasal loss before x (PGmcNasalLossBeforeX)

```
define PGmcNasalLossBeforeX Ctx([
    {*a} -> {*ą̄} || _ EnglishStarNasal {*x},
    {*i} -> {*ī} || _ EnglishStarNasal {*x},
    {*u} -> {*ū} || _ EnglishStarNasal {*x},
    {*á} -> {*ą̄} || _ EnglishStarNasal {*x},
    {*í} -> {*ī} || _ EnglishStarNasal {*x},
    {*ú} -> {*ū} || _ EnglishStarNasal {*x}
] .o. [
    EnglishStarNasal -> 0 || _ {*x}
]);
```

The rule performs the three parts of the change together: it lengthens the vowel, it marks the lengthened low vowel as nasalized, and it then removes the conditioning nasal. The low vowel is written **ą̄* and the high vowels are written **ī* and **ū*, because the nasality of the high vowels has no further consequence: they go on to develop exactly as the inherited long **ī* and **ū* do (Campbell 1959, 47, §121). The nasality of the low vowel is carried forward because its later fate depends on it.

The corpus witnesses both branches of the rule. The high branch is **fúnxstiz* ‘fist’, which becomes **fúxsti* ‘fist’ and, after the loss of **x* before the cluster in SC028 PNWGmcPreconsonantalXLoss, gives Old English *fȳst* ‘fist’. The long vowel of Old High German *fūst* ‘fist’, Dutch *vuist* and German *Faust* shows that the word belongs here and to no later law (Kroonen 2013, 160). The rule also supplies the **xst* cluster on which the loss of **x* before a consonant operates, so the two stand in a feeding relation.

The low branch is **θánxtē* ‘thought’, the preterite of the verb ‘to think’, whose principal parts are reconstructed as **þankijaną* ‘to think’, **þanhtē* ‘thought’, **þanhtaz* ‘thought (past participle)’

with the nasal still standing before the fricative (Ringe 2017, 281; 2017, 136). Here the rule yields **θą̄xtē* ‘thought’, and the nasalized low vowel is later rounded by SC104 EAFNasalizedLowRounding to give Old English *þōhte* ‘thought’. This is the form the handbooks themselves cite for the Anglo-Frisian rounding of the vowel produced here (Fulk 2018, 55, §4.1). It does not enter the loss of **x* before a consonant, in which respect it differs from ‘fist’, so the fricative survives to the surface and the word displays the vowel history alone.

4 Chapter 2. From Proto-Northwest Germanic to Proto-West Germanic

4.1 Historical interval

This chapter covers the sound changes that took place in the proto-language shared by the West Germanic languages — Old English, Old Frisian, Old Saxon, Old High German, and Old Dutch — before the individual languages diverged. The starting reconstruction is Proto-Northwest Germanic (PNWGmc), the hypothetical common ancestor of North Germanic and West Germanic together; the ending reconstruction is Proto-West Germanic (PWGmc), the immediate common ancestor of the West Germanic languages specifically.

4.2 Scope and internal diversity

Changes in this chapter are not all equally pan-West-Germanic in scope. They may be grouped broadly as follows:

Northwest Germanic innovations (shared by both North and West Germanic): innovations in the unstressed vowel system, certain final-syllable vowel changes, and selected consonant cluster simplifications. Changes labelled `NWGmc` in the CAPR rule names fall here, though rule prefixes are not always reliable guides to historical scope.

Proto-West-Germanic innovations (shared within West Germanic but not in North Germanic): the cluster of morphological and phonological changes that distinguish Old English, Old High German, Old Saxon, and Old Frisian from Old Norse. Changes labelled `PWGmc` in the CAPR rule names generally fall here. They include early apocope rules, certain consonant assimilations, and the West Germanic gemination of consonants before `*j`.

4.3 Major changes

The chapter opens with the root-noun nominative `*-z` loss (SC096), the generalization of ending-less nominatives through the athematic consonant stems, complete before Proto-West Germanic: none of the West Germanic daughters shows any ending in this class (Ringe and Taylor 2014, 118). It is the earliest of the three historically distinct final-`*z` developments; the other two (SC020 and SC097) open Chapter 3.

The unstressed **ai* > **ē* development (SC014 PNWGmcUnstressedAiMonophthongization) is one of the widespread Northwest Germanic vowel shifts. Unstressed dative-singular and strong-adjective plural endings provide examples of monophthongization and merger with inherited unstressed **ē*. Ringe and Taylor distinguish this broad correspondence from the more questionable claim of a single completed Proto-Northwest Germanic event (Ringe and Taylor 2014, 25–26). The stressed monophthongization is treated separately under SC004 EAFAiMonophthongization.

The adjacent SC015 PNWGmcILowering concerns stressed first-syllable **i*, not further unstressed-vowel reduction. Its following-vowel hypothesis has traditional authority, but its comparative regularity and precise consonantal conditioning remain disputed. The section compares vowel-conditioned and consonant-conditioned accounts and explains the provisional restrictions of the current implementation (Campbell 1959, 43, §§114–116; Fulk 2018, 56–57; Hill 2017, 149–58).

The West Germanic consonant changes of this chapter — j-gemination (SC010), early i-apocope (SC006), coronal-w assimilation (SC008), and related rules — represent the most productive phonological territory for the CAPR derivations. They feed a large proportion of the distinctive consonant clusters of Old English. Handbooks vary in exactly how they group and name these changes (Campbell 1959, sects 404, 406; Hogg 1992a, 71, §4.11 (held 2011 reissue)).

Coronal assimilation also supplies the geminate in *fēower* ‘four’ and *ēow* ‘you’, before inherited short-vowel-plus-geminate-glide reanalysis. SC031 OEWWSimplification now represents that earlier development, not an unrestricted late deletion. The resulting diphthong-plus-glide se-

quence is inherited by later English realization; j-created and singleton paths remain distinct (Ringe and Taylor 2014, 41–42, 65–66; Campbell 1959, 45–47).

The quoted definitions use `Ctx(...)` to carry selected sentence context outside the segmental operation. It is computational transport, not a reconstructed segment or an additional sound law. Lexical accent, sentence stress and phonological-word finality are separate: the early high-vowel-loss account requires a heavy syllable in a sentence-unstressed, phonologically final word, not simply the absence of an acute. Ordinary evaluation selects strong-final citation context; the explicit weak-final selection for *ēow* is discussed with SC098 PWGmcUnstressedWordFinalIApocope (Ringe and Taylor 2014, 55, 57–58).

The nasal spirant corridor (SC026–SC027), treated in Chapter 3 at its cascade position, illustrates a type of change common in historical grammars of the "Ingvaeonic" or "North Sea Germanic" area: nasals disappear before voiceless fricatives, with compensatory vowel lengthening (Campbell 1959, sects 462–463; Hogg 1992a, 56, §3.14 (held 2011 reissue)). The CAPR model splits this into two ordered steps to make the vowel effect computationally tractable; the book prose explains that split against the handbook tradition, which typically presents the change as a single process.

Chapters in this part of the book follow the executable cascade order, which models the reconstructed chronology itself. Several rules that carry `PWGmc` labels — final bare-`*a` loss (SC041), surviving bimoric `*ō` unrounding (SC042), and Sievers-law syncope (SC050) — execute later in the cascade and are therefore presented in Chapter 4, where their individual sections discuss their historical stage labels. Conversely, one rule with a West Saxon label, the palatal-glide rule (SC016), is an orthographic rule of the written surface: it executes after the Old English orthography stage and is presented in Chapter 5.

One historically Proto-Germanic change, Gm-simplification (`SC002 PGmcGmSimplification`), precedes everything in this chapter as a support stage of the cascade. It is documented in the book-entry plan and its literature dossier confirms the source base is narrow (two lexical families: **draugma-* 'dream' and **taugma-* 'team'; (Kroonen 2013, 101, 511)). A reader-facing section for SC002 awaits a stronger explanatory source base and is not yet assembled in the reader-facing sequence.

4.4 A note on source terminology and subgrouping

The literature uses several partly overlapping stage labels for this period:

- Northwest Germanic: the node uniting North and West Germanic.
- Proto-West Germanic: the node uniting only the West Germanic languages.
- North Sea Germanic or Ingvaeonic: a proposed subgroup within West Germanic covering Old English, Old Frisian, and Old Saxon (and sometimes Old Low Franconian), sharing certain innovations over a broader area.
- Anglo-Frisian: a narrower proposed subgroup linking only Old English and Old Frisian.

These labels are not always used consistently across sources. Ringe and Taylor are cautious about reconstructing a discrete Proto-West-Germanic node (Ringe and Taylor 2014, 50–55). Campbell notes that many of the "West Germanic" shared features could alternatively be treated as parallel developments rather than common inheritance (Campbell 1959, sects 1–5).

CAPR uses `PWGmc` and `NWGmc` as organizing labels for this chapter without claiming to have settled all questions about West Germanic subgrouping. Changes that appear in the literature under "Ingvaeonic" labels but affect the Old English–to-Proto-Germanic derivation chain are treated here as late expressions of the same West Germanic developmental period unless existing CAPR dossier research specifically argues for Anglo-Frisian or English-specific placement.

4.5 Rule names

Most CAPR rules in this chapter carry names beginning with `NWGmc` or `PWGmc`; the reformulated inherited-glide rule retains its older `OE` identifier. These names are stable internal identifiers. A name beginning with `NWGmc` does not guarantee that the change is exclusive to Northwest

Germanic, and a name beginning with PWGmc does not guarantee that it is absent from North Germanic. The historical analysis in each sound-change section takes priority over the name prefix.

5 Root-noun nominative *-z loss

5.1 Historical discussion

The athematic consonant stems — the "root nouns" of the handbooks — attached the nominative-singular marker directly to a consonant-final root, and the Proto-Germanic outcome of that collision is genuinely uncertain. Ringe gives the consonant-stem nominative ending as zero, *-z*, or possibly *-s*, and states plainly that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306, §4.3.4). His own paradigm tables carry the uncertainty into print: the nominative of 'foot' appears as "fōts? (fōs?)", while 'mouse' is plain **mūs*, its expected extra sibilant already absorbed by degemination (Ringe 2017, 149, 313). Ringe and Taylor repeat the same three-way agnosticism — the root-noun nominative "either ended in *-s* or *-z*, or was endingless" (Ringe and Taylor 2014, 28, §2.3.1).

The dictionary traditions encode this situation in different notations, and the differences are conventions of citation rather than competing claims of fact. Orel prints morphologically explicit nominatives with final *-z* across the whole class: **bōkz* 'book', **gansz* 'goose', **lūsz* 'louse' (Orel 2003, 52, 126, 252). Kroonen cites the same words as bare stems or endingless forms, **bōk-* and **gans-* (Kroonen 2013, 71–72, 168–69), and Kluge/Seebold print a third variant with voiceless *-s*, as in **bōks* 'Buch' (Kluge 2011, 158). Bammesberger shows why the marker is nonetheless real: for voiced-final stems the overt nominative *-z* is positively reconstructible — **burgz* (Gothic *baúrgs*), **frijōndz* (Gothic *frijonds*) — even though **fōt-z* is "phonotaktisch kaum denkbar", and in West Germanic the ending simply "fiel es ab" (Bammesberger 1990, 190–92, §8.2.3.1). CAPR retains Orel's morphologically explicit forms as its inputs precisely because they record the inflectional marker whose fate this rule describes.

The three focal words are only superficially parallel. The root-final *s* of 'louse' is itself an extension of **luw-* on the model of 'mouse' (Bammesberger 1990, 195, §8.3). For 'goose', Szemerényi's lengthening would give a Proto-Indo-European nominative **ǵʰanss* > **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196, §8.3; Kroonen 2013, 168–69). 'Book' preserves a plain obstruent-final root. What unites them is not a single phonetic history but membership in a paradigm class that generalized the endingless nominative.

The branch evidence dates and localizes that generalization. Gothic keeps its sibilant throughout the class (*baúrgs*, *nahts*, *reiks*) (Fulk 2018, 165–66, §§7.26–7.27). Old Norse redistributes the ending morphologically: masculine root nouns keep *-r* (*fótr*), feminines are endingless (*nótt*, *geit*), while the vocalic-stem feminines *kýr*, *sýr*, *ær* retain *-r* from *-z* with R-umlaut (Fulk 2018, 167, §7.28; Bammesberger 1990, 192–93). West Germanic alone is uniform: there was "no ending in PWGmc, as none of the daughters exhibits any" (Ringe and Taylor 2014, 118, §3.4). Fulk supplies the mechanism: Szemerényi's law removed the nominative sibilant after sonorant-final stems, and endinglessness then spread analogically through the class (Fulk 2018, 143, §7.2). The development is therefore best understood as a morphological generalization enacted differently in each branch — absolute in West Germanic, consonant- and gender-conditioned in North Germanic, absent in Gothic — rather than as one exceptionless sound law.

This change is distinct from the two later final-*z developments. It was complete before Proto-West Germanic, whereas the loss of *-z* in unstressed syllables (SC020 EAFFinalZDeletion) is itself a Proto-West Germanic change: polysyllabic consonant-stem nominatives such as **fadurz* 'father' — and, in this corpus, **frijōndz* 'friend', **melukz* 'milk', and **mēnōþz* 'month' — kept their ending into Proto-West Germanic and lost it there in an unstressed syllable (Ringe and Taylor 2014, 44–45, §3.1.1). The still later northern loss of *-z* in stressed monosyllables (SC097 MonosyllabicFinalZLoss) affects vowel-final monosyllables like **hwaz* and does not touch the consonant-final root nouns at all, whose ending was gone long before.

5.2 SC096. Root-noun nominative *-*z* loss (`RootNounNomZLoss`)

```
define RootNounNomZLoss Ctx([{*z} -> 0 ||
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
        EnglishStarVocalic+
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.]);
```

The rule deletes word-final **z* after a consonant in a monosyllable. Three claims of different kinds meet here and must be kept apart. The historical claim is morphological: the nominative-singular ending was lost in the athematic root-noun class, so that this one paradigm cell came to lack its marker — a development complete before Proto-West Germanic (Ringe and Taylor 2014, 118, §3.4). The lexical claim belongs to the dictionaries: Orel's citation forms, which supply the corpus inputs, print that marker explicitly as *-z* in **bōkz*, **flauxz*, **gansz*, and **lūsz* (Orel 2003, 52, 105, 126, 252). The executable statement is neither of these but a computational proxy for them: 'delete word-final **z* after a consonant in a monosyllable'. It is not proposed as a Proto-Germanic sound law; it earns its place only because every form the corpus submits to the morphological development is a consonant-final monosyllable, so the narrow phonological statement covers the class exactly. Four corpus derivations witness the rule, each yielding its expected Old English outcome: PGmc **bōkz* 'book' yields OE *bōc* 'book', **gánsz* 'goose' yields *gōs* 'goose', **lūsz* 'louse' yields *lūs* 'louse', and **fláuxz* 'flea' yields *flēah* 'flea'. Should the corpus ever acquire a consonant-final monosyllable in **-z* that is not a root-noun nominative, the proxy and the morphology would come apart, and the rule would need to be re-scoped; the project's regression tests pin the firing population to exactly these four words so that any fifth firing forces that adjudication rather than passing silently.

The rule applies at the head of the English line, before SC009 PWGmcIjContraction. That ordering is fixed by the identity of the process rather than by a wrong form: contraction turns the polysyllabic **frījōndz* 'friend' into a monosyllable, and if the root-noun rule applied after contraction it would capture **friundz* — yet the ending of 'friend' survived into Proto-West Germanic and fell in an unstressed syllable, the change described under SC020 EAFFinalZDeletion (Ringe and Taylor 2014, 44–45, §3.1.1). Because both rules delete the same segment, moving this rule later changes no Old English output; the early placement keeps the derivation of 'friend' aligned with the historical account rather than with an accident of the cascade.

Negative controls behave as the morphology predicts. Stressed monosyllables whose **z* follows a vowel — the domain of the later northern change — do not meet the post-consonantal environment. Medial **z* is untouched and remains available for rhotacism (SC003 EAFRhotacism): PGmc **déuzą* 'deer' still yields OE *dēor* 'deer' with its rhotacized medial consonant.

6 Early Northwest Germanic vowel developments

6.1 Historical discussion

These two developments affect different parts of the vowel system. SC014 PNWGmcUnstressedAiMonophthongization monophthongizes unstressed **ai*; SC015 PNWGmcILowering represents the disputed lowering of stressed short **i*. Their grouping here does not identify them as successive stages of unstressed-vowel reduction.

6.2 Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the widespread monophthongization of unstressed **ai* to a long mid vowel, largely merging with inherited unstressed **ē*. Their examples include final dative-singular and nonfinal inflectional syllables. They nevertheless question a single completed Proto-Northwest Germanic merger: Runic evidence suggests that some unstressed diphthongs survived later than others (Ringe and Taylor 2014, 25–26). The stressed development of **ái* to **ā* is treated separately as SC004 EAFAiMonophthongization.

6.3 SC014. Monophthongization of unstressed **ai* (PNWGmcUnstressedAiMonophthongization)

```
define PNWGmcUnstressedAiMonophthongization Ctx([
    {*ai} -> {*ē}
]);
```

The dative-singular endings span **spánnai* ‘span’ and meed **mízdai* ‘meed’ carry this change; both give a final **ē*. If SC014 PNWGmcUnstressedAiMonophthongization is delayed until after SC072 OEUnstressedLongVowelShortening, the **ē* is no longer present for shortening, so PGmc **spánnai* ‘span’ yields †*spannē* rather than expected OE *spanne* ‘span’. This shows that SC014 PNWGmcUnstressedAiMonophthongization must come before SC072 OEUnstressedLongVowelShortening in the modeled sequence.

6.4 SC015. Lowering of stressed short **i* (`PNWGmcILowering`)

6.4.1 The traditional account and its limits

The traditional formulation lowers stressed **i* to **e* before a mid or low vowel in the following syllable. It parallels the better-attested lowering of **u* in SC017 PNWGmcULowering. Campbell’s §114 explicitly states this front-vowel change, while emphasizing its scant English evidence: OE *nest* ‘nest’, *wer* ‘man’, and *spec* ‘bacon’ beside *spic* ‘bacon’. His §§115–116 distinguish the more regular back-vowel development and blocking before nasal-plus-consonant groups. This is an account of accented vowels, distinct from the later unstressed developments of SC074 OEMedUnstressedILowering1 and SC075 OEMedUnstressedILowering (Campbell 1959, 43, §§114–116).

Fulk likewise accepts vowel-conditioned lowering, interpreting its uneven distribution through earlier alternations and subsequent paradigmatic levelling, especially the elimination of lowered vowels. He regards a Northwest Germanic date as methodologically preferable, without excluding a Proto-Germanic beginning. His treatment supplies substantial authority for the central hypothesis represented here, although not for every consonantal restriction in CAPR. The comparative irregularity is itself part of his analysis, not evidence that the traditional account was universally accepted (Fulk 2018, 56–57, §4.3).

6.4.2 Objections to general vowel-conditioned lowering

Lloyd’s strongest objection is the class I participle: inherited short **i* survives before the same non-high participial suffix that accompanies lowered **u* in class II. He challenges the proposed morphological links used to explain away this difference. In nouns, he questions whether the

reconstructed high-vowel endings could have supplied enough forms to restore **i* throughout a paradigm. Uninflected OHG *hina* ‘hence’ and *nidana* ‘from below’ make indiscriminate appeals to case-based levelling still less persuasive (Lloyd 1966, 738–41).

Cercignani retains a role for assimilation and analogy but disputes uniform conditioning and chronology across North and West Germanic. His earlier defence of an inherited phonemic opposition between **i* and **e* concerns a different proposition from the existence of a particular lowering rule. Crucially, his later paper explicitly warns that the preceding-velar restriction reported for North Germanic cannot simply be transferred to West Germanic: Old High German provides counterevidence. He also recognizes the difficulty of explaining the inherited adverbs and fish vocabulary by analogy alone (Cercignani 1979, 75, 82; 1980, 129–31).

Ringe and Taylor question a general early West Germanic lowering. For OE *nest* ‘nest’ and *wer* ‘man’, they leave open whether Proto-Germanic already had **e*. They distinguish the more extensive southern West Germanic lowering before labial and velar obstruents from these widely shared correspondences. Their analogical interpretation of particular strong-verb transfers depends on identifiable paradigm coincidences. These empirical distinctions deserve consideration without making independent daughter innovations the default explanation of every difficulty (Ringe and Taylor 2014, 34–36).

6.4.3 Hill’s consonant-conditioned alternative

Hill argues that the following consonants offer a more explanatory generalization than the next vowel. His strongest proposed environments are tautosyllabic **z*, the cluster **st*, and **r*. Thus PGmc **iz* ‘he’, **hiz* ‘this here’, and **mizdōn* ‘reward’ supply the first environment; the nest and man words supply the others. The contrasting preservation before **sk* occurs in inherited vocabulary, including PGmc **wiskō* ‘wisp’, and cannot readily be dismissed as a late change of inflectional class. The following table summarizes the principal comparisons (Hill 2017, 153–56).

Earlier form	Comparative evidence	Diagnostic value
PGmc **nistan* ‘nest’	OE *nest* ‘nest’	Compatible with both following-vowel and **st* conditioning.
PGmc **wiraz* ‘man’	ON *verr* ‘man’, OE *wer* ‘man’, Gothic *wair* ‘man’	Supports lowering before **r*; Gothic breaking obscures its date.
PGmc **mizdōn* ‘reward’	OE *meord* ‘reward’, *mēd* ‘reward’, OHG *miata* ‘reward’	Supports a history with lowering before **z*, alongside separate consonant and quantity changes.
PGmc **fiskaz* ‘fish’	ON *fiskr* ‘fish’, OE *fisc* ‘fish’	Retained **i* before **sk*, despite the non-high ending.
**bitanaz* ‘bitten’, **biðanaz* ‘awaited’	OE *biten* ‘bitten’, *biden* ‘awaited’	Class I participles resist the proposed general vowel-conditioned lowering.

The inherited root vowel of OE *nest* ‘nest’ also has a comparative etymology independent of the English outcome: Kroonen connects it with the downward adverb and the zero grade of the root ‘sit’. Hill’s Northwest Germanic **nęstijana* ‘to nest’ and **lęstiz* ‘skill’ illustrate how later i-mutation could restore a high vowel after earlier consonant-conditioned lowering. Such a specified sound change is stronger evidence than a generic invocation of analogy. Nevertheless, **z*, **st*, and **r* do not by themselves define one established natural class: their unification as a single process remains a hypothesis (Kroonen 2013, 391; Hill 2017, 153–56).

6.4.4 Difficult counterevidence

The participial comparison requires particular care. Hill prints ON *biðinn* ‘awaited’, but Fulk discusses *beðinn* ‘awaited’ and expressly says that the high-vowel variant does not occur. Seebold

prints the corresponding *beðenn* ‘awaited’, allowing either regular lowering or, preferentially, formal interference from the verb ‘bid’. Fulk challenges the latter explanation because independent early Norse evidence for that interference is lacking; the English parallels he mentions are late and never participial. Hill’s argument that influence from a long-vowel infinitive should also have introduced length is substantive, but the printed discrepancy does not settle the historical alternatives (Hill 2017, 151–52; Seebold 1970, 94–95; Fulk 2018, 58–59, especially p. 59, n. 6).

OHG *irh* ‘chamois leather’ is a potential counterexample to unrestricted lowering before **r*. Hill acknowledges this and prefers a loanword analysis. Kroonen instead connects the leather and he-goat vocabulary with wider European comparanda and reconstructs **irha-* ‘he-goat’. These are competing etymological premises. The inherited analysis would require an explanation under an unrestricted pre-**r* law. CAPR’s present formulation instead excludes the intervening velar fricative. Calling the word a loan solely to protect a rule would beg the question (Hill 2017, 156, n. 30; Kroonen 2013, 270).

Later phonology can also conceal a genuine difference between early formulations. In the thistle comparison, a consonant-conditioned trial lowers the root of selected **θístilas* ‘thistle’, whereas the incumbent leaves it high. Later i-mutation restores **i* in the former path, and both finish as OE *þistles* ‘thistle’. Agreement of final forms therefore does not establish agreement about the earlier history. This illustrates the kind of convergence invoked in Hill’s derivative comparisons, without proving every proposed instance of it (Hill 2017, 153).

6.4.5 Lowering and the history of ‘here’

Hill’s positive chronological argument begins with early PGmc **hir* ‘here’, formed like **hwar* ‘where’ and **þar* ‘there’. He proposes lowering to **hẹr* ‘here’, followed by monosyllabic lengthening to **hẹ̄r* ‘here’. His dotted vowel indicates a close mid value; the long result must not automatically be identified with inherited **ē₁*. Lengthening first would instead produce a long high vowel, for which the required general early lowering lacks independent support (Hill 2017, 148–49, 156–58).

The comparison depends on more than the vowel of Gothic *hēr* ‘here’. Hill relates the long-vowel monosyllables to disyllabic formations with short vowels. OHG *hiar* ‘here’ contrasts with *hĕra* ‘hither’, while the inherited disyllabic pronoun **hwarja-* ‘which’ could precede lengthening of its monosyllabic base. The short vowel in Gothic *ƕarjis* ‘which’ consequently does not require late lengthening of the adverb ‘where’. Hill also argues that North Germanic adverbial variants reflect the interaction of originally distinct local and directional formations (Hill 2017, 146–48).

Three inferences should be separated. Shared North and West Germanic reflexes support an early lowering before **r*. A Proto-Germanic date remains possible because Gothic raising could conceal earlier **e* before **z*, while Gothic breaking makes the man word chronologically ambiguous. Hill’s additional argument places lowering before a lengthening which he reconstructs as Proto-Germanic, thus positively dating the pre-**r* change. That conclusion depends on his adverbial reconstruction and does not independently date lowering before **z* or **st*. Ringe and Taylor retain alternative explanations for the adverb, including a different pronominal vowel and lengthening under deictic stress (Hill 2017, 156–58; Ringe and Taylor 2014, 36–37).

6.4.6 The implemented restriction

```
define PNWGmcILowering Ctx([
    {*i} -> {*e}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel,
    {*í} -> {*é}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel
]);
```

The left boundary makes the affected vowel the first vowel of the word, approximating stressed first-syllable lowering. Both plain and acute-marked short **i* are handled; absence of an acute is not interpreted as sentence stresslessness. A non-high vowel must follow one or more members of the defined nonnasal coronal class: **d, l, r, s, t, z, ð, θ*. Intervening nasals, labials, velars, and **j* are excluded. Every consonant in the word-initial interval must also be nonvelar. The operation therefore excludes an onset velar anywhere in that interval, not just an immediately preceding velar.

The following-vowel condition has direct traditional authority; the exact consonantal restrictions have more uneven support. Hill's **st/sk* contrast does not establish transparency of every admitted coronal. Cercignani explicitly restricts the reported preceding-velar condition to its North Germanic evidence. Fulk's corresponding note is likewise about North Germanic. Howell and Salmons' account of intervening consonants principally concerns i-umlaut and its failure; it offers a phonetic comparison, not direct authority for this lowering expression (Hill 2017, 153–55; Cercignani 1980, 130; Fulk 2018, 57, n. 3; Howell and Salmons 1997, 96–97).

In particular, OE *hlid* 'lid' is protected by the onset prohibition in the current selected derivation. Removing that prohibition alone produces †*hled*. Independent English evidence for the general onset restriction has not been established. A more selective right-hand environment, such as Hill's proposed consonantal conditions, would leave the dental-stop context untouched without that prohibition. This identifies a weakness in the present approximation, while leaving the historical status of a replacement to be decided.

6.4.7 The three live witnesses and conditional chronology

The current rule changes exactly three selected forms:

Selection	Selected input	Root-vowel change	Final output
Meed	**mízdai* 'meed'	**í* > **é* before **zdē*	OE *meorde* 'meed'
Nest	**nístą* 'nest'	**í* > **é* before **stą*	OE *nest* 'nest'
World	**wír-àldu* 'world'	**í* > **é* before **rà*	OE *weorold* 'world'

These witnesses have different evidential weight. OE *nest* 'nest' supports lowering but distinguishes neither the following-vowel hypothesis nor Hill's **st* hypothesis. The history of OE *meord* 'reward' and *mēd* 'reward' supports lowering before **z*, with separate histories of consonant loss or rhotacism and vowel quantity. CAPR's selected **mízdai* 'meed' is a dative input, not Hill's nominative **mizdōn* 'reward'. The two must not be used as interchangeable whole-word reconstructions (Kroonen 2013, 370, 391; Hill 2017, 153).

OE *weþer* 'wether' instead has the selected masculine **wéθruz*, with inherited **e*. Kroonen's **weþru-* 'yearling' and Orel's masculine **weþruz*, alongside alternative **weþraz*, support that reconstruction. Gothic *wiþrus* 'lamb, young ram' provides positive evidence for the *u*-stem. This noun does not require lowering and is not a live application of SC015 PNWGmcILowering (Kroonen 2013, 584; Orel 2003, 459–60).

OE *weorold* 'world' likewise depends on the chosen analysis. The selected **wír-àldu* 'world' is a mixed-stage transponent, combining earlier root vocalism with a later ending. Ringe and Taylor give an already **e*-bearing Northwest Germanic **weraldiz* 'world', followed by **weraldu* 'world' and **weruld* 'world' (Ringe and Taylor 2014, 326).

With selected **wír-àldu* 'world' held fixed, delaying the incumbent beyond SC036 OEInterStressRaising produces †*wuruld* in place of OE *weorold* 'world'. SC036 OEInterStressRaising has raised the weak vowel, removing the incumbent's non-high trigger. The observation is genuine but conditional on both input and rule: the vowel-independent Hill-style trial still lowers before **r* after SC036 OEInterStressRaising, while the root-**e* control **wér-àldu* 'world' produces the expected form under every tested placement, even without SC015 PNWGmcILowering. This displacement therefore does not independently establish that the historical lowering preceded interstress raising.

6.4.8 Interim interpretation

Early stressed **i*-lowering remains a serious comparative hypothesis, with a substantial traditional account and a consequential consonantal alternative. A hybrid history would need independently justified component changes and specific evidence for any subsequent levelling. The present restricted executable is retained provisionally because neither the full replacement domain nor the affected lexical premises are sufficiently settled. Its successful outputs demonstrate computational compatibility, not the correctness of every historical condition. Explicit competing formulations make those premises testable without silently changing lexical inputs to secure the preferred result.

7 Unstressed **a*-raising before final **m*

7.1 Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

7.2 SC005. Unstressed **a*-raising before final **m* (`PNWGmcAToUBeforeM`)

```
define PNWGmcAToUBeforeM Ctx([
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
     ↪  ({*z})? .#.
]);
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 PNWGmcULowering, PGmc **skúldramiz* ‘shoulders’ yields †*sċoldrum* rather than expected OE *sċuldrum* ‘shoulders’; earlier placements converge on the expected output. The scope of the change is established by inflectional evidence across multiple paradigm types: a-stem dative plural ON *dǫgum* ‘days’, OE *dagum* ‘days’, OS *dagun* ‘days’, OHG *tagum* ‘days’, beside Gothic *dagam* ‘days’; strong-adjective dative singular ON *góðum* ‘good’, OE *gōdum* ‘good’, OS *gōdum* ‘good’, beside Gothic *godamma* ‘good’ (OS also shows variant forms gōdumu and -un); and first-plural present ON *berum* ‘we carry’, OHG *berumēs* ‘we carry’, beside Gothic *baíram* ‘we carry’. Across these sets, North/West Germanic shows unstressed *-um* where Gothic preserves *-am*. The derivation of *sċuldrum* ‘shoulders’ supplies a CAPR ordering witness for the relative chronology, but the cognate set for ‘shoulder’ does not contribute comparative evidence for the rule’s historical scope.

8 Northwest Germanic lowering of long *ē*

8.1 Historical discussion

Proto-Germanic **ē₁* (the long mid vowel of PIE origin, as against the later **ē₂*) split East Germanic from the rest of the family. Gothic keeps a mid vowel, written □e□, in *gadēþs* ‘deed’, *slēpan* ‘to sleep’, *mēna* ‘moon’, and *jēr* ‘year’, while Norse and all of West Germanic show a low vowel, as in Old Norse *ráða* ‘to consider, to advise’, *láta* ‘to let go, to allow’, *ár* ‘year’, *nál* ‘needle’, Old High German *tāt* ‘deed’, *slāfan* ‘to sleep’, *jār* ‘year’, *māno* ‘moon’, Old Saxon *dād* ‘deed’, *slāpan* ‘to sleep’, *jār*. Ringe and Taylor assemble the comparative set and reconstruct a Northwest Germanic sound change **ē₁ > *ā* (Ringe and Taylor 2014, 11–13). The change is directly dated by the earliest epigraphy. The Early Runic accusative *mākija* ‘sword’ (beside Gothic *mēkeis* ‘sword’) already shows □a□ in the second half of the second century AD (Ringe and Taylor 2014, 12), and the a-rune likewise writes the stressed reflex in the Opedal stone’s *swestar* ‘sister’ (Stiles 2017, 4). Loanwords borrowed into Sami from early Proto-Scandinavian point the same way. Lid showed that the root vowel of Sami *mānno* ‘moon’, Proto-Germanic **ē* as in Gothic *mēna*, “had gone over to *ā* before the word was borrowed into Sami”, and adduced the parallel Sami *saððo* ‘bran’ beside Proto-Nordic **sāðō* from **sēðō-*, Old Norse *sáð* ‘bran’ (Lid 1952, 238). This is Sami borrowing from Proto-Scandinavian, so it dates and localizes the *ā* stage in the North; its application to West Germanic rests on the daughter correspondences above.

The change affected stressed syllables only. Ringe and Taylor restrict it explicitly: unstressed **ē* did not lower but was eventually shortened, as in **fadēr > *fader* ‘father’ (Ringe and Taylor 2014, 13, and p. 147); in the runic material the unstressed final vowel of *faþiR* ‘father’ has already merged with **-i* (Stiles 2017, 4). Within stressed syllables, however, the lowering was unconditioned: it applied before nasals exactly as elsewhere, and Ringe and Taylor accordingly print the intermediates **mānō* ‘moon’, **mānōþ-* ‘month’, and **spānuz* ‘spoon’ (Ringe and Taylor 2014, 11). The later, dialectally narrower fates of this **ā* — fronting in the North Sea area when oral, rounding when nasalized — are separate sound changes treated in their own chapters.

The reconstruction of the intermediate value is disputed. The objection is an old one: Bennett observed that the received account obliges Gothic to move *ē* to *ǣ* and back to *ē*, and the non-West-Saxon dialects and Old Frisian to run through *ē > ǣ > ā > ǣ > ē*, “with no apparent agreement among the languages and no discernible phonological trend”, and that the Proto-Germanic *ǣ* is on that account preserved directly nowhere (Bennett 1950, 232–33, 235). In its place he proposed that the backing to *ā* began in the north of the Germanic homeland and spread southward, never reaching Gothic in the east or Anglian and Frisian in the west, so that West Saxon *ǣ* is a retention of the intermediate stage rather than a re-fronting (Bennett 1950, 234–35). Fulk argues in the same direction on modern evidence: the Northwest Germanic vowel was a low front **ǣ*, retained unchanged in Anglo-Frisian, with the backing to *ā* of Norse and inner West Germanic a separate areal development spreading northward from Upper German territory; on that reading the runic □a□ spellings write a front [æ:] for which the futhark had no better grapheme (Fulk 2018, 60–61, §4.6). Campbell is deliberately noncommittal, finding the **ā* stage “tempting to assume, though not definitely demonstrable” (Campbell 1959, 50–51, §§128–129). Ringe and Taylor answer with the lengthened place-adverbs *þǣr* ‘there’ and *hwǣr* ‘where’, whose front vowels are most naturally the output of fronting applied to a back **ā* (Ringe and Taylor 2014, 13–14). The present model adopts the two-step reconstruction while recording that the alternative remains live. Campbell is deliberately noncommittal, finding the **ā* stage “tempting to assume, though not definitely demonstrable” (Campbell 1959, 50–51, §§128–129). Ringe and Taylor answer with the lengthened place-adverbs *þǣr* ‘there’ and *hwǣr* ‘where’, whose front vowels are most naturally the output of fronting applied to a back **ā* (Ringe and Taylor 2014, 13–14). The present model adopts the two-step reconstruction while recording that the alternative remains live.

8.2 SC024. Lowering of stressed long *ē* (`PNWGmcLongELowering`)

```
define PNWGmcLongELowering Ctx([
    {*é} -> {*ā}
]);
```

The rule reads the stressed tier **ǽ* only, in keeping with the stress restriction; unstressed **ē*, as in **fadēr* ‘father’, is left for the unstressed-shortening rules of the Old English stage and never lowers. No segmental environment is imposed: nasal forms such as **mǽnōþz* ‘month’ and **spǽnuz* ‘spoon’ pass through **mānōþ-* and **spānuz* on their way to *mōnaþ* ‘month’ and *spōn* ‘spoon’, exactly as reconstructed by Ringe and Taylor (Ringe and Taylor 2014, 11).

The rule stands near the head of the cascade, before the genuinely West Germanic innovations such as early *i*-apocope, **ij*-contraction, and *j*-gemination — the placement follows from the dating: the second-century runic evidence puts the lowering among the earliest Northwest Germanic developments, well before the changes that separate West Germanic from Norse. Its output **ā* is consumed much later in the cascade by SC102 EAFHiatusWInsertion, SC025 EAFLongANasalRounding, and SC101 EAFLongAFronting: if the lowering is instead displaced after those rules, **mǽnōθz* ‘month’ yields †*mānaþ* rather than OE *mōnaþ* ‘month’, **skǽpą* ‘sheep’ yields †*sċāp* rather than *sċēap* ‘sheep’, and **sǽaną* ‘to sow’ never acquires the hiatus-filling *w* of *sāwan* ‘to sow’. Within the cascade no earlier boundary has been demonstrated; the second-century runic attestation supplies the absolute dating.

9 Early i-apocope

9.1 Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geogu̯þ* ‘youth’ (Brunner 1965, sects 145–146; Ringe and Taylor 2014, 141). Campbell’s *dugup* ‘troop’ and *geogup* ‘youth’ examples belong to the same pattern (Campbell 1959, sect. 332).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

9.2 SC006. Early i-apocope (`PWGmcEarlyIApocope`)

```
define PWGmcEarlyIApocope Ctx([
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
]);
```

The absence of umlaut in *geogu̯þ* ‘youth’ provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLong-Diphthong, PGmc **skáwōθi* ‘shows’ yields †*sċēaweþ* rather than expected OE *sċēawaþ* ‘shows’.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

10 Final *ō-lowering before *r

10.1 Historical discussion

Ringe and Taylor separate two points here: a broader shortening of vowels before word-final **r* in unstressed syllables (for which kinship **r*-stems such as PGmc **fadér* > PWGmc **fader* are a key diagnostic), and the specific **ō*-before-**r* development needed here (Ringe and Taylor 2014, 58–59). The direct lexical witnesses for **ō* in that environment are two independent etyma: PGmc **fedwōr* 'four' with WGmc reflexes OE *fēower* 'four', OFris *fiuwer* 'four', OS *fiuwar* 'four'; and PGmc **watōr* 'water' with OE *wæter* 'water'.

The rule is historically secure but narrow: final or pre-final **ō* before word-final **r*. The clearest evidence remains concentrated in the `four` and `water` material. No broader environment for **ō* is attested.

10.2 SC007. Lowering of final bimoric *ō before *r (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering Ctx([
    {*ō} -> {*a} || _ {*r} .#.
]);
```

OE *wæter* 'water' reveals why lowering must precede SC043 EAFBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* 'water' yields †*water* rather than expected OE *wæter* 'water': brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. Comparative support for the **ō*-before-**r* rule comes from the two lexical witnesses **fedwōr* 'four' and **watōr* 'water' and their WGmc reflexes; kinship **r*-stems belong to the broader pre-**r* shortening context, not to a direct **ō* > **a* control. Within CAPR, *wæter* 'water' is the form that establishes ordering before brightening. No broader lowering of **ō* is attested.

11 Coronal-w assimilation

11.1 Historical discussion

Ringe and Taylor treat the assimilation of **dw* and **zw* to **ww* as a shared Proto-West-Germanic innovation supported by one example of each input cluster (Ringe and Taylor 2014, 56–57; Stiles 1985, 89–94). The **dw* example is the numeral 'four': PGmc **feðwor* (Gothic *fidwor*) → WGmc **fewwar* → OE *fēower*, Old Frisian *fiuwer*, Old Saxon *fiuwar*. The **zw* example is the second-person plural pronoun, where two oblique case forms show the change: acc./dat. PGmc **izwiz* (Gothic *izwis*) → OE *eow*, Old Frisian *iu*, Old Saxon *iu*, OHG *iu*; and gen. Ringe and Taylor's PGmc **izweraz* (Gothic *izwara*) → OE *eower*, OHG *iuwer* (Ringe and Taylor 2014, 56). Stiles discusses the same pronominal material using his own reconstruction conventions and explicitly treats Gothic *izwara* among the relevant comparanda (Stiles 1985, 89–94). These two case forms belong to a single pronominal paradigm, not to two independent etyma.

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but the evidence for each cluster is confined: the numeral alone supplies the **dw* instance, and the oblique case forms of the second-person plural pronoun supply the **zw* instance. Both clusters are now witnessed in the corpus: 'four' for **dw*, and 'you' — selected in its dat.(-acc.) plural cell **izwiz* — for **zw*, deriving through **iwwi*, apocopated **iww*, to OE *ēow* 'you' (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, sect. 8.3, pp. 204–205).

11.2 SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation Ctx([
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
]);
```

OE *fēower* 'four' exposes a feeding relation: coronal assimilation creates the **ww* input of the early reanalysis in SC031 OEWWSimplification, yielding the diphthongal intermediate before later English realization. Ringe and Taylor explicitly give the coronal-assimilation and glide-reanalysis sequence (Ringe and Taylor 2014, 41–42). The earlier displacement result belonged to the superseded late **ww*-deletion proxy and does not by itself predict the result of displacing the reformulated early operation.

The numeral fixes that relative order. The pronoun now fixes a second one: assimilation must precede rhotacism (SC003 EAFRhotacism). The **z* of **izwiz* stands between vowel and **w*; had rhotacism applied first, it would have produced †*irwiz*, from which OE *ēow* 'you' can never be derived. The executable cascade composes the assimilation well before rhotacism, and the corpus derivation of *ēow* 'you' fails if the two are reversed. 'Four' remains the sole **dw* witness and the sole source of the coronal-assimilation → *ww*-simplification ordering constraint. The earlier boundary of the assimilation remains undetermined.

12 Inherited short-vowel-plus-geminate-glide reanalysis

12.1 Historical discussion

Inherited geminate glides after short vowels underwent reanalysis in Proto-West Germanic: the first glide became the offglide of a diphthong, while the second remained consonantal. This is not simply deletion of one *w*, and it is not the later English realization of the new diphthong. Ringe and Taylor distinguish those histories explicitly; Fulk specifies the short-vowel environment ((Ringe and Taylor 2014, 41–42, 65–66, 171–75; Fulk 2018, 117)).

Coronal assimilation feeds the change in *fēower* 'four' and *ēow* 'you': SC008 PWGmcCoronalWAssimilation supplies the geminate that becomes *eu*+*w* or *iu*+*w*. Inherited geminates also supply the histories of *ċēowan* 'chew', *dēaw* 'dew' and *hēawan* 'hew'. The resulting diphthongs enter the separate English realization described under SC032 OEDiphthongLeveling ((Ringe and Taylor 2014, 41–42, 65–66, 172–75; Campbell 1959, 45–47)).

Homorganic *uww* requires a separate quantity decision. We adopt Ringe and Taylor's preferred long-*u* account for *sċūwa* 'shadow', but retain their qualification of its evidence rather than treating the quantity as independently certain. Hypothetical inherited long-vowel-plus- *ww* sequences are not additional reconstructed witnesses: the held description specifies inherited geminates after short vowels. The parallel inherited *jj* development is independently governed and is not changed here ((Ringe and Taylor 2014, 65–66; Fulk 2018, 117)).

12.2 SC031. Inherited short-Vww reanalysis (`OEWWSimplification`)

```
define OEWWSimplification Ctx([
    {*a} {*w} {*w} -> {*au} {*w},
    {*á} {*w} {*w} -> {*áu} {*w},
    {*e} {*w} {*w} -> {*eu} {*w},
    {*é} {*w} {*w} -> {*éu} {*w},
    {*i} {*w} {*w} -> {*iu} {*w},
    {*í} {*w} {*w} -> {*íu} {*w},
    {*u} {*w} {*w} -> {*ū} {*w},
    {*ú} {*w} {*w} -> {*ū} {*w}
]);
```

The transport wrapper carries an independently selected sentence context outside the segmental rewrite; it adds no reconstructed sound and does not condition the reanalysis. Acute and unmarked spellings are notation variants, not alternatives in sentence stress. The retained executable name likewise does not determine the historical stage.

The modeled position after coronal assimilation and before j-gemination is a representative serialization that keeps inherited/coronal-created geminates distinct from the subsequently created j-path. It is not a newly proved strict chronology between the whole events. *Hīeġ* 'hay' and reconstructed West Saxon strew retain the separately adjudicated SC029 OEAwwjResolution path. *Hīew* 'hue' has distinct j-created reanalysis at SC033 OEEwLongDiphthong and long-io realization at SC110 OEJGlideIO. The retained SC106 OEJWWSimplification compatibility operation is no longer fed by hue ((Campbell 1959, 45–47; Ringe and Taylor 2014, 173, 250)).

For you, early *iu*+*w* now reaches SC098 PWGmcUnstressedWordFinalIApocope. Its weak-final context, not the former presence of literal *ww*, licenses apocope before mutation. Earlier source-backed reanalysis thus coexists with genuine prosodic conditioning rather than an output-restoring segment proxy ((Ringe and Taylor 2014, 41–42, 55, 57–58)).

13 *ij*-contraction in *friend*

13.1 Historical discussion

Ringe and Taylor describe a change of **ijo* to **iu* in the ancestor of *friend*, with the pathway PGmc **frijōnd-* (Gothic *frijonds*) → PWGmc **friund* → OE *frēond*, Old Frisian *frīund*, Old Saxon *friund*, Old High German *friunt* (Ringe and Taylor 2014, 62). The same source immediately warns that the **ijo* sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62). Luick (printed p. 118) notes that *iu* generalised within several *j*-stem paradigms through a related but differently conditioned loss of *j*, but does not supply a second example of the exact stressed **ijo* sequence (Luick 1914--1940, 118).

The change concerns a rare sequence attested only in the **frijōnd-* etymon and cannot safely be generalized into a broadly productive rule.

13.2 SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

```
define PWGmcIjContraction Ctx([
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
]);
```

Only the **frijōnd-* etymon tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* ‘friend’ yields †*friund* rather than expected OE *frēond* ‘friend’; moving contraction earlier within the tested range changes no output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond the **frijōnd-* etymon, precisely the reservation made by Ringe and Taylor.

14 West Germanic j-gemination

14.1 Historical discussion

Fulk treats West Germanic consonant gemination before *j after a short vowel as a regular development and illustrates it with forms such as OE *settan* ‘set’ and *lecgan* ‘lay’ (Fulk 2018, 127, §6.15).

The change applies specifically after a short vowel before **j*, not to geminate consonants generally.

14.2 SC010. West Germanic j-gemination (PWGmcJGemination)

```
define PWGmcJGemination Ctx([
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
]);
```

OE *nett* ‘net’ fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátją* ‘net’ yields †*nete* rather than expected OE *nett* ‘net’. Earlier movement of gemination changes no output.

The chronology is phonologically transparent: the consonant must geminate before **j* ceases to be consonantal. The witness establishes no earlier boundary.

15 Syllabic j after final-vowel loss

15.1 Historical discussion

Ringe and Taylor state directly that after final unstressed *a and *ą were lost, postconsonantal *j became syllabic *i, with outcomes behind OE *here* ‘army’ and *rice* ‘kingdom’ (Ringe and Taylor 2014, 46).

The sources establish the development, although the lexical evidence supplies little independent support for its position. Its scope is postconsonantal j, not high-vowel vocalization generally.

15.2 SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

```
define PWGmcSyllabicJ Ctx([
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
]);
```

The same PGmc **nátją* ‘net’ witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGemination yields †*nete* rather than expected OE *nett* ‘net’; moving it later changes no output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* ‘army’ and *rice* ‘kingdom’ as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

16 Simplification of the cluster **xs* before a consonant

16.1 Historical discussion

The change treated here is narrow. When **x* stood before **s* and that **s* was itself followed by a further consonant, the **x* was lost and the cluster was reduced. Campbell states the rule in exactly these terms, that when a consonant follows, **xs* becomes *s*, and he illustrates it with *wæstm* ‘growth’ and *wæsma* ‘growth’ beside *weaxan* ‘to grow’, and with Northumbrian *sesta* ‘sixth’ beside West Saxon *siexta* (Campbell 1959, 170, §417). Brunner gives the same statement and adds that the following consonant may be *j* as well as a true obstruent, citing *nēosian* ‘to visit’, *þīsl* ‘pole’, *wæsma* and *wæstm* (Brunner 1965, 184, §221.2). Bülbring formulates it as a pre-English change and is careful to say that it held as a rule, with exceptions (Bülbring 1902, 215, §527).

The cluster that the change requires is a specific one, and two neighbouring clusters show that the restriction is real. Where **xs* stood with no consonant after it, the **x* was not lost at all. It survived long enough to cause breaking and then hardened to the sound written *x*, that is [ks], as in *fox* ‘fox’, *siex* ‘six’, *weaxan*, *oxa* ‘ox’ and *fleax* ‘flax’ (Campbell 1959, 170, §416). Where **x* stood before a single consonant it was likewise kept. Campbell observes that once **xs* had become [ks], the only group in which **x* still stood before a voiceless consonant in earliest Old English was **xt*, and that this group remained, as in *feohtan* ‘to fight’, *miht* ‘might’, *niht* ‘night’ and *sōhte* ‘sought’ (Campbell 1959, 186, §464). The contrast is attested outside English as well. Old High German and Old Saxon *lastar* ‘reproach’ comes from **laxstra-* with the **x* lost before **st*, while Old English *leahtor* comes from **laxtra-* with the **x* kept before **t* (Campbell 1959, 170, §417).

The change is not Proto-Germanic. Gothic keeps the **h* of this cluster in *bi-niuhsjan* ‘to spy out’ and *saihsta* ‘sixth’, so the loss must be later than the separation of Gothic (Ringe and Taylor 2014, 157–58). Campbell reports the loss from the whole West Germanic area and from North Germanic as well, citing Old Norse *ísl* ‘axle’ and *nýsa* ‘to investigate’, Old Saxon *weslon* ‘to exchange’, *wastum* ‘growth’ and *niustan*, and Old High German *niusen* ‘to try’ (Campbell 1959, 170, §417). Ringe and Taylor set out the same comparative material and reach a more guarded conclusion. They derive Proto-Germanic **niuhsijanǫ* through Proto-West Germanic **niusjan* to Old English *nēosan* ‘to seek out, to visit’, and **sehstō* to Northumbrian *sesta*, and they take the Old Saxon agreement in *wastum*, *thisla* and *niusian* to show a shared northern West Germanic change. Against that they weigh the competition between *þixl* ‘pole’ and *þīsl* ‘pole’ in early Mercian, the survival of **x* in *eaxl* ‘shoulder’ from **ahslu*, and the retention in Old High German *sehsto* and *dihsala*. Their conclusion is that the **h* was lost, possibly variably and possibly only in some dialects, when two or more consonants followed, and that the loss may have been in part a parallel development in the diverging Northwest Germanic dialects (Ringe and Taylor 2014, 157–58). The existence of the change is therefore secure while its exact date and extent are not, and the rating given here reflects that division.

Two chronological anchors are available. Ringe and Taylor place the loss after the Proto-West Germanic syncope of **-CijV-*, since it is that syncope which brings the **s* and the **j* of **niuhsjan* together (Ringe and Taylor 2014, 157). They place it before breaking, observing that the undiphthongized vowels of *wæstm* ‘growth’ and *þīsl* ‘pole’ can be accounted for only by supposing that these **h* were lost before breaking took place (Ringe and Taylor 2014, 158). The rule is stated between those two points.

One witness in this collection undergoes the change. Proto-Germanic **funxstiz* ‘fist’ reaches the rule as **fúxstiz*, the long vowel having been produced by the Proto-Germanic loss of a nasal before **x* described in SC103 PGmcNasalLossBeforeX. The cluster **xst* is then reduced to **st*, and the word continues to Old English *fȳst* ‘fist’. The comparative set for this particular word is West Germanic throughout, with Old Frisian *fēst* ‘fist’, Old Saxon and Old High German *fūst* ‘fist’, Dutch *vuist* and German *Faust* (Kroonen 2013, 160; Orel 2003, 157). Neither Gothic nor Old Norse preserves a reflex of it, so the word bears on the domain of the change rather than on its date.

The word *thought* is the control. Proto-Germanic **θánxtē* also passes through SC103 PGmcNasalLossBeforeX, which leaves **θąxtē*, but the **x* there stands before a single **t* and not before **s*, so the present rule does not touch it and Old English has *þōhte* ‘thought’ with its *h* intact. The pair

fist and *thought* reproduces within this collection the same contrast that *lastar* and *leahtor* show across the West Germanic languages.

The relation between the two rules should not be overstated. The earlier rule does supply the cluster that this one simplifies, but the order of the two is established by their stages and not by any word in this collection. If the present rule were stated first, *fist* would still reach *fyst* ‘fist’, because removing the **x* from **funxstiz* leaves a nasal before *s*, and that nasal is removed with compensatory lengthening by the North Sea Germanic nasal-spirant law. The outcome is overdetermined, and the chronology rests on Gothic *bi-niuhsjan* ‘to spy out’ and *saihsta* instead.

16.2 SC028. Simplification of **xs* before a consonant (PNWGmcPreconsonantalXLoss)

```
define PNWGmcPreconsonantalXLoss Ctx([
    {*x} -> 0 || _ {*s} EnglishStarConsonant
]);
```

The **s* in the structural description carries the whole weight of the rule. Without it the rule would delete **x* before any two consonants, and it would then wrongly remove the first element of a geminate **xx* before **j*, where Old English in fact has *hliehhan* ‘to laugh’ with the geminate written *hh* (Campbell 1959, 186, §464). The rule fires on *fist* and on no other word in this collection, and it leaves *fox*, *six*, *wax*, *flax* and *ox* untouched, as it must, along with *thought*, *fight*, *night*, *light*, *might*, *knight*, *fright* and *wight*.

The implementation name retains an older description of the rule and will be brought into line in a separate pass.

17 *lþ*-voicing

17.1 Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan* ‘fold’, *beald* ‘bold’, *wuldor* ‘glory’, and *gylden* ‘golden’ (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold* ‘gold’, and *feld* ‘field’ (Campbell 1959, 169, §414).

The comparative evidence supports *lþ* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

17.2 SC012. Northern West Germanic *lþ*-voicing (`EAFLThVoicing`)

```
define EAFLThVoicing Ctx([
    {*θ} -> {*d} || {*l} _
]);
```

The `field`, `fold`, `gold`, and `wold` families preserve **lþ* to **ld*, but none dates the change against a neighboring rule. Every output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lþ* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

18 Dental hardening

18.1 Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative *ð became stop *d in all positions (Ringe and Taylor 2014, 43).

The change is systemic across early West Germanic and extends beyond any one lexical family.

18.2 SC013. Dental hardening (`PWGmcDentalHardening`)

```
define PWGmcDentalHardening Ctx([
    {*ð} -> {*d}
]);
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

19 Endingless *aw*: the regular straw branch

19.1 Historical discussion

The inherited nominative/accusative **stráwą* loses its final ending and vocalizes resulting final *w*, giving **strau*. Ordinary English diphthong development then produces reconstructed **strēa*. Attested *strēaw* ‘straw’ instead combines that long diphthong with *w* restored from inflected forms; the long nucleus also spreads into those forms. The regular free simplex and short obliques are explicitly unattested. The compound *strēaberige* ‘strawberry’ preserves the endingless branch, but is not substituted for an attested free noun (Campbell 1959, 46–47, 232–33; Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 172–73, 387).

The model keeps attested *strēaw* ‘straw’ as its comparison and produces regular reconstructed **strēa*: this is an explicit mismatch. The fragment below combines final-vowel loss and ensuing vocalization. Its narrow matcher does not make general final-vowel loss *w*-conditioned.

19.2 SC111. Endingless *aw* fragment (`PWGmcEndinglessAw`)

```
define PWGmcEndinglessAw [
    {*a} {*w} {*ą} -> {*au} || _ .#. ,
    {*á} {*w} {*ą} -> {*áu} || _ .#.
];
```

The ordinary *au* product feeds SC030 OEAuBrightening and SC032 OEDiphthongLeveling, not the retained show proxy SC034 OEAwLongDiphthong. The early holding site is compatible with the source’s West Germanic history, not evidence for strict relative order against unrelated dental hardening (Ringe and Taylor 2014, 172–73).

20 Northwest Germanic u-lowering

20.1 Historical discussion

Northwest Germanic lowered **u* to **o* when the following syllable contained a non-high vowel. Campbell describes the change and lists *ġeoc* ‘yoke’ among its regular outcomes (Campbell 1959, 42–43, §115); Fulk gives the same word as a standard example — “OIcel. ok, OE geoc, OHG joh beside juh and OS juk” — and notes the paradigmatic alternation between lowered and unlowered stems that the conditioning produced (Fulk 2018, 56, §4.3). A word-initial **j* does not block the change: the blocking effect of *j* concerns only a consonantal *j* standing between the target vowel and the conditioning vowel, as in the class I weak verbs of the *cnyssan* ‘strike’ type (Fulk 2018, 56, §4.3). Ringe and Taylor accordingly reconstruct the Proto-West Germanic paradigm of ‘yoke’ with the lowering applied (Ringe and Taylor 2014, 129).

Regular corpus witnesses include ġeoc ‘yoke’, *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’. Where the following syllable kept a high vowel the lowering did not apply, as in *ġeoguþ* ‘youth’, whose root *u* survived (Brunner 1965, 64–65, §92.1).

Regular lowering does not imply that every attested inherited word preserves the lowered vowel. The masculine noun rust ‘rust’ is a lexical residue: selected PGmc **rústaz* ‘rust’ regularly predicts **rost* ‘rust’. Orel’s alternative **rustō* ‘rust’ gives the same lowered root; his masculine alternative is independently supported by Ringe (Orel 2003, 309; Ringe 2017, 247–48). Since **frústą* ‘frost’ regularly gives forst ‘frost’, a general *-st-* exemption is unwarranted. The retained *u* remains unexplained, alongside the other retained-*u* cases such as *bucc* ‘buck’, *fugol* ‘bird’, *wulf* ‘wolf’ and *wulle* ‘wool’; their consonantal environments do not establish a general replacement law (Campbell 1959, 43, §115; Ringe and Taylor 2014, 27–34). The lowering law is retained; its non-high conditioning class includes nasalized long **ǭ*, not just oral **ō*. The independently reconstructed accusative **wullǭ* therefore lowers just as the nominative **wullō* does under the preferred chronology (Ringe 2017, 242–43). This does not merge either ending with trimoric **ô*. The rust lexical discussion distinguishes the selected masculine citation, Orel’s alternative and diagnostic paradigm constructions.

20.2 SC017. Lowering of **u* before following non-high vowels (PNWGmcULowering)

```
define PNWGmcULowering Ctx([
    [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
    ]
]);
```

The implementation lowers the first nucleus through a consonantal interval without **j*; the immediately following consonant must not be nasal. This literal guard is not a claim that every nasal anywhere in the interval blocks the historical process. Fulk distinguishes tautosyllabic and heterosyllabic nasal environments (Fulk 2018, 56).

The lowering feeds the much later West Saxon palatal-glide spelling (SC016 OEWsPalatalGlide): the *o* that the scribes wrote in *ġeoc* ‘yoke’ is the output of this change, so PGmc **júką* ‘yoke’ passes through **joką* on its way to the attested spelling (Fulk 2018, 56, §4.3; Ringe and Taylor 2014, 129). If lowering is placed after SC019 PNWGmcFinalLongORaising, selected **núsō* ‘nose’ yields †*nusu* rather than expected *nosu*, PGmc **skúflō* ‘shovel’ yields †*sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* ‘sorrow’ yields †*surg* rather than expected *sorg*. These are conditional computa-

tional consequences, not independent proof that those paradigms underwent no levelling. CAPR provisionally prefers lowering before final raising because it regularly accounts for these low-root forms without additional, individually unproved analogical generalizations. Nose already involves earlier stem remodelling, and the retained high vowels of wool and love remain a genuine cost of that preference. The main comparison of Fulk, Ringe and Taylor, and Stiles, including the Runic evidence and the plausible but unproved nominative-generalization account of wool, is in SC019 PNWGmcFinalLongORaising.

21 Stressed monosyllable *ō-raising

21.1 Historical discussion

Campbell treats the development of final accented ō to ū in stressed monosyllables directly, with the familiar outcomes behind *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' (Campbell 1959, 47, §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

21.2 SC018. Raising of final stressed monosyllabic **ō* (PNWGmcStressedMonosyllableORaising)

```
define PNWGmcStressedMonosyllableORaising Ctx([
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
]);
```

Campbell's *cū* 'cow', *hū* 'how', and *tū* 'two' establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 PNWGmcStressedMonosyllableORaising with neighboring changes leaves every output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

22 Raising of final unstressed long **ō*

22.1 Historical discussion

Final unstressed oral bimoric **ō* became short **u* in Northwest Germanic. The qualification matters: oral **ō*, nasalized **ǭ* and trimoric **ô* were not interchangeable endings. Ringe and Taylor distinguish the first two developments explicitly; Fulk also distinguishes the quantities when reconstructing the endings (Ringe and Taylor 2014, 15–16, 58–60; Fulk 2018, 82–86, 155–56).

The nominative singular **wullō* 'wool' and accusative singular **wullǭ* are independently reconstructed by Ringe. Their distinct endings must survive the comparison even when a later paradigm levels the root vowel (Ringe 2017, 242–43). Likewise, an oral vowel followed by **z* was not yet word-final. These distinctions explain both the scope of raising and its interaction with later loss.

22.2 SC019. Raising of final unstressed long **ō* (PNWGmcFinalLongORaising)

```
define PNWGmcFinalLongORaising Ctx([
    [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
    ]
]);
```

The preceding vowel and consonants select a polysyllabic input. The distinct stressed-monosyllable development is retained separately in SC018 PNWGmcStressedMonosyllableO-Raising; it is not absorbed into this unstressed ending law.

22.2.1 Lowering before raising: the preferred account

Fulk argues that *scolu* 'troop, shoal', from **skulō*, requires lowering of the root before final **ō* became **u*. If raising had already removed the non-high conditioner, an unchanged nominative would retain root **u* (Fulk 2018, 56). CAPR provisionally follows this order: SC017 PNWGmcU-Lowering, then final oral raising. It regularly gives the low roots of *nosu* 'nose', *sċofl* 'shovel' and *sorg* 'sorrow', rather than requiring a later root-vowel generalization for each selected form.

This is an argument from explanatory economy, not from the number of matching outputs. Fulk supplies the direct *scolu* argument; the comparison of its explanatory costs with the other selected paradigms is CAPR's assessment. The witnesses are not equally independent. *Nosu* already depends on an earlier remodeled zero-grade stem beside the comparative **nasō-*; it is not an untouched inherited nominative (Kroonen 2013, 383; Campbell 1959, 43, §115). For *sċofl* 'shovel', Luick also infers an earlier high-root doublet, **scufel*, beside *scofel* 'shovel'; that inference makes an alternating paradigm credible but does not establish its direction of levelling (Luick 1914--1940, 105–6, §78). The comparative formations of shovel and sorrow support the selected short-vowel stems, not a directly attested early English levelling operation (Orel 2003, 347; Kroonen 2013, 493). The preferred sequence avoids those additional operations, but it does not thereby explain every retained high vowel.

22.2.2 Raising before lowering: a coherent alternative

Stiles expressly places final **ō* raising before a-umlaut of **u* (Stiles 2012, 45). Ringe and Taylor explain how *wull* 'wool' and *lufu* 'love' could preserve a nominative root **u* by generalizing it through a paradigm whose other cells had lowered **o*; they also leave open genuine exceptions (Ringe and Taylor 2014, 32). With the nasal accusative correctly recognized, the alternative has the following regular consequences. The table gives reconstructed sound-law predictions, not unattested forms promoted to attestations.

Paradigm	Lowering before raising	Raising before lowering
Wool: nominative / accusative	**woll* / **wolle*	**wull* / **wolle*
Love: nominative / accusative	**lofu* / **lofe*	**lufu* / **lofe*
Shoal: nominative / accusative	**sċolu* / **sċole*	**sċulu* / **sċole*

The inherited ending opposition follows Ringe's reconstructed paradigm and the independently described ō-stem inflections; it is not a special ending invented to rescue wool (Ringe 2017, 242–43; Fulk 2018, 154–56). Under earlier raising, nominative-to-oblique generalization can account for attested *wull* and *lufu/lufe* 'love'. Conversely, generalization of the lowered oblique root can account for *scolu* without replacing its nominative ending. The same possibility exists for nose, shovel and sorrow. Neither particular direction is directly witnessed in the reconstructed early English paradigms.

There is independent evidence that levelling occurred: Ringe and Taylor discuss early *Holtijaz* and the daughter and ox paradigms, while Stiles distinguishes conditioned Runic forms from analogical generalizations (Ringe and Taylor 2014, 33–34; Stiles 2012, 47–48). That establishes the mechanism's credibility, not its occurrence in every inconvenient lexeme. Continental *skola* is not an untouched nominative control either: Fulk describes accusative-ending generalization into the nominative in Old Saxon and Old High German (Fulk 2018, 154–55). Thus his *scolu* argument is strong only conditional on the inherited nominative root not having been remodeled. It does not refute Ringe and Taylor's phonologically coherent wool account.

There is a further lexical caution: Fulk elsewhere calls *scolu* 'host' a loan from Latin *schola* 'school', whereas his chronology argument expressly starts from inherited **skulō* (Fulk 2018, 47, n. 1, 56). These analyses cannot silently supply independent confirmation of one another. The comparison here evaluates his inherited analysis.

The balance nevertheless favors retaining lowering first as a working account: several independently selected low-root forms follow without additional, individually unproved levelling. Wool and love remain its real explanatory cost. Ringe and Taylor's alternative is plausible but unproved, not a demonstrated error. Analogy belongs to the separately argued history of a paradigm, not to the exceptionless sound laws.

22.2.3 What the Runic evidence dates

Fulk cites *horna* 'horn' on the Gallehus horn, about 400, for lowering and *waritu* 'I write' on the Järsberg stone, about 450, for final oral raising (Fulk 2018, 56, 82). These are separate positive chronological bounds. An occurrence of raising around 450 does not establish that it began after 400. Nor does a written final *o* in another inflection establish retention of oral bimoric **ō*: nasalization, quantity and a following consonant must first be identified. Stiles's treatment of the Runic distribution also requires distinguishing conditioned realization, phonemicization and analogical extension (Stiles 2012, 47–48, 54–60). No new general revaluation law follows from those distinctions.

Fulk's note cites Ringe and Taylor on lowering and final **-az* loss, not as an explicit rebuttal of their wool explanation (Fulk 2018, 57, n. 1; Ringe and Taylor 2014, 27–34). Stiles predates their book. Luick discusses alternation and rejects a proposed labial protection law using regular low-root counterexamples; Campbell and Bülbring list retained-high-vowel cases (Luick 1914--1940, 104–6, §§77–79; Campbell 1959, 43, §115; Bülbring 1902, 45–46, §116). None of these checked passages supplies our exact comparative economy argument. No specific later criticism or defence was verified in the bounded literature search; this is not a claim of novelty or of its absence from all scholarship. The Campbell and Bülbring passages remain text-checked rather than newly certified against original images.

22.2.4 Final consonants and later ending reduction

In **rástōz* 'rest', **z* protects oral **ō* from this raising. After SC020 EAFFinalZDeletion, the newly final long vowel can undergo SC042 PWGmcSurvivingBimoricOUnrounding and later fronting

and shortening, yielding *ræste* ‘rest’. Moving raising after final consonant loss wrongly exposes that vowel and gives **rast* instead. This relation is retained separately; qualifying the lowering/raising comparison does not weaken it.

Nasal **ǭ* does not raise here. Its later reduction yields the accusative ending in **wolle*; the current serialization carries the nasal symbol to that reduction rather than claiming that nasality historically lasted until then. The genitive alternatives must also remain distinct: Ringe and Taylor reconstruct bimoric **-ōz*, where Fulk argues for trimoric **-ôz* (Ringe and Taylor 2014, 59, 299; Fulk 2018, 155). The trimoric genitive’s complete pathway remains unmodeled; it is not silently replaced by a bimoric genitive or covered by the present nasal-vowel repair.

23 Chapter 3. From Proto-West Germanic to Anglo-Frisian

23.1 Historical interval

This chapter covers the sound changes that occurred after the Proto-West Germanic period and before, or during the emergence of, the specifically English line. The starting reconstruction is Proto-West Germanic; the end point is the Anglo-Frisian ancestor required by the model's historical tree. Its existence and the identification of its innovations are separate questions. Its inventory must be reconstructed from conditioned laws and daughter chronologies, not from the traditional subgroup label alone.

23.2 A necessary terminological caution

The title of this chapter uses "Anglo-Frisian" as an organizing historical concept. That choice requires an explicit qualification.

The scholarly literature uses several overlapping terms for this developmental period:

- North Sea Germanic and Ingvaeonic: labels used by some scholars for a proposed subgroup comprising Old English, Old Frisian, and Old Saxon (or more narrowly, Old English and Old Frisian only). The innovations associated with this label — especially the nasal spirant changes and certain vowel developments — are sometimes described as diffusion rather than shared inheritance (Bremmer 2009, 24–27, 126–28).
- Anglo-Frisian: a label used specifically for the Old English / Old Frisian branch, or for innovations shared between the two languages. Its use presupposes a tighter relationship between English and Frisian than between either and Old Saxon.
- Proto-Anglo-Frisian (PAF): the strongest interpretation, positing a discrete reconstructed common ancestor for Old English and Old Frisian specifically. This is the position of Ringe and Taylor, who reconstruct a PAF stage between Proto-West Germanic and Proto-Old-English (Ringe and Taylor 2014, 54–68).

CAPR requires a discrete Anglo-Frisian ancestral node. This does not guarantee that every traditionally associated innovation is inherited from it, or exclusive to English and Frisian. A broader innovation can already be present in its input; similar daughter outcomes can also result from independent events. Authors' diffusion interpretations remain accurately attributed, but they cannot resolve an incompatible chronology within CAPR's ordered tree (Versloot 2017, 302–19; Bremmer 2008, 284, 292–93).

23.3 A worked historical tree test

If a supposed shared innovation obligatorily follows a daughter-only change in Frisian which English never underwent, it cannot occupy the common stem under those premises. Either the ordering or scope premise must change, the innovation happened independently, or the apparently identical events are historically different. The required ancestor is not removed.

Campbell reconstructs Frisian au contraction before ordinary fronting; Goblirsch instead proposes a shared diphthong followed by Frisian stress shift and contraction (Campbell 1939, 91, 104; Goblirsch 1991, 17, 20–21). The daughter-only event is a predecessor in the first account and a follower in the second. The second is compatible with sharing, but does not attest the reconstructed Frisian intermediate. Surface agreement, compatibility with inheritance, compelled parallelism under premises, and unresolved history must therefore remain distinct judgments.

23.4 The working account and why it is used

The English reconstruction adopted as a working scaffold is the conventional account developed by Campbell and critically refined by Ringe and Taylor. It distinguishes inherited long vowels from the long vowels produced by diphthong contraction, and it treats fronting, breaking, restoration and mutation as conditioned phonological developments. That scaffold is useful because it

specifies contrasting inputs and intermediate states, not because one author's authority decides the history (Campbell 1959, 52–53, 60–72; Ringe and Taylor 2014, 170–73, 215–37).

The comparative specialists test particular load-bearing premises. Campbell's argument about English and Frisian fronting, Versloot's runic interpretations and Goblirsch's proposed lost Frisian diphthong do not form one automatically compatible package (Campbell 1939, 90–91, 104; Versloot 2017, 295–97, 318; Goblirsch 1991, 17–21). Where an alternative changes the inherited inventory or the identity of an event, it must be evaluated as that alternative, not inserted into the conventional chain without its premises.

CAPR implements regular sound laws. Prosodic and phonological conditions may distinguish environments, but lexical diffusion, grammatical categories and individually selected exceptions cannot repair a contradiction. Paradigm remodeling and borrowing may affect which historical form a lexical entry represents; they are not thereby conditions on a sound law. This methodological choice is distinct from faithfully reporting an author's different theoretical account.

23.5 Long vowels, diphthongs and the ancestral inventory

The inherited-long contrast is central. On the conventional reconstruction, inherited oral long **ā* has already acquired a front quality when English stressed **ai* completes its development to a new back **ā*. Otherwise the new vowel should share the inherited vowel's fronting. Ringe and Taylor formulate this as a relation between temporal milestones: inherited-long fronting must be well under way before contraction completes. They allow overlap; a serial implementation is not proof that the changes occupied disjoint periods (Campbell 1939, 90; Ringe and Taylor 2014, 170).

The inherited vowel itself is disputed. Fulk's retained-front reconstruction does not require precisely the same earlier backing and re-fronting as Ringe and Taylor's account. That is a disagreement about the input inventory, not a small adjustment to the position of an otherwise identical rule (Fulk 2018, 60–61; Ringe and Taylor 2014, 10–13). Nielsen also proposes a different causal relation between contraction and long-vowel restructuring, allowing co-occurrence rather than imposing the conventional strict sequence (Nielsen 2001, 514–16, 521).

Luick's proposed fronted-ai path is a further alternative, not an observed intermediate series. His conjecture is **æe* > **æə* > **æa* > **ā*; the middle offglide is schwa. Campbell rejects that explanation, but the rejection is not an independent observation that such a phonetic path is impossible (Luick 1914--1940, 132–33; Campbell 1939, 90, n. 3). The inherited-long and secondary-long histories must consequently remain separate even where later mutation makes their final vowels converge.

Nasal environments require another distinction. Nasalization, loss of a nasal with compensatory lengthening, rounding and eventual merger are not one event. The nasalized input can escape an oral-fronting law before its later rounded outcome is established. A shared final vowel therefore does not place every substep at the same ancestral node (Bremmer 2009, 24–27; Ringe and Taylor 2014, 142–46).

The following table specifies the relevant inventory questions at a conservative ancestral cut. It is a working reconstruction, not a newly adjudicated full phoneme inventory. Earlier West Germanic developments can be inherited at this cut without being exclusively Anglo-Frisian innovations.

Input class	Conservative working ancestral state	English daughter consequence	Remaining premise
Inherited oral long vowel	Front long vowel under the conventional account	Subsequent dialectal and palatal treatment	Earlier back versus retained-front input
Nasal long vowel	Distinguished from the oral long vowel	Rounding and merger must be dated separately	Which nasal substeps precede the cut

Input class	Conservative working ancestral state	English daughter consequence	Remaining premise
Stressed ai	Retained diphthong before completed English contraction	Secondary back long vowel, eligible for later mutation	Runic interpretation and any limited ancestral onset
Ordinary short non-nasal a	Unfronted at this conservative cut	Ordinary English fronting	Daughter contraction before fronting
Inherited au	Not identified with completed English or Frisian outcome	English front element and later offglide development	Identity with ordinary short fronting
Supported inherited Vww classes	Adopted West Germanic a/e/i-ww reanalysis; separate homorganic uww quantity	Later English diphthong realization, not inherited English surface vowels	Qualified long-u shadow account; j-created classes remain separate
Velars and palatal tendencies	No blanket completed assibilation is stipulated	Class-specific daughter histories	Articulation, cutoff and merger are distinct

The evidence underlying these qualifications is component-specific (Ringe and Taylor 2014, 41–42, 65–66, 170–73; Versloot 2017, 295–97, 318; Laker 2007, 167–84). The table does not assign inherited-short fronting to the stem merely because the node is called Anglo-Frisian.

The inherited-glide component is now implemented rather than left as a possible future decomposition. SC031 OEWWSimplification belongs to the earlier West Germanic account discussed in Chapter 2: short nonhomorganic **Vww* gives **Vuw*, with retained consonantal **w*; homorganic **uww* separately gives long **ūw*. The latter follows Ringe and Taylor's preferred but qualified shadow analysis. These earlier products can be inherited at the Anglo-Frisian node without placing the later English long diphthongs there. The selected sentence context for early apocope is independent of the segmental reconstruction and of lexical accent (Ringe and Taylor 2014, 41–42, 55, 57–58, 65–66, 171–75; Fulk 2018, 117).

23.6 Alternative cuts and their tree consequences

A more expansive common-stem reconstruction is possible only with different premises. In Luick's account, participation of the diphthong nucleus in fronting can coexist with a conjectured later Frisian development. In Goblirsch's account, a common fronted au diphthong precedes Frisian stress shift and contraction. Both require historical intermediates whose existence must be argued, not obtained from the name of the subgroup (Luick 1914--1940, 130–33; Goblirsch 1991, 17, 20–21).

Proposed common-stem content	Required premise	Conditional consequence
Inherited-long restructuring, but retained ai and short a	Completed English contraction is later	Compatible conservative cut
Ordinary short fronting	Its secure predecessors can also occur before the split	Excluded if daughter-only English contraction obligatorily precedes it
Shared fronted-au development	Frisian stress shift/contraction follows that development	Compatible in Goblirsch's account, not proof of inheritance
Shared consonantal palatalization	Corresponding consonant classes and necessary predecessors are genuinely identical	Unresolved; apparent similarities do not settle all layers

Campbell's English chain makes the second row especially consequential: if completed contraction is English-only and ordinary fronting must follow it, the latter cannot be inherited from the common stem. The corresponding English and Frisian frontings are then parallel under those premises. Changing a disputed premise can change that conclusion, but cannot remove the required ancestor (Campbell 1939, 90–91; Versloot 2017, 295–97, 318).

23.7 Palatalization is not a single subgroup character

A palatal articulation can precede phonemic contrast, and the productive cutoff of a conditioning process can precede its eventual assibilated reflex. Fricative-g merger with j is different again. Geminate and postnasal stops, singleton fricatives, k, h and sk must not be treated as one dated change simply because later spellings look palatal (Hogg 1979, 100–111; Laker 2007, 167–68, 175–84).

The shared-palatalization hypothesis must therefore identify which layer is inherited. A daughter-specific predecessor can exclude a completed component from the stem without excluding every earlier articulatory tendency. Conversely, removing an argument against sharing establishes compatibility, not that one common event actually occurred (Laker 2007, 175–84). The vowel diphthongization following initial palatals is another event; Luick explicitly distinguishes the similar Frisian and English outcomes (Luick 1914--1940, 162–63).

The English implementation now distinguishes singleton palatal fricatives from gg/ng stops and adopts postmutation fricative merger as a working account (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32). Hogg's objection remains explicit (Hogg 1979, 102–11). This daughter account does not select a complete ancestral inventory, exclude every shared early articulatory tendency, or date stop affrication from an eventual written reflex.

23.8 What the runes date

An inscription dates a written object. A sound-change terminus additionally requires the reading, etymology, linguistic affiliation and relation of the grapheme to phonetic or phonemic structure. The interpretation of the Caistor inscription as retaining ai and the interpretation of Frisian retention are consequential but conditional (Versloot 2017, 295–97, 318). Waxenberger likewise distinguishes early allophonic mutation from later phonemicization and lacks immediate runic evidence for every short-vowel split (Waxenberger 2019, 63–74).

No one calendar date is therefore attached here to every change described as Anglo-Frisian. The English discussion that follows starts from the working inventory and makes its necessary daughter developments explicit. Canonical classification, preferred reconstruction and an unresolved component are kept distinct until individual adjudication.

23.9 Major changes and their historical basis

23.9.1 West Germanic rhotacism (SC003)

The medial change of `*z` to `*r` in environments such as **déuzaz* 'deer', **xúrdaz* 'hoard', and **líznōjaną* 'learn' is historically a post-Proto-West-Germanic development. Ringe and Taylor argue that rhotacism was not inherited from Proto-Northwest Germanic and was not uniform within West Germanic (Ringe and Taylor 2014, 52, 98, 102). Crist separates this change explicitly from the deletion of word-final `*z` and argues that rhotacism must follow the deletion rules (Crist 2001, 104–6; 2002, 1, 4). Bammesberger gives the standard Old English–facing summary: `*z` yielded `*r` in intervocalic position but was generally lost in final position (Bammesberger 1992, 39).

The CAPR rule is named `EAFRhotacism`, placing it in the Early Anglo-Frisian corridor, CAPR's operational post-Proto-West-Germanic stage on the English line; the reader-facing chapter label describes the change as a West Germanic rhotacism.

23.9.2 Word-final *z deletion (SC020) and the three final-*z developments

The loss of word-final *z is not one process but three historically distinct developments, and this chapter contains two of them. The central one, SC020, is the Proto-West Germanic loss of *z in unstressed syllables, seen in forms such as **rástōz* 'rest (nom.sg.)' and stated for the whole branch by Ringe and Taylor (Ringe and Taylor 2014, 44–45); Crist's analysis distinguishes it both from the earlier NWGmc changes and from the later narrower Ingvaeonic deletion rules (Crist 2002, 1, 4). Earlier still, the consonant-stem root nouns had generalized endingless nominatives before Proto-West Germanic (SC096, Chapter 2). Later, and only in the north, *z was lost in stressed monosyllables with compensatory lengthening (SC097, this chapter); the southern dialects instead retained and rhotacized it. The standard handbooks confirm the West Germanic deletion in general terms: Bammesberger gives a clean statement that Germanic *z is generally lost in final position (Bammesberger 1992, 39); the three-way division refines those summaries rather than contradicting them.

The CAPR rule for the unstressed loss is still named EAFFinalZDeletion, an identifier that predates the restaging of the change to Proto-West Germanic; the name is retained as a stable identifier pending a global renaming pass, and the historical stage recorded in the staging metadata takes priority over the name prefix. SC020 remains presented in this chapter, beside rhotacism, because the two changes jointly determine the fate of every remaining *z.

23.9.3 Completed English ai contraction (SC004, treated in Chapter 4)

The English outcome of stressed/root **ai* is **ā*. Versloot distinguishes an early velar-conditioned Frisian contraction from later non-velar treatment, with mutation and delabialization intervening (Versloot 2017, 302–9, 316). His wave account is an author's interpretation, not CAPR's own genealogy. CAPR now places completed English contraction on the daughter branch, while preserving the distinct possibility of a narrower conditioned ancestral onset (Campbell 1939, 90–91; Versloot 2017, 295–97, 318). This local decision does not select every component of the ancestral inventory.

The current SC004 EAFAiMonophthongization models English contraction after SC101 EAFLongAFronting. The new ai-derived **ā* escaped the earlier fronting of inherited oral **ā*; Campbell states that inference and Ringe and Taylor retain it (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70). Soul provides another implementation boundary at interstress raising, not the sole historical argument or proof of inherited contraction.

The unstressed development *ai > *ē (in final and nonfinal syllables) is a separate and earlier Proto-Northwest Germanic change (SC014), discussed in Chapter 2; its corpus witnesses are the dative-singular endings of span (**spánnai* 'span' > *spanne* 'span') and meed (**mízdai* 'meed' > *meorde* 'meed').

23.9.4 Anglo-Frisian brightening (SC043, treated in Chapter 4)

The fronting of low **a* to **æ* outside nasal environments is traditionally called Anglo-Frisian brightening. The name does not establish one inherited event. It executes later in the cascade than the changes of this chapter, so its full section appears in Chapter 4; it is introduced here because it anchors the "Anglo-Frisian" label that names this period. Campbell gives the classical statement: "By a very early change Prim. Gmc. a > æ in OE and OFris. when not followed by a nasal consonant" (Campbell 1959, 52, §131). The traditional name also appears in modern comparative treatments; it does not independently establish a shared event (Fulk 2018, 72–73).

The change is notable for what follows it: OE Breaking presupposes the fronted input; OE a-Restoration partially undoes it in back-vowel environments. The three-change sequence is the conventional English working account, but its components and conditioning remain individually arguable. Versloot proposes restricted early fronting and late e-breaking; final forms can coincide under front-and-restore and never-front histories (Versloot 2025, 123, 125–28, 131–35).

Campbell notes that English and Frisian may not simply reflect one undifferentiated shared prehistoric event, and Ringe and Taylor leave open whether the wider spread of fronted outcomes

happened mainly on the continent or in Britain (Ringe and Taylor 2014, 60–62). CAPR's implementation treats the stressed English component separately from the retained unstressed and final-vowel representations.

The completed ordinary stressed component is now characterized on the English daughter under Campbell's conventional working chain (Campbell 1939, 90–91; 1959, 52–53). Ringe and Taylor's diphthong-nucleus objection remains explicit (Ringe and Taylor 2014, 170–75); earlier restricted ancestral tendencies are not excluded. English plain-a/au process identity is distinct from inherited stem identity, and their current serialization is retained. The unstressed and final-vowel contributions are not assigned the same date from this argument. This local adoption does not select the complete ancestral inventory.

23.10 Cascade vs. historical order in this chapter

The sections follow executable serialization so that derivations are inspectable. That order is not an independently demonstrated total historical chronology. The broader final-z loss, later northern loss and rhotacism are distinguished by their actual environments (Ringe and Taylor 2014, 44–45; Crist 2002, 1, 4).

Earlier-stage proxies remain in this editorial corridor. The cluster change represented by SC022 PNWGmcMnDissimilation and final-n loss represented by SC023 PNWGmcNStemNLoss are classified as Common Germanic/Proto-Germanic despite retained name prefixes (Fulk 2018, 121; Ringe 2017, 101–3). The retired unstressed-o raising is not an active member. Book order, computational position and historical stage must remain separate.

24 West Germanic final *z-deletion

24.1 Historical discussion

Word-final *z in unstressed syllables was lost in Proto-West Germanic. Ringe and Taylor state the change for the whole branch and illustrate it with the nominative plural **dagōz* > **dagō* and the consonant-stem nominative **fadurz* > **fadur*, noting that the ending is lost after consonants as well as after vowels (Ringe and Taylor 2014, 44–45, §3.1.1). Crist's handout formulates the same development and its Ingvaeonic sequels (Crist 2002, 2, §§5–6). The change is pan-West-Germanic, not specifically Ingvaeonic: every West Germanic daughter shows the loss, and the Frienstedt comb inscription *kaba* < **kambaz* 'comb' (c. 250–300 CE) supplies early epigraphic confirmation (Fulk 2018, 25, n. 1).

The conditioning segment is specifically the voiced sibilant *z, never *s: Ringe and Taylor's near-minimal pair of nominative singular **dagaz* > **dag* beside genitive singular **dagas*, which keeps its sibilant into Old English *dæġes*, shows that the change reads the Verner voicing distinction (Ringe and Taylor 2014, 212, §6.1). Where the handbooks disagree about whether a given ending had *-*s* or *-*z* — as for the nominative plural *-*ōz* — the disagreement matters directly to whether this rule applies (Ringe and Taylor 2014, 115–16, §4.2.1).

This is the middle of three historically distinct final-*z developments, and Ringe and Taylor explicitly separate it from the later loss in stressed monosyllables, citing Crist's demonstration that they are two changes (Ringe and Taylor 2014, 44–45, §3.1.1). Earlier, the consonant-stem (root-noun) nominatives of monosyllables had already generalized endinglessness before Proto-West Germanic (SC096 RootNounNomZLoss), so forms like **bōkz* 'book' never reach this rule with their marker intact. Later, and only in the north, *z was lost in stressed monosyllables with compensatory lengthening (SC097 MonosyllabicFinalZLoss); the present rule leaves stressed monosyllables untouched. Older accounts that grouped all of these under one loss of final *z, such as Campbell's, are superseded by this three-way division (Campbell 1959, 166).

At the boundary with Chapter 2's Northwest Germanic sequence, the derivation of *ræste* 'rest' shows that final **ō*-raising (SC019 PNWGmcFinalLongORaising) must precede this rule: raising applies to *-*ō* but not to *-*ōz*, whose final vowel is still sheltered by the sibilant when raising runs (Ringe and Taylor 2014, 15–16, 24). On the later side, Ringe and Taylor order the loss of *z before

the loss of word-final bare *-*a*, since **dagaz* first becomes **daga* and only then **dag* (Ringe and Taylor 2014, 45–46, §3.1.2).

24.2 SC020. West Germanic final *z-deletion (`EAFFinalZDeletion`)

```
define EAFFinalZDeletion Ctx([{*z} -> 0 ||
    .#. ?* EnglishStarVocalic
        [EnglishStarConsonant | EnglishPalatalConsonant]+
        EnglishStarVocalic ?* _ .#.,
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
        EnglishStarVocalic+
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.]);
```

The rule deletes word-final *z in unstressed syllables, stated through two environments. The first clause covers polysyllables, where the final syllable of these corpus forms is unstressed: this is the ordinary case, with 110 corpus derivations, such as PGmc ***bárdaz*** ‘beard’ on its way to OE ***beard*** ‘beard’ and ***rástōz*** ‘rest’ on its way to ***ræste*** ‘rest’. The second clause covers post-consonantal *z in monosyllables. By the time this rule runs, SC096 RootNounNomZLoss has already removed the genuine root-noun nominative endings, so the only form reaching the second clause is ***fríjōndz*** ‘friend’, contracted to monosyllabic ***fríundz*** by SC009 PWGmcIjContraction; its ending, like that of ***fadurz***, stood in an unstressed syllable when the Proto-West Germanic change applied and so belongs here rather than to the root-noun development (Ringe and Taylor 2014, 44–45, §3.1.1). Stressed monosyllables ending in vowel plus *z meet neither clause and are left for SC097 MonosyllabicFinalZLoss.

The chronology of word-final *z-loss is unusually well delimited: ***ræste*** ‘rest’ supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 PNWGmcFinalLongORaising, PGmc ***rástōz*** ‘rest’ yields †***rast*** rather than expected OE ***ræste*** ‘rest’. After SC040 OEMedUnstressedULowering, PGmc ***bébruz*** ‘beaver’ yields †***befro*** rather than expected ***befer*** ‘beaver’, PGmc ***kwéðuz*** ‘cud’ yields †***cwedo*** rather than expected ***cwedu*** ‘cud’, and PGmc ***félθuz*** ‘field’ yields †***feldo*** rather than expected ***feld*** ‘field’, alongside eight other newly failing rows. Final *z*-loss therefore follows SC019 PNWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The ***rástōz*** ‘rest’ derivation fixes the local relation to SC019 PNWGmcFinalLongORaising. The distant boundary at SC040 OEMedUnstressedULowering shows only that word-final *z-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final *ō-raising.

25 Early apocope in unstressed words

25.1 Historical discussion

Alongside the regular loss of word-final short high vowels in third syllables (SC006 PWGmcEarlyIApocope), Ringe and Taylor identify a second, earlier apocope: “Short high vowels were also lost after heavy syllables in unstressed words” (Ringe and Taylor 2014, 57–58, §3.1.4). The two laws must not be conflated. Fully stressed disyllables kept their final **-i* long enough to cause i-umlaut — OE *ġiest* ‘guest’ < **gastiz* and *fȳr* ‘fire’ < **fūri* require exactly that survival (Ringe and Taylor 2014, 55, §3.1.4) — whereas words that carried no sentence stress lost the vowel already in Proto-West Germanic.

The conditioning is prosodic. Like Verner’s law, the change is governed by accent: it applied in words unstressed in the sentence, and its apparent exceptions are systematic, not sporadic. Forms such as OE *ymbe* ‘around’ and OHG *umbi* kept their final vowel because, as Ringe and Taylor observe, proclitics “were not phonologically word-final” and so stood outside the environment altogether (Ringe and Taylor 2014, 57–58, §3.1.4). Sentence-level accent placement therefore decides which sandhi variant each daughter language continues, and doublets across the family reflect the stressed and unstressed sentence forms of the same word — regular sandhi, not lexical diffusion.

The diagnostic witness is the second-person plural pronoun. Ringe and Taylor print the Proto-West Germanic form as a doublet: PGmc **izwiz* (Gothic *izwis*) → **iwwi* (by SC008 PWGmcCoronalWAssimilation and the loss of final **z*, SC020 EAFFinalZDeletion) → PWGmc **iuwi* ~ **iuw* (Ringe and Taylor 2014, 41–42, §3.1.1). Old English continues the apocopated, unstressed variant, and Ringe and Taylor’s proof is the vocalism itself: “OE iow ‘you (dat. pl.)’ definitely does [show early apocope] (since it does not exhibit i-umlaut)” (Ringe and Taylor 2014, 57–58, §3.1.4). Had the **-i* survived, i-umlaut (SC055 OEIUmlaut) would have fronted the diphthong; West Saxon *ēow* ‘you’ beside early West Saxon and Northumbrian *īow* ‘you’ shows the normal unumlauted development (Campbell 1959, sect. 702, p. 283).

25.2 SC098. Early apocope in unstressed words (PWGmcUnstressedWordFinalIApocope)

```
define PWGmcUnstressedWordFinalIApocope [
    [ [{*i}|{*u}] -> 0 || .#. {*ᵘ} ?* [
        [EnglishStarLongVowel | EnglishStarLongDiphthong] EnglishStarConsonant*
        | EnglishStarShortVowel EnglishStarConsonant EnglishStarConsonant+
    ] _ .#. ]
    .o. [{*ᵘ} -> 0]
    .o. [
        [{*ᶜ}:0 [EnglishStarAlphabet - {*ᶜ}]* 0:{*ᶜ}]
        | [EnglishStarAlphabet - {*ᶜ}]*
    ]
];
```

Sentence stress and phonological finality are now explicit, independently selected context. They do not alter the PGmc reconstruction, the selected segmental input or its historical stage. The computational marks in the rule distinguish weak-final context from weak-nonfinal context; they are not reconstructed phonemes and are absent from displayed forms. Ordinary evaluation explicitly chooses strong-final citation context. You selects the independently source-discussed weak-final variant, not an unstressed classification inferred from a pronoun label or missing acute.

The worked history is **izwiz* → **iwwiz* by assimilation → **iuwiz* by SC031 OEWWSimplification → **iuwi* by final *z*-loss → weak-final **iuw* by this rule → OE *ēow*. The operation requires a heavy syllable and a phonologically final short high vowel: long vowel/diphthong weight or a short vowel followed by a closing consonant cluster, not literal *ww*. Its condition therefore survives glide reanalysis. Loss of final *z* still feeds it, and removal of *i* bleeds SC055 OEIUmlaut.

Its PWGmc placement precedes the later northern loss of stressed-monosyllabic final *z* (SC097 MonosyllabicFinalZLoss) ((Ringe and Taylor 2014, 41–42, 55, 57–58)).

The staged strong contexts retain the high vowel in the histories of *ġiest* ‘guest’ and *fȳr* ‘fire’. A weak-final and context yields *and* ‘and’, whereas the proposed proclitic context retains the vowel in *ymbe* through the later apocope corridor. The right-boundary annotation is removed only after SC063 OEHighVowelApocope has had the opportunity to apply. These controls implement the defended exceptionless account; they do not claim direct observation of prehistoric sentence stress or lexical optionality ((Ringe and Taylor 2014, 55, 57–58)).

The corresponding English realization of *iu*+*w* belongs to SC032 OEDiphthongLeveling, not a late vocalization of retained *ww*. The model’s retained-i counterfactual produces an umlauted result after intervening w-loss; its exact terminal spelling is a computational prediction, not an attested strong OE form. The attested weak-final target and the source’s mutation argument must not be replaced by that counterfactual ((Ringe and Taylor 2014, 41–42, 57–58, 173–75; Campbell 1959, 283)).

26 Northern monosyllabic final *z-loss

26.1 Historical discussion

Long after the Proto-West Germanic loss of final *z in unstressed syllables (SC020 EAFFinalZDeletion), the northern West Germanic dialects lost word-final *z in stressed monosyllables as well, with compensatory lengthening of a short nucleus. Ringe and Taylor's witness set is OE *mā* 'more' < **maiz*, the pronouns *wē* 'we', *ġē* 'you', *mē* 'me', *þē* 'thee', *hē* 'he', *hwā* 'who' < **hwaz*, and — hedged in their own print with question marks — *cū* 'cow' (Ringe and Taylor 2014, 86, §3.3.1). The southern dialects retained the sibilant and rhotacized it: Old High German *mir*, *wir*, *mēr*, *er* answer the Old English endingless forms, which is why Fulk counts this loss among the diagnostic Ingvaeonic features (Fulk 2018, 18, n. 6). Ringe and Taylor explicitly treat this as a change separate from the Proto-West Germanic unstressed loss, citing Crist's demonstration that the two must be distinguished (Ringe and Taylor 2014, 44–45, §3.1.1).

The scholarship disagrees about the exact conditioning, and the disagreement is worth recording. An older account, represented by Campbell and going back to Luick, derived the endingless pronouns from unaccented sentence variants rather than from a regular sound change (Campbell 1959, 166; Luick 1914--1940, 819). Ringe and Taylor reject that analysis because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86, §3.3.1). Crist formulates an Ingvaeonic rule in which *z is lost after front vowels, with compensatory lengthening, covering preconsonantal cases as well; his data contain no word-final back-vowel monosyllables, so forms like **hwaz* and the ancestor of *cū* fall outside what his statement can decide — a documented gap rather than a refutation (Crist 2002, 1, 4, §§1, 10). Kilday narrows the preconsonantal subcase to Old Saxon and Old Frisian while accepting the word-final monosyllabic loss for Old English, contrasting regular *meord* 'reward' with the loanword-influenced *mēd* 'reward' (Kilday 2024, 1–3). CAPR adopts Ringe and Taylor's quality-neutral formulation because it alone generates the back-vowel witnesses, while noting that the front-vowel forms are compatible with both analyses.

Apparent counterexamples are analogical, not phonological: OE *dēor* 'deer', *ār* 'oar', and *gār* 'spear' show final *-r* from levelling out of inflected forms where the sibilant was word-internal and regularly rhotacized, not from retention of word-final *z (Ringe and Taylor 2014, 86, §3.3.1, n. 24). The change precedes rhotacism (SC003 EAFRhotacism), which Ringe and Taylor place last in this sequence of northern developments (Ringe and Taylor 2014, 87, §3.3.1).

The corpus now witnesses this change directly. The interrogative pronoun 'who' is selected in its nominative singular masculine cell, PGmc **hwaz* (Gothic *hwas*), precisely the form Ringe and Taylor cite for this loss (Ringe and Taylor 2014, 86, §3.3.1): the rule lengthens the short nucleus and deletes the sibilant, giving **hwā*, whence OE *hwā* 'who'. The resulting back vowel never undergoes Anglo-Frisian brightening — "**hwǣ* does not exist", as Campbell puts it (Campbell 1959, sect. 125, p. 49; Brunner 1965, sect. 137 Anm. 1, p. 129) — so the derivation ends with the attested form. Other members of Ringe and Taylor's witness set remain outside the corpus because it selects oblique or plural cells for them — 'cow' and 'meed', for instance, enter the cascade in inflected forms whose *z, where present, is word-internal.

26.2 SC097. Northern monosyllabic final *z-loss (MonosyllabicFinalZLoss)

```
define MonosyllabicFinalZLoss [
    {*a} -> {*ā}, {*á} -> {*ā},
    {*e} -> {*ē}, {*é} -> {*ḗ},
    {*i} -> {*ī}, {*í} -> {*ī́},
    {*o} -> {*ō}, {*ó} -> {*ō},
    {*u} -> {*ū}, {*ú} -> {*ū}
        || .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
            _ {*z} .#.
] .o. [
    {*z} -> 0 ||
        .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
```

```
            EnglishStarVocalic+ _ .#.
];
```

The rule first lengthens a short nucleus standing immediately before word-final **z* in a monosyllable, then deletes the **z* after any vowel in a monosyllable. The principal synthetic controls run the change on Ringe and Taylor's own witnesses, fed to the rule in their chronologically correct intermediate shapes. Short-nucleus inputs show both halves of the change at once: **hwaz* yields **hwā* and **hiz* yields **hī*, each with loss of the sibilant and compensatory lengthening (Ringe and Taylor 2014, 86, §3.3.1). A form whose nucleus is already bimoric skips the lengthening step and simply loses the sibilant: **maiz* yields **mai* at this stage, with its diphthong intact; the attested OE *mā* 'more' arises only later, when the stressed monophthongization (SC004 EAFAiMonophthongization) takes **ai* to **ā*. The word 'cow', whose history Ringe and Taylor themselves print with question marks and whose analysis remains disputed, is deliberately not used as a principal control; a long-vowel input of that shape (**kūz* yielding **kū*) merely repeats what **maiz* already demonstrates, and the word's evidentiary weight is discussed in the historical dossier rather than leaned on here.

The corpus derivation of *hwā* 'who' now fixes this rule's position empirically as well as philologically. It stands after SC020 EAFFinalZDeletion, since the two losses are historically distinct changes with the unstressed loss earlier (Ringe and Taylor 2014, 44–45, §3.1.1). Rhotacism (SC003 EAFRhotacism) follows in the executable cascade as in the historical account: Ringe and Taylor place rhotacism after this loss, at the end of the sequence of **z*-losses (Ringe and Taylor 2014, 87, §3.3.1), and the cascade composes it immediately after this rule, so a sibilant removed here can never surface as *-r* — were the order reversed, **hwaz* would rhotacize to **hwar* and 'who' could never be derived; the negative controls below confirm this. Consonant-final monosyllables are untouched: their nominative **-z*, where it ever existed, was eliminated before Proto-West Germanic under SC096 RootNounNomZLoss. Word-internal **z*, as in PGmc **déuzą* 'deer' on its way to OE *dēor* 'deer', does not meet the environment of this rule and duly rhotacizes.

27 West Germanic rhotacism

27.1 Historical discussion

Bammesberger states that Germanic *z yielded *r in intervocalic position in Old English, while final *z was generally lost (Bammesberger 1992, 39). Ringe and Taylor argue that this merger of *z with *r was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic *z-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

The internal identifier SC003 EAFRhotacism places the change in CAPR's Early Anglo-Frisian corridor, the operational post-Proto-West-Germanic stage on the English line; historically the change is a West Germanic rhotacism, later than Proto-Germanic. It is also distinct from SC020 EAFFinalZDeletion, which removes final *z before the surviving medial consonant becomes *r.

27.2 SC003. West Germanic rhotacism (`EAFRhotacism`)

```
define EAFRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc **líznōjaną* 'learn' yields †*lirnian* rather than expected OE *liornian* 'learn', PGmc **líznōθi* 'learns' yields †*lirnaþ* rather than expected *liornaþ* 'learns', PGmc **líznô* 'learn' yields †*lirna* rather than expected *liorna* 'learn', and PGmc **mízdai* 'meed' yields †*merde* rather than expected OE *meorde* 'meed'. Moving rhotacism earlier within the tested range changes no output.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. The lower boundary rests on the historical analyses cited above: Ringe and Taylor put rhotacism at the end of the sequence of *z-losses — "first *z was lost in a variety of environments …, then all surviving *z became *r" (Ringe and Taylor 2014, 87, §3.3.1) — and Crist observes that the deletions distinguish *z from *r and so must precede the merger (Crist 2002, 2–3). The rule is accordingly ordered after the three final-*z losses (SC096 RootNounNomZLoss, SC020 EAFFinalZDeletion, and SC097 MonosyllabicFinalZLoss) and before breaking, so that the derivation follows the reconstructed chronology.

28 *mn*-dissimilation

28.1 Historical discussion

In the inherited *n*-stem paradigm the zero-grade oblique cells brought *m* and *n* into direct contact, and in that adjacent cluster the labial nasal dissimilated to a labial spirant: *mn* > *βn* (surfacing as *fn*). Old Norse preserves the older paradigmatic distribution, with the labial confined to the oblique cluster (*himinn* 'heaven' beside dative *hifni*); Old English and Old Saxon generalized it. Fulk treats the cluster change among developments common to Germanic, while warning that its surface results are irregular and that reverse *bn* > *mn* is later well attested in Northwest Germanic (Fulk 2018, 121, §6.11). The relevant *heofon* 'heaven' and *mōnaþ* 'month' material is discussed by Campbell (Campbell 1959, 189, 195, §§470, 484).

The underlying cluster change is therefore late Proto-Germanic / Common Germanic, not securely a pan-Northwest-Germanic innovation; `PNWGmc` remains a stable executable identifier only. The lexical evidence does not constrain a positive local cascade position.

28.2 SC022. Dissimilation of adjacent *mn* (`PNWGmcMnDissimilation`)

```
define PNWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _ {*n}
];
```

The rule fires only where *m* stands directly before *n*. It supplies the labial of *stefn* 'stem, trunk' from the *mn*-cluster of the *stamn*-family, and it is the historical change behind the labial of *heofon* 'heaven', which was generalized from the oblique cluster into the vowel-bearing stem before the Old English vocalic changes. (An earlier cross-syllable formulation that labialized an intervocalic *m* before a later nasal has been retired: it simulated paradigm levelling rather than a sound law.)

Moving SC022 PNWGmcMnDissimilation earlier or later leaves every output unchanged. Its executable place in this holding zone is therefore editorial/computational, while its historical classification rests on the handbook account of *mn*-dissimilation.

29 Word-final *n*-loss

29.1 Historical discussion

The change isolated here is far older than its position in the cascade suggests: it is the general (pre-)Proto-Germanic loss of word-final **n*, with nasalization of the preceding vowel, in polysyllables. Ringe's proof set for the law spans the whole grammar — nouns such as **yugón* > **juką* 'yoke', pronouns such as **tón* > **þanǭ*, and even the verb form **dedǭ* 'I did' — so it is general phonology, not a fact about any one declension (Ringe 2017, 101–3). Gothic *tuggo* shares the weak nominative-singular outcome, and the nasalized reflex **-ǭ* remained contrastive into Proto-West Germanic before yielding OE *-e* (Ringe and Taylor 2014, 54–55, 58–59).

Within the present corpus the change surfaces in exactly one shape: the weak nouns are cited in the stem form **-ōn-*, and this rule carries them to the Proto-Germanic nominative singular in **-ǭ*, as in **túngōn* > **túngǭ* > *tunge* 'tongue', alongside *eorþe* 'earth', *heorte* 'heart', *nǣdre* 'adder', and thirteen further weak nouns. The masculine weak nominative singular in trimoric **-ô* never had a final **-n* to lose, and Proto-Germanic **sebun* 'seven', **nigun* 'nine', and **tehun* 'ten' kept their *-n* by lexical analogy among the numerals (Ringe 2017, 103); the rule's narrow **-ōn* environment leaves all of these correctly untouched.

29.2 SC023. Loss of word-final **n* after **ō* (`PNWGmcNStemNLoss`)

```
define PNWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

The verb *dōn* 'do' supplies the negative, counterfeeding witness for the chronology. PGmc **dōną* 'do' passes this rule untouched; only SC047 OEHeavySyllableNasalApocope later strips the final **ą* and creates a new word-final **-ōn*. That secondary **-n* survives into *dōn* precisely because the old loss was no longer active: if SC023 PNWGmcNStemNLoss is displaced after the apocope, it consumes the new nasal and the derivation collapses entirely (+*?*).

The retained *-n* of *dōn* 'do' therefore supplies a terminus ante quem for the loss — it must be dead before the apocope — while the seventeen weak nouns above are its positive witnesses; the lower boundary remains unattested within the cascade, as befits a change already complete in Proto-Germanic.

30 Nasal spirant changes

30.1 Historical discussion

Germanic lost nasal consonants before voiceless fricatives twice, in two changes that are easily confused because their outcomes look alike. Both replace a sequence of vowel, nasal and fricative with a long nasalized vowel and the fricative. They differ in date, in geography, and in which fricatives they affect.

The earlier change is common Germanic and is treated in the opening chapter of this book, on the Proto-Germanic loss of a nasal before **x*. Its results are shared by every daughter language, and only **a*, **i* and **u* occur in its input (Campbell 1959, 44, §119; Fulk 2018, 55, §4.1; Ringe 2017, 149–50, §3.2.7).

The later change belongs to the dialects bordering the North Sea, that is to Old English, Old Frisian and Old Saxon, the group traditionally called Ingvaeonic. Here the groups **mf*, **ns* and **nþ* likewise reject the nasal with compensatory lengthening and nasalization (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11; Brunner 1965, 176, §186.1; Luick 1914--1940, 276, §301.1). Ringe and Taylor call it the most obvious phonological innovation of the northern dialects and list some thirty examples, among them **gans* ‘goose’ and **jugunþi* ‘youth’ (Ringe and Taylor 2014, 139–41). Campbell describes it as a later change similar to the common Germanic one, and Fulk as comparable to it; neither treats them as the same law.

The two are told apart by what the languages outside the North Sea area show. The common Germanic change left no nasal anywhere, so Old High German has *fūht* ‘damp’ and *fūst* ‘fist’ exactly as Old English does. The later change was confined to the north, so the cognates of its witnesses keep the nasal: Old High German *fimf*, *gans*, *ander*, *jugund* answer Old English *fīf* ‘five’, *gōs* ‘goose’, *ōþer* ‘other’, *ġeoguþ* ‘youth’. On this test **funhsti-* ‘fist’, whose long vowel is shared by Old High German *fūst*, Dutch *vuist* and German *Faust*, belongs to the earlier change and not to the North Sea law at all (Kroonen 2013, 160).

Where the vowel was **a*, the long nasalized vowel that this law produced was afterwards rounded to *ō* in Anglo-Frisian, which is why Old English has *gōs* ‘goose’, *tōþ* ‘tooth’ and *ōþer* ‘other’. That rounding is a third change again, and it is the same rounding that gives *fōn* ‘seize’ and *þōhte* ‘thought’ from the common Germanic law and *mōna* ‘moon’ and *spōn* ‘chip’ from inherited long *ā* before a surviving nasal; Campbell states that it reached all three sources at one and the same time (Campbell 1959, 50, §128 n. 1; Brunner 1965, 33, §26; Fulk 2018, 72, §4.11). It is treated in the chapter on the rounding of the long nasalized low vowel. That the vowel was rounded and did not simply merge shows that its nasality survived the law that created it: as Fulk observes, it did not fall together with the *ā* that came from **ai* (Fulk 2018, 55, §4.1). Ringe and Taylor take the nasalization to have remained subphonemic until it was lost separately in each daughter (Ringe and Taylor 2014, 141). Old Saxon shares the loss of the nasal and the nasalization, and rounds only variably, which is why the law itself is described as North Sea Germanic and the rounding as Anglo-Frisian (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11).

The North Sea Germanic law is a single connected sound change: in every handbook account the nasal is lost *with* compensatory lengthening, and the lengthening is the compensation for the loss. It is stated here as two rules only because the vowel must be adjusted while the conditioning nasal is still present, before the nasal can be removed. The order of SC026 EAFNasalSpirantLengthening before SC027 EAFNasalSpirantLoss is a requirement of the statement, not evidence for two successive historical stages.

30.2 SC026. North Sea Germanic nasal-spirant law, first step (EAFNasalSpirantLengthening)

```
define EAFNasalSpirantLengthening [
    {*a} -> {*ą} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*á} -> {*ą} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
```

```
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarNSGmcSpirant,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarNSGmcSpirant
];
```

The environment is nasal plus **f*, **þ* or **s*. The fricative **x* is excluded: nasal loss before **x* is the earlier change stated as SC103 PGmcNasalLossBeforeX. Campbell names the groups **mf*, **ns* and **nþ*, Fulk says the change affects **mf*, **ns* and **nþ*, Sievers and Brunner name the fricatives **f*, **þ* and **s*, and no example in Ringe and Taylor's list contains **x* (Campbell 1959, 47, §121; Fulk 2018, 72, §4.11; Brunner 1965, 176, §186.1; Ringe and Taylor 2014, 139–41). As in the earlier change, only **a*, **i* and **u* occur. The outcome of **a* is the long nasalized **ą̄*, which SC104 EAFNasalized-LowRounding later rounds; the outcomes of **i* and **u* are written **ī* and **ū*, their nasality having no further consequence (Campbell 1959, 47, §121).

Two witnesses apply in the present corpus. PGmc **gánsz* 'goose' becomes **gą̄ns* 'goose', and PGmc **júgunθ* 'youth' becomes **júgūnθ* 'youth'. In *ġeoguþ* 'youth' the syllable carrying the lengthened vowel is unstressed, and the length is given up again by the later shortening of unstressed syllables; Sievers and Brunner note the same course in *beraþ* 'they carry' from **beranþi* 'they carry' through **berōþ* 'they carry' (Brunner 1965, 176, §186.1 Anm. 3; Luick 1914--1940, 276, §301.1).

If the rule is stated after the loss of the nasal, PGmc **gánsz* 'goose' yields †*ġeas* in place of *gōs* 'goose', and PGmc **júgunθ* 'youth' yields †*ġeogoþ* in place of *ġeoguþ* 'youth'. This shows only that the vowel must be adjusted before its conditioning nasal is removed. It does not establish a date for either operation, and no earlier or later boundary is claimed here.

30.3 SC027. North Sea Germanic nasal-spirant law, second step (EAFNasalSpirantLoss)

```
define EAFNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarNSGmcSpirant
];
```

The nasal is removed in the environment that conditioned the lengthening, giving **gą̄s* 'goose' and **júgūθ* 'youth'. The rule completes the statement of the single change begun in SC026 EAFNasalSpirantLengthening; the two are not independent sound laws. The converse test, stating the loss first, merely reproduces the same two wrong forms. Nothing in the material fixes a later boundary for the loss.

31 The hiatus-breaking *w* of the verba pura

31.1 Historical discussion

The small class of Germanic strong verbs whose roots ended in a vowel, the *verba pura*, reached Northwest Germanic with a morphologically expected hiatus. Ringe and Taylor reconstruct Proto-Germanic **sēanąˀ* ‘to sow’ (Gothic *saian*), which the Northwest Germanic lowering of **ē₁* carried to **sāanąˀ*; the West Germanic languages then “exhibit innovative consonants that eliminated” the hiatus (Ringe and Taylor 2014, 12). The repair differs by branch, and the difference dates and localizes the change. Old English and Old Frisian inserted *w*, as in Old English *sāwan* ‘to sow’ and Old Frisian *sāwinge* ‘sowing’ beside *grōwinge* ‘growth’, while Old Saxon and Old High German used *j* (*sāian* ‘to sow’, *sāen* ‘to sow’, *sājen* ‘to sow’) (Ringe and Taylor 2014, 12, 151). The insertion of *w* is therefore an Anglo-Frisian development and the consonant of *sāwan* is not inherited; it must not be projected back into the protoform.

The origin of that *w* is neither inherited nor morphological. Þórhallsdóttir, in the standard treatment of the problem, rejects in turn the derivation of the West Germanic *w* from a Proto-Indo-European *u*-perfect (Þórhallsdóttir 1993, 115–17), from a **ue/o*-present (Þórhallsdóttir 1993, 117–19), and from the transfer of a preterite *w* into the present (Þórhallsdóttir 1993, 121–22). The *w* “rather has an inner-Germanic phonological explanation” (Þórhallsdóttir 1993, 136). It began as an ordinary phonetic glide filling the hiatus, and it began in one narrow environment, before **u*. The cells in question, as Bremer had already seen, are the first singular and first plural of the present indicative together with the plural of the reduplicated preterite in **-um*, **-uþ*, **-un* (Þórhallsdóttir 1993, 120). Þórhallsdóttir sets out the pre-Old English present indicative with the hiatus still intact, first singular **sā’u*, second **sā’is*, third **sā’iþ*, first plural **sā’um*, second plural **sā’iþ*, third plural **sā’anþ*, and observes that “the glide *w* was a natural hiatus filler before the *u* of the endings of the 1st person singular and plural” (Þórhallsdóttir 1993, 127). The preterite stem **seuwun* arose by the same insertion before the **u* of the plural endings (Þórhallsdóttir 1993, 126).

From those few cells the *w* was carried through the whole paradigm by analogy. Three forces converge. Levelling proceeded out of the **u*-cells themselves; the preterite stem **seuw, *seuwun* “would have assisted in extending the new *w* to all forms of the present paradigm” and to the preterite participle; and verbs with an etymological **w*, above all *flōwan* ‘to flow’ from the root **pleu-*, exerted their own attraction (Þórhallsdóttir 1993, 127). The result, complete already in pre-Old English, is the uniform pattern **sāwan*, **seuw*, **seuwun*, **sāwan-* (Þórhallsdóttir 1993, 136). Old Frisian inherits the same generalized type, so that its *w* and the Old English *w* are one Anglo-Frisian event (Þórhallsdóttir 1993, 130, 134). The scattered *w*-forms of Old High German are a separate and later inner-German development and are not to be identified with it (Þórhallsdóttir 1993, 119, 123).

This layering matters for how the rule below should be read. Old English citation forms are infinitives, and the infinitive **sā’an* is precisely one of the cells that never received *w* by the phonological rule, so the *w* of *sāwan* ‘to sow’ is analogical. The rule therefore models the outcome of the generalization, and its environment, any vowel after *ā*, is a citation-form proxy for a paradigm-wide result. It is not a reconstruction of the conditioning of the original insertion.

The chronology is fixed on both sides by Ringe and Taylor. The *w* must postdate the West Germanic loss of intervocalic **w*, or the new glide would itself have been swept away (Ringe and Taylor 2014, 151, n. 9); and it “must have occurred early enough to prevent fronting of **ā*” in pre-Old English (Ringe and Taylor 2014, 151). The whole reason *sāwan* ‘to sow’, *cnāwan* ‘to know’, *blāwan* ‘to blow’, and *māwan* ‘to mow’ keep their back vowel is that the *w* was already in place when the North Sea Germanic fronting applied. Since it is the generalized *w* that stands in the infinitive, this later boundary constrains the generalization itself. The rule accordingly stands before the fronting in the cascade, even though on the present corpus the two orders happen to produce the same outputs, since the fronting rule as implemented does not touch a prevocalic *ā* in any case; the ordering encodes the historical chronology, and no corpus-internal contrast demonstrates it. That chronology rests on Ringe and Taylor alone, for Þórhallsdóttir’s chapter nowhere discusses the fronting of **ā* and supplies no independent evidence for the order.

31.2 SC102. Generalized hiatus-breaking *w* after long *ā* (EAFHiatusWInsertion)

```
define EAFHiatusWInsertion [
    [..] -> {*w} || {*ā} _ EnglishStarVocalic
];
```

The rule is fed by SC024 PNWGmcLongELowering, which creates the *ā*-initial hiatus it repairs; ***sǽaną*** ‘to sow’ passes through ***sāaną*** to ***sāwaną*** on its way to *sāwan* ‘to sow’. Displaced before the lowering, the rule can never apply, since the root vowel is still ***ē*** and no hiatus after *ā* exists, and the derivation loses its consonant altogether.

Its output in turn feeds the blocking environment of SC101 EAFLongAFronting: the inserted *w* is precisely what shields the *ā* of *sāwan* ‘to sow’ from fronting, as treated in that chapter. The corpus carries *sāwan* as the diagnostic witness of the class; *cnāwan* ‘to know’, *blāwan* ‘to blow’, *māwan* ‘to mow’, *wāwan* ‘to blow (of wind)’, and *þrāwan* ‘to turn’ instantiate the same derivation (Ringe and Taylor 2014, 151).

32 North Sea Germanic nasalization of long *ā* before a nasal

32.1 Historical discussion

In the dialects along the North Sea coast the long low vowel **ā* — centrally the vowel produced from **ē₁* by the Northwest Germanic lowering — was nasalized when a nasal consonant followed and survived. The nasalized vowel was afterwards rounded in Anglo-Frisian, which is why Old English has *mōna* ‘moon’, *mōnaþ* ‘month’ and *spōn* ‘spoon’ against Old High German *māno* ‘moon’, *mānōd* ‘month’, *spān* ‘spoon’ and Old Norse *máni* ‘moon’, *mánaðr* ‘month’, *spánn* ‘spoon’. Campbell describes the split of Germanic **ǣ¹* before nasals in these terms and identifies the vowel that the rounding operated on as a nasalized and unrounded **ą̄* [Campbell (1959), p. 50, §127; p. 50, §128 n. 1]. Fulk places the rounded outcome of **ǣ* before nasals among the Anglo-Frisian changes and derives it from an earlier nasalized vowel (Fulk 2018, 72–73, §4.12).

The nasalization and the rounding have different geographies, and separating them resolves an apparent disagreement in the handbooks. Ringe and Taylor state that stressed low vowels were nasalized in the northern West Germanic dialects, Old Saxon among them (Ringe and Taylor 2014, 142, §5.1.2); Old Saxon accordingly shows *ōdar* ‘other’ and *sōd* ‘true’ beside unrounded *quān* ‘wife’ and *sāno* ‘immediately’ (Ringe and Taylor 2014, 150–51). What Old Saxon shares is the nasalization; what it shares only in part is the rounding, and for the vowel inherited from Proto-Germanic it does not share the rounding at all (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). The nasalization stated here is therefore North Sea Germanic, and the rounding treated in the chapter on the long nasalized low vowel is Anglo-Frisian.

The nasalization is the conditioned counterpart of the fronting of oral **ā* treated in the fronting chapter: a following nasal gives nasalization and eventual rounding, and its absence gives fronting. Together the two exhaust the fate of the old low vowel in this area. The comparative material does not order the two branches against each other, and no such ordering is claimed here.

32.2 SC025. Nasalization of long *ā* before nasals (EAFLongANasalRounding)

```
define EAFLongANasalRounding [
    {*ā} -> {*ą̄} || _ EnglishStarNasal
];
```

The rule consumes the **ā* created by SC024 PNWGmcLongELowering: displacing the lowering after this rule leaves the nasalization without an input, and **mḗnōθz* ‘month’ surfaces as †*mānaþ* in place of OE *mōnaþ* ‘month’, **spḗnuz* ‘spoon’ as †*spān* in place of *spōn* ‘spoon’. Its output is consumed in turn by SC104 EAFNasalizedLowRounding, which supplies the rounded vowel that the two words actually show.

Equally important is what the rule must precede. The monophthongization of **ai* in SC004 EAFAi-Monophthongization creates a new long *ā*, and that vowel was never nasalized and never rounded before nasals: *stān* ‘stone’ and *hām* ‘home’ keep *ā*. Stating the monophthongization before this rule makes the new vowel eligible for nasalization and hence for rounding, and the cascade then wrongly yields †*stōn* and †*hōm*. This is the same chronological inference Campbell draws for the fronting: the treatments of the old low vowel were complete, or at least under way, before *ai*-monophthongization supplied a new one (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70).

33 Anglo-Frisian rounding of the long nasalized low vowel

33.1 Historical discussion

Three separate developments described in earlier chapters end in the same sound: a long, nasalized, low vowel. The oldest is the Proto-Germanic loss of a nasal before **x*; the second is the North Sea Germanic nasal-spirant law; the third is the nasalization of the inherited long *ā* before a nasal that survived. In Old English and Old Frisian all three surface as *ō*, and Campbell states the unification without qualification: the nasalized vowel became identical with the *ō* inherited from Proto-Germanic already in prehistoric Old English, and the same change affected the nasalized vowel of the Proto-Germanic law and the nasalized vowel of the Ingvaeonic law at one and the same time (Campbell 1959, 50, §128 n. 1). One later sound change therefore accounts for all three, and the appearance of three independent roads to *ō* is an illusion created by looking only at the Old English surface.

That the intermediate vowel was nasalized and unrounded is shown by a form in which it was shortened very early. Campbell's example is *samcucu* 'half alive', where the shortened reflex is *a* and never *o*; the vowel that the earlier laws produced was therefore a nasalized **ą̄*, and the rounding is a distinct and later event (Campbell 1959, 50, §128 n. 1). Sievers and Brunner describe the same nasalized long vowel and its rounded Old English outcome [Brunner (1965), p. 33, §26; p. 58, §64].

The geography is Anglo-Frisian. Old Saxon nasalized its stressed low vowels along with the rest of the northern West Germanic area (Ringe and Taylor 2014, 142, §5.1.2), and it shares the loss of the nasal in both of the earlier laws; the categorical, systematic rounding, however, it does not share. For the Proto-Germanic nasalized low vowel Old Saxon retains *ā* consistently, and for the vowel created by the nasal-spirant law it has *ā* or *ō* according to word and dialect, while Old English and Old Frisian have *ō* throughout (Campbell 1959, 44, §119; Fulk 2018, 72, §4.11). That partial and lexically variable Old Saxon rounding is comparative evidence bearing on the innovation rather than participation in it. Ringe treats the rounding as a parallel development of the diverging northern dialects and locates it in the northernmost of them, that is in Anglo-Frisian (Ringe 2017, 149–50, §3.2.7; Ringe and Taylor 2014, 142, §5.1.2). Luick groups the whole set of changes among those peculiar to the Anglo-Frisian dialect group and notes that the nasalized vowel later gave up its nasality (Luick 1914--1940, 276, §301.1). The rounding is accordingly Anglo-Frisian, and the nasalization that feeds it is the wider North Sea Germanic property.

Ringe and Taylor confirm that a single rounding covers all the sources: the rounding affected the nasalized low vowels of the nasal-spirant law, the nasalized low vowels of the accompanying list, and the reflexes of Proto-Germanic **/anh/* alike (Ringe and Taylor 2014, 142, §5.1.2). The condition on the input is nasality and nothing else, which is why the long *ā* that Old English later won from **ai* escapes: that vowel was oral, and it came into being after the rounding had run its course. Fulk makes this the central argument for the long survival of the nasality, since the nasalized vowel developed to *ō* and did not fall together with Old English *ā* from **ai* (Fulk 2018, 55, §4.1).

33.2 SC104. Rounding of the long nasalized low vowel (EAFNasalizedLowRounding)

```
define EAFNasalizedLowRounding [
    {*ą̄} -> {*ō}
];
```

The rule is unconditioned, since the vowel it operates on exists only where one of the three earlier changes created it. Its inputs arrive from SC103 PGmcNasalLossBeforeX, from SC026 EAFNasalSpirantLengthening and from SC025 EAFLongANasalRounding, and each of those three stands in a feeding relation to it: stated before any one of them, the rule leaves that source's nasalized vowel untouched and the cascade returns no form at all for its witnesses.

Four lexemes in the present corpus reach Old English through this rule, and between them they

witness all three sources. **gánsz* ‘goose’ arrives as **gą̄s* ‘goose’ from the North Sea Germanic law and gives *gōs* ‘goose’; **mḗnōθz* ‘month’ arrives as **mą̄nōþ* ‘month’ and gives *mōnaþ* ‘month’; **spḗnuz* ‘spoon’ arrives as **spą̄nu* ‘spoon’ and gives *spōn* ‘spoon’. The Proto-Germanic law contributes **θánxtē* ‘thought’, the preterite of the verb ‘to think’, which arrives as **θą̄xtē* ‘thought’ and gives *þōhte* ‘thought’. That is the very form cited for this rounding, beside Old Frisian *thochte*, and it is the evidence that the vowel did not fall together with the *ā* of *stān* ‘stone’ (Fulk 2018, 55, §4.1; Campbell 1959, 44, §119). The other firing of the Proto-Germanic law, the high vowel of *fȳst* ‘fist’, does not reach this rule at all.

The counterpart is what the rule leaves alone. **stáinaz* ‘stone’ and **xáimaz* ‘home’ acquire their long *ā* from SC004 EAFAiMonophthongization, which follows this rule, and their vowel was never nasalized; they surface as *stān* ‘stone’ and *hām* ‘home’. Placing the monophthongization before the nasalization and the rounding makes that vowel eligible and the cascade then yields †*stōn* and †*hōm*, which is the chronological inference Campbell draws for the treatments of the old low vowel generally (Campbell 1959, 52–53, §132; Ringe and Taylor 2014, 169–70).

34 North Sea Germanic fronting of long *ā*

34.1 Historical discussion

Long after the Northwest Germanic lowering of **ē₁* to **ā*, the dialects of the North Sea coast fronted the surviving oral **ā* to a low front vowel: West Saxon *ǣ* in *dǣd* ‘deed’, *slǣpan* ‘to sleep’, *lǣtan* ‘to let’, *rǣdan* ‘to read’, and *mǣl* ‘meal’, against Anglian, Kentish, and Old Frisian *ē* (*dēd* ‘deed’, *slēpa* ‘to sleep’, *jēr* ‘year’) (Ringe and Taylor 2014, 146–50; Campbell 1959, 50–51, §128). Old Saxon and Old High German keep the back vowel (Old Saxon *dād* ‘deed’, Old High German *tāt* ‘deed’, *slāfan* ‘to sleep’, *lāzan* ‘to let’), but sporadic Old Saxon spellings in □e□ show that the fronting lapped unevenly into Old Saxon territory (Ringe and Taylor 2014, 150). The change is therefore a North Sea Germanic development rather than an exclusively Anglo-Frisian one, though only Old English and Old Frisian carry it through systematically.

The fronting affected stressed, non-nasalized **ā*; nasalized **ą̄* was instead rounded, as treated in the rounding chapter. Ringe and Taylor further establish a conditioning by a following **w*. Before **w* plus a back or non-high vowel the **ā* was retained: the clearest witnesses are the *verba pura* — *sāwan* ‘to sow’, *cnāwan* ‘to know’, *blāwan* ‘to blow’, *māwan* ‘to mow’, *þrāwan* ‘to turn’ — together with *clāwu* ‘claw’ (Ringe and Taylor 2014, 151); the simplest hypothesis, which Ringe and Taylor adopt from Hogg, is that fronting never occurred in that environment (Ringe and Taylor 2014, 151; Hogg 1992a, 81). Before **w* plus a high front vocalic, by contrast, the fronting did apply: **lēwijaną* ‘to betray’ (Gothic *lēwjan* ‘to betray’, Old High German *gilāen* ‘to betray’) yields West Saxon *lǣwan* ‘to betray’, and the same environment appears in *eltǣwe* ‘entire’ and *brǣw* ‘eyelid’ (Ringe and Taylor 2014, 150). Both sides of the condition are encoded in the rule below and witnessed in the corpus. Bennett states the retaining environment independently from the West Saxon evidence alone, and slightly more broadly: West Saxon “shows *ǣ* as a regular isolative development of IE *ē* but has *ā* before *w* or *g* plus a back vowel”, with the paradigmatic alternation *wǣg* ‘wave’ beside plural *wāgas* ‘waves’ (Bennett 1950, 235, n. 6). The rule below encodes the *w* environment, for which the corpus supplies witnesses on both sides; the parallel *g* environment is left for separate treatment.

Whether this fronting restored a front vowel that had earlier been backed, or whether — as Fulk argues — the North Sea dialects simply retained an old front **ǣ* that was never backed at all, is the same dispute recorded in the lowering chapter (Fulk 2018, 60–61, §4.6; Campbell 1959, 50–51, §§128–129). On the retention analysis this chapter’s change dissolves into the non-event of staying put; the present model follows Ringe and Taylor’s two-step reconstruction, on the strength of the runic evidence for an early [a:] and the place-adverbs *þǣr* ‘there’ and *hwǣr* ‘where’ (Ringe and Taylor 2014, 13–14).

34.2 SC101. Fronting of long *ā* outside nasal and *w* environments (EAFLongAFronting)

```
define EAFLongAFronting [
    {*ā} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal - {*w}],
    {*ā} -> {*ǣ} || _ {*w} EnglishIUmlautTrigger
];
```

The first clause fronts **ā* before any oral consonant other than **w*; the second admits the fronting before **w* exactly when a high front vocalic follows. The two corpus witnesses of the *w* condition form a minimal contrast. **sḗaną* ‘to sow’, whose hiatus-filling *w* is supplied by SC102 EAFHiatusWInsertion, reaches this rule as **sāwaną* and is left unfronted, surfacing as *sāwan* ‘to sow’; were the *w*-block removed, the cascade would deliver †*sǣwan* instead. **lḗwijaną* ‘to betray’, whose inherited *w* is followed by **i*, is fronted by the second clause to **lǣwijaną* at this rule’s own stage — before, and independently of, the much later i-umlaut — and surfaces as *lǣwan* ‘to betray’. The rule consumes the **ā* created by SC024 PNWGmcLongELowering; displaced before the lowering, it has nothing to front, and **skḗpą* ‘sheep’ surfaces as †*sċāp* rather than OE *sċēap* ‘sheep’, **jḗrą* ‘year’ as †*ġār* rather than *ġēar* ‘year’, **slḗpaną* ‘to sleep’ as †*slāpan* rather than *slǣpan* ‘to sleep’.

Two later boundaries carry real historical content. First, the fronting must precede the completion of SC004 EAFAiMonophthongization: the *ā* that arose from **ai* was never fronted — *stān* ‘stone’, *hām* ‘home’, *lāþ* ‘hostile’, *rāp* ‘rope’, *tācn* ‘token’, *gāst* ‘spirit’ all keep the back vowel. Campbell draws exactly this chronological inference (Campbell 1959, 52–53, §132), and Ringe and Taylor endorse it as cogent (Ringe and Taylor 2014, 169–70); in the present cascade the inference is enforced by rule order, and displacing the fronting after the monophthongization wrongly yields †*lǣþ*, †*rǣp*, †*tǣcn*, †*sǣwol*, and †*ġēast*. Historically the two changes may well have overlapped in time; the discrete ordering is the grammar’s way of stating that inherited *ā* had been fronted before the new *ā* arose.

Second, the fronting must precede SC056 OEWsPalatalDiphthongization, which operated on the already-fronted vowel: *ǣ* > *ēa* after the palatals, as in *sċēap* ‘sheep’ and *ġēar* ‘year’ (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1). Displaced after the diphthongization, the cascade yields †*sċǣp* and †*ġǣr* instead.

35 Chapter 4. From Anglo-Frisian to Old English

35.1 Historical interval

This chapter follows the English daughter from the required Anglo-Frisian ancestor through prehistoric English to attested Old English. It includes changes earlier than the manuscript period, not merely changes within it. West Saxon is the principal target; each comparison still requires its actual dialect, paradigm cell and evidential status.

35.2 Scope and dialect variation

Not every change in this chapter has pan-Old-English scope. Some changes — most notably ordinary West Saxon palatal diphthongization and parts of the back-mutation history — are specifically West Saxon or more broadly southern Old English phenomena. The diphthong corridor also represents earlier West Germanic feeders; a West Saxon final outcome does not date every feeder to West Saxon (Ringe and Taylor 2014, 41–42, 65–66, 171–73, 215–17). (The West Saxon palatal-glide spellings, SC016, belong to the written surface of Old English and are treated in Chapter 5.)

The CAPR derivations target West Saxon Old English citation forms as the default comparator. Changes that belong to other dialects, or that are absent from West Saxon, may appear in lexical entries as comparanda rather than as derivational steps.

The distinction between West Saxon and Anglian comparanda requires the actual text and dialect, not merely a regional label. Hogg discusses the limits of the traditional divisions and the evidence for the principal textual varieties (Hogg 1992a, 3–8, §§1.5–1.12).

35.3 Chapter structure

The changes in this chapter fall into several natural historical subgroups, though the boundaries between them are not always sharp:

Prehistoric English contraction and fronting: Completed stressed **ai* > **ā* is now adopted on the English daughter in SC004 EAFAiMonophthongization. The new long vowel is distinguished from inherited oral long vowels; the source milestones allow overlap with their earlier restructuring (Campbell 1939, 90–91; Ringe and Taylor 2014, 170–71). This placement leaves possible conditioned ancestral onset and the runic interpretation qualifications explicit (Versloot 2017, 295–97, 318).

Early Old English changes linked to the Anglo-Frisian inheritance: Changes that feed directly on, or are closely related to, Anglo-Frisian brightening (SC043), whose section opens the vowel corridor of this chapter. The **awj* resolution, **au* fronting and diphthong operations currently execute before ordinary-a brightening. They cannot be described as consuming its output. Their histories combine earlier glide reanalysis with English diphthong realization (Campbell 1959, 44–47; Ringe and Taylor 2014, 65–66, 171–73). The conventional breaking/restoration account instead presupposes fronted **æ*. English plain-a/au process identity and inherited stem identity remain separate questions.

A coordinated formalization of ordinary **a* fronting and **au* fronting with its completion has now been tested over the selected lexical material. Its twenty au paths wait until the ordinary fronting corridor and then converge with the current derivations before breaking. This establishes computational compatibility, not historical event identity. The production serialization remains unchanged, and the unstressed and surviving-long-final fronting components are not dated from this test. The completed ordinary stressed component is now independently characterized on the English daughter. Its English-episode interpretation remains explicitly dependent on its contraction and nucleus premises (Campbell 1939, 90–91; 1959, 52; Ringe and Taylor 2014, 170–75).

The earlier unstressed contribution is not the complete unstressed law: SC070 OEUnstressedFrontingEarly separately implements fronting before heterosyllabic nasals, whereas the coda-nasal history protects other endings (Campbell 1959, 140–41). The retained final-vowel helper

connects unrounding's carried quantity to later shortening and merger; it does not establish an independent historical long-a fronting (Ringe and Taylor 2014, 58–59, 299–300).

Old English consonantal changes: Velar palatalization (SC052), palatalization of *sk (SC051), j-cluster coalescence (SC057), and related changes produce the characteristically Old English consonant phonemes. Hogg discusses these as OE consonant changes with class-specific conditions and chronology (Hogg 1979, 90–111).

Old English i-umlaut and its context: The i-umlaut changes vowels under a following high front vocoid; paradigm alternations are evidence for that phonological conditioning, not grammatical conditions on the law (Campbell 1959, 69–72, §§190–197; Ringe and Taylor 2014, 222). Ordinary West Saxon palatal diphthongization precedes mutation in the handbook account, while the later treatment of some mutation products after *sċ* follows it (Fulk 2018, 74; Campbell 1959, 68–69; Ringe and Taylor 2014, 215–17, 235). The adopted cascade now places the ordinary component before mutation and retains the late approximation afterward. Gift enters with the earlier-raised vowel, following Ringe's explicitly PGmc reconstruction (Ringe 2017, 135, pp. 151–153). Its lexical section discusses the published e-alternatives and the confidence in this choice.

Late Old English syllabic reduction and apocope: High-vowel apocope (SC063), medial syncope (SC065), and the cluster of late unstressed-vowel changes (SC069–SC078) represent the later stage of Old English phonological history, when the syllabic structure of the language began to shift toward the more reduced profile of Middle English.

35.4 The English partial chains

The working historical scaffold has several connected chains, not one uniquely demonstrated total order. Their evidential strength differs. An explicit handbook derivation, a contrast between original and secondary vowels, and the survival of a modeled output are not interchangeable arguments (Ringe and Taylor 2014, 170–73, 215–37; Hogg 1979, 100–110).

Historical relation	Evidence and interpretation	Limitation
Coronal assimilation before inherited ww reanalysis	Four and the second-person pronoun have assimilated inputs to vocalization	Does not date every j-created geminate
Inherited-long restructuring before completed ai contraction	New ai-derived long a escapes inherited-long fronting	Temporal overlap is possible
Completed ai contraction before ordinary short fronting	Conventional Campbell chain	Depends on the diphthong-nucleus premise
Ordinary fronting before breaking/restoration	Conventional English reconstruction	Never-fronted and restored outputs can coincide
Breaking before ordinary palatal diphthongization before mutation	Broken inputs and worked handbook vowel histories	Does not date every consonantal palatal layer
Mutation before the later sc treatment	Ai-derived mutation products such as sheath	Exact later conditioner remains unresolved
Initial productive palatal cutoff before mutation-created fronts	Unrounded mutation vowels are the stronger controls	Medial fricative merger is a separate problem

The first three relations are supported and qualified by the earlier vowel accounts (Ringe and Taylor 2014, 41–42, 65–66, 170–73; Campbell 1939, 90–91). The palatal and mutation relations require their own evidence (Luick 1914--1940, 162–63; Fulk 2018, 74; Ringe and Taylor 2014, 215–17, 235; Hogg 1979, 100–110). The table states the defended historical targets; it does not claim that every present executable proxy already realizes them faithfully.

35.5 Early glide history and the English outcome

The diphthong corridor includes inherited ww, assimilated ww and geminates created later before j. These inputs must be distinguished before assigning one date to their long-vowel outputs. Ringe and Taylor describe inherited glide reanalysis separately from the English realization of its products (Ringe and Taylor 2014, 41–42, 65–66, 171–73). The model now separates the earlier reanalysis from English realization.

Witness	Adopted earlier checkpoint	Later English outcome	What it tests
Four	Assimilated ww is reanalyzed as eu plus w	fēower ‘four’	Assimilation feeder and retained glide
Hew	Inherited a plus ww gives au plus w	hēawan ‘hew’	Reanalysis distinct from English realization
You	Vocalized iu plus w with a surviving final i before apocope	ēow ‘you’	Prosody/apocope compatibility, not merely quantity
Hue	Later j-gemination creates a distinct glide input	hīew ‘hue’	Negative control for the earlier inherited-ww subset
Hay	Later awj history remains separate	hīeġ ‘hay’	Gemination and secondary glide resolution

These are existing lexical witnesses, not new corpus admissions. The current spelling differences also require their actual dialectal interpretation (Campbell 1959, 44–47; Ringe and Taylor 2014, 41–42, 57–58, 171–73). The reconstructed West Saxon strew target is a computational control, not an additional attested form.

The completed decomposition replaces the old consonantal-ww apocope proxy with the historical condition: a short final high vowel after a heavy syllable in a sentence-unstressed, phonologically final word. You selects that weak-final context independently of its unchanged PGmc reconstruction. Thus **iuwi* loses the final vowel to give **iuw* before mutation. A proclitic context retains the vowel, as in *ymbe* ‘around’; weak-final *and* ‘and’ and stressed *ġiest* ‘guest’ / *fȳr* ‘fire’ check the positive and negative conditions (Ringe and Taylor 2014, 41–42, 55, 57–58; Campbell 1959, 283).

The computational retained-i counterfactual for you yields a predicted †*īei*, not a source-backed strong Old English spelling: the existing w-loss before i intervenes before mutation. It tests the represented alternative context, not an additional attestation. Unmarked evaluation explicitly selects strong-final citation context; absence of an acute is never used to infer sentence stress. All selected final outputs remain unchanged.

The later realization paths are now explicit. Chew, dew, four, hew and you complete their earlier products through SC032 OEDiphthongLeveling. SC033 OEEwLongDiphthong supplies the separate j-created offglide reanalysis for *hīew* ‘hue’. The selected input is inherited PGmc **xíwją*; its path is **xíwwją* to **xíuwją*, then long **xīowją* through SC110 OEJGlideIO, before mutation gives the West Saxon long *īe* reflex. Ringe and Taylor explicitly give the inherited i and long-io chain while leaving its phonemic analysis unclear (Ringe and Taylor 2014, 53, 250). Orel’s and Kroonen’s e reconstructions are real alternatives, and Fulk disputes traditional geminate dismantling; these qualifications are discussed in the lexical account (Orel 2003, 171–72; Kroonen 2013, 224; Fulk 2018, 59, 71–72). Knee instead selects the regular dative *cneowe* ‘knee (dat.sg.)’: its short pre-ending diphthong comes from SC044 OEBreaking, not singleton lengthening. The long endingless *cnēo* ‘knee’ and analogically restored final *w* in *cnēow* ‘knee’ remain a separate paradigm comparison; later long obliques must not be confused with the selected short cell (Campbell 1959, 232–33; Hogg 2011, 21–22, 86; Ringe and Taylor 2014, 187–88, 387). SC111 PWG-

mcEndinglessAw now supplies straw's regular endingless **strau*, feeding ordinary English realization to reconstructed **strēa*. The comparison remains attested *strēaw* 'straw': its restored final *w* belongs to paradigm levelling, so this regular derivation explicitly mismatches the target. The sources mark the free regular simplex and short obliques as unattested; the compound *strēaberige* 'strawberry' is not substituted for the selected noun (Campbell 1959, 232–33; Brunner 1965, 49, 110; Ringe and Taylor 2014, 172–73, 387). SC034 OEAwLongDiphthong retains only the three show cells as a disclosed proxy. Dictionary singleton stems and the inherited-geminate handbook analysis remain distinct, unresolved premises (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49). The unchanged SC106 OEJWWSimplification remains a technical compatibility operation, but hue now reaches it with singleton w and no longer feeds it. The before-j placement of early reanalysis is a representative serialization, not a new strict historical chronology claim (Campbell 1959, 45–47; Ringe and Taylor 2014, 65–66, 171–75).

35.6 Ordinary fronting, breaking and restoration

The preferred conventional account identifies ordinary non-nasal short-a fronting and fronting of au's first element as one English episode. The later offglide development is distinct. Early spellings record the fronted element, but do not measure the simultaneity of two laws (Campbell 1939, 91; 1959, 52–53; Luick 1914--1940, 130–31). English episode identity is not inherited Anglo-Frisian event identity. Nor does it identify the stressed-short process with every unstressed or long-final clause in the present model.

Breaking then has vowel, quantity, consonantal and dialectal conditions. Restoration in a following back-vowel environment can remove a previously fronted vowel. A final restored vowel alone cannot distinguish that history from original blocking of fronting (Hogg 1979, 90–97; Versloot 2025, 104, 123). The conventional account is retained as the working baseline because of the combined argument, not because the more elaborate intermediate path is inherently preferable.

Versloot's late Anglian alternatives must be compared in their exact domains. His mutation-created e argument and the unbroken neighboring environments are relevant to a restricted e-breaking component, not to a wholesale postmutation relocation of every West Saxon breaking law (Versloot 2025, 126–28, 131–35). A different dialectal target cannot be silently substituted for a West Saxon lexical target. Similarly, Frisian closed-syllable breaking is not English rC breaking merely because both are called breaking (Bremmer 2009, 33–35, 37).

35.7 Palatal consonants and the key–day problem

The useful chronological contrast is between original conditioning fronts and front vowels created by mutation. Rounded mutation products alone may support a weaker conclusion, a cutoff before later unrounding. Unrounded mutation-created fronts more directly test whether initial velar palatalization was still productive (Hogg 1979, 100–103; Laker 2007, 167–68). This cutoff does not by itself date assibilation or every earlier articulatory tendency.

Hogg's key and day paradigms identify the disputed merger premise. The initial key consonant survives a mutation-created unrounded front vowel; the day paradigm does not show the mutation that would follow if its new medial palatal element were already equivalent to inherited j. These statements concern particular historical cells, not arbitrary exceptions to a sound law (Hogg 1979, 102–10).

Diagnostic	Required distinction	What is not established
Key oblique vowel	Original fronts versus mutation-created fronts	Date of all palatal articulation
Day oblique vowel	Inherited j versus a palatal fricative at mutation	Unanimous agreement on merger timing
Geminate/postnasal g	Stop-class history versus singleton fricative	One universal g-to-j change

Diagnostic	Required distinction	What is not established
h	Breaking trigger versus subsequent palatal/weakening history	Identity with voiced-fricative chronology
sk	Own consonantal law versus later sc vowel treatment	A single combined palatalization event

Hogg considers several resolutions without endorsing one as demonstrated (Hogg 1979, 103–11). Ringe and Taylor, however, explicitly adopt merger after mutation (Ringe and Taylor 2014, 204). CAPR now implements that regular working account, preserving Hogg's phonetic objection rather than claiming consensus. Singleton g is a fricative, including initially; gg/ng are stops (Fulk 2018, 130–32).

At Hogg's pre-palatal, pre-OE checkpoint, key has **kājœ* and day **dœɣœ*. The inherited j in the former causes mutation, producing **kājœ* without reactivating initial-k palatalization. In the latter, the newly palatal fricative remains ȷ and the vowel stays æ: **dœȷœ*. Native realization then gives *cāġe* 'key' and *dæġe* 'day' (Hogg 1979, 105; Ringe and Taylor 2014, 204). Premature merger instead supplies a j trigger and wrongly raises the day vowel. These oblique cells are non-corpus diagnostics, not new selected PGmc reconstructions; the existing day nominative cannot substitute for them.

The complete key suffix exposed a second defect: unrestricted inherited-j normalization would erase its retained glide. The repaired SC082 OEIntervocalicJVocalization excludes non-high long front monophthongs. Its residual weak-suffix normalization is a telescoped representation, not a universal historical VjV vocalization law (Hogg 2011, 283–86; Ringe and Taylor 2014, 228).

The fricative merger has its own visible SC109 OEPalatalFricativeMerger. Its late execution after suffix raising is a tested computational holding zone, not a precisely proved historical date. Likewise stop ʧ/ʤ are eventual-reflex proxies, not early affricate claims. Cluster coalescence and a dotted written reflex cannot identify all these classes as one historical sound (Ringe and Taylor 2014, 203–4; Laker 2007, 167–68; Fulk 2018, 131–32).

35.8 Three different mutation and diphthongization arguments

Guest, sheath and gift do not establish the same chronological relation. The guest derivation passes through ordinary palatal diphthongization before mutation in both the handbook account and the adopted model. The former model raised a simple vowel before diphthongizing it. Both paths reach *ġiest* 'guest', so their final agreement does not select the earlier path (Ringe and Taylor 2014, 216).

The following worked comparison abstracts from consonant notation and retains the vocalic trigger where it matters. For guest, the source starts with PGmc **gastiz* 'guest', continued as PWGmc **gasti* 'guest'. The vowel history is **a* > **œ* > **ea* > **ie*; final loss of the trigger follows the mutation it caused (Ringe and Taylor 2014, 216, 287).

Checkpoint	Handbook and adopted guest path	Superseded modeled guest path
Ordinary fronting	Short a becomes æ	Short a becomes æ
After palatal conditioning is established	Original front vowel is available to ordinary PD	Original front vowel is available, but PD remains later
First disputed vowel operation	æ becomes ea by ordinary PD	æ becomes e at mutation
Second disputed vowel operation	ea becomes ie at mutation	e becomes ie in the bundled PD proxy
Written result	ġiest 'guest'	ġiest 'guest'

The agreement in the last row is precisely why the intermediate rows must be tested. This table does not treat the implementation's palatal-consonant symbol as proof of a uniquely dated phonetic merger (Hogg 1979, 103–10).

Witness	Source-supported distinction	Present model consequence
Guest	Ordinary palatal diphthongization feeds mutation	Adopted æ > ea > ie replaces the output-equivalent old path
Sheath	Ai-derived ā mutates; a later sc layer can affect its product	Its unchanged late clause is separately retained as an approximation
Gift	Earlier inherited e-to-i raising is distinct from English mutation	Selected PGmc i is outside ordinary diphthongization
Sheep/year	Ordinary palatal diphthongization without a mutation trigger	Controls for the ordinary component
Cow/lung	Mutation changes the productive consonantal environment	Controls for cutoff, not diphthongal mutation

For sheath, *sċēaþ* 'sheath' beside *sċœþ* 'sheath' belongs to the separately described later group. Reporting both outcomes does not authorize lexical optionality in the law; the exact phonological layer and conditioner still require a defended specification (Campbell 1959, 68–69; Ringe and Taylor 2014, 235). The incremental repair preserves the literal late clause as a visible approximation, not a lexical exception or a newly established conditioner.

The sheath comparison instead follows the ai-derived long vowel: **ai* > **ā* > **œ̄*, with the final step supplied by mutation. The separately discussed later sc treatment can then give **ēa*. The vowel available to the earlier ordinary process and the vowel created later by mutation are therefore different historical inputs, which the adopted model now separates (Campbell 1959, 68–69; Ringe and Taylor 2014, 235).

For gift, Ringe reconstructs PGmc **giftiz* 'gift', with the Old English plural *ġifta* 'wedding', and describes the earlier raising separately (Ringe 2017, 135, pp. 151–153). Ringe and Taylor explicitly distinguish that early change from the much later English vowel history (Ringe and Taylor 2014, 220). The selected input now follows that earlier-raised reconstruction. The choice is lexical and source-led, not a new raising rule justified by output fit. Orel and Kluge–Seebold print e-vowel alternatives (Orel 2003, 130; Kluge 2011, 359); the lexical section compares their evidence with the e-grade presentations and Ringe's qualified PGmc dating rather than claiming unanimous agreement.

35.9 Quantity, reduction and the written surface

Later apocope and syncope depend on prosody and syllable structure; earlier input conventions and surviving triggers must be inspected before their effects are used as dates for other changes. Morphological cells provide evidence for such environments, not grammatical conditions on phonological rules (Ringe and Taylor 2014, 57–59, 299–300). Inherited quantity, compensatory lengthening, diphthong quantity and later shortening must likewise remain separate dimensions.

The transition to Chapter 5 is a transition to the written surface, not proof that an orthographic change is a newly dated sound change. A single spelling can represent different historical consonant inputs, and a changed spelling need not change the phonetic derivation. The source discussions and worked lexical entries distinguish implemented laws from working reconstructions and localized unresolved components.

35.10 Executable serialization and unresolved interfaces

Sections follow executable order to make derivations inspectable, not because that serialization proves a unique total history. Final agreement can conceal different intermediate paths. Some proxies execute here with earlier historical classifications; renaming identifiers would not resolve that substantive distinction:

- SC041 (PWGmc Final Bare-*a Loss) and SC042 (Surviving Bimoric *ō Unrounding) carry Proto-West Germanic labels but execute in this stretch of the cascade.
- SC064 (NWGmc *-n Stem *n Loss) carries a Northwest Germanic label but executes after OE High-Vowel Apocope (SC063).
- SC049 (PGmc B Allophony) carries a Proto-Germanic label but executes here.

35.11 Sources

Campbell's *Old English Grammar* is the primary source for the dating and scope of individual changes in this chapter (Campbell 1959, 52–72). Hogg provides critical reassessments of palatal conditioning and its relative chronology (Hogg 1979, 90–112). Ringe and Taylor supply the most detailed relative-chronology analysis for the earlier portion of the chapter, through back-mutation (Ringe and Taylor 2014, 169–73, 215–37). Fulk's *Comparative Grammar* provides additional coverage for phonological conditioning (Fulk 2018, 58–61, 72–74). For individual changes, source-specific citations appear in the relevant sound-change sections.

36 English stressed ai contraction

36.1 Historical discussion

Inherited stressed **ái* yields **ā* in English. Ringe and Taylor discuss this development separately from the earlier unstressed contraction. The distinction between inherited long vowels and new **ā* requires inherited-long fronting to have been well under way before contraction completed; it does not prove that the two developments could not overlap (Ringe and Taylor 2014, 170–71). Campbell gives the stricter conventional sequence and places contraction before ordinary short-vowel fronting (Campbell 1959, 52–53). Later i-mutation of the new long vowel is a separate conditioned development, not that ordinary fronting (Campbell 1959, 69).

Versloot proposes a wave account of the regional outcomes, but CAPR does not adopt diffusion as a solution to the comparative tree problem. His readings of early English and Frisian inscriptions are relevant evidence independently of that mechanism, conditional on their provenance, etymology and phonetic interpretation (Versloot 2017, 295–97, 318). The required Anglo-Frisian ancestral node is retained. CAPR places completed English contraction on the daughter branch, while leaving a possible earlier conditioned onset on the common stem distinct. This is the defended working placement under the cited runic and comparative premises, not proof against every ancestral onset (Campbell 1939, 90–91; Versloot 2017, 295–97, 318). The inherited-long relation permits temporal overlap; the retained executable order does not assert a strict order between entire events.

The live selected-corpus census has twenty-three applications, all carrying stressed **ái*. Loam's selected **láimą* 'loam' is explicitly a pre-Old-English model input, not an independent Proto-Germanic witness. The raw corpus's additional roe reconstruction has no attested target and is excluded from that census. The unstressed development **ai* > **ē* is the separate earlier change SC014 PNWGmcUnstressedAiMonophthongization.

36.2 SC004. English stressed ai contraction (`EAFAiMonophthongization`)

```
define EAFAiMonophthongization [
    {*ái} -> {*ā}
];
```

The soul form fixes the relation to interstress raising. If the monophthongization is delayed until after that change, PGmc **sáiwalō* ‘soul’ yields †*sāwel* rather than expected OE *sāwol* ‘soul’. An earlier placement changes no output. This shows that SC004 EAFAiMonophthongization must come before SC036 OEInterStressRaising in the modeled sequence.

The unstressed development **ai* > **ē* in final and nonfinal syllables is a separate and earlier change; see SC014 PNWGmcUnstressedAiMonophthongization.

37 Awj resolution and the English brightening of au

37.1 Historical discussion

Two changes stand between the Proto-Germanic diphthong **au* and its Old English reflex *ēa*. The earlier one repairs a West Germanic gemination and restores a diphthong that the gemination had obscured. The later one fronts the first element of every **au*, whether inherited or newly created, and belongs with Anglo-Frisian brightening.

The two are separate developments with separate domains. The first concerns a handful of words in which **w* stood before **j*. The second concerns the whole Old English history of **au*, for which *lēaf* ‘leaf’, *strēam* ‘stream’ and *brēad* ‘bread’ are ordinary witnesses, and to which the first change merely adds two more inputs.

37.2 Historical discussion of the resolution of **awj*

Old English *hīeġ* ‘hay’ and *strīeġan* ‘strew’ go back to forms in which **w* preceded **j*. West Germanic doubled every consonant except **r* before **j* after a short syllable, and **w* was an ordinary member of that law (Campbell 1959, 167, §407). Proto-Germanic **hawja-* therefore appears as West Germanic **hauuj* (Campbell 1959, 46, §120.2). Campbell writes the sequence as *auj* > *auuj* > *auj*, with the diphthong restored before the Old English developments begin.

Ringe and Taylor reach the same result and explain why it is possible (Ringe and Taylor 2014, 53, §3.1.3). They find the gemination of **wj* clearest where the preceding vowel was **i*, as in Proto-Germanic **niwjaz* ‘new’ and **siwjaną* ‘sew’, which give Old Saxon and Old High German *niuwi* and *siuwen*. Gemination was reversible, since it merged nothing and altered no underlying form, so the sequence Northwest Germanic **awj* to West Germanic **[aw’w’]* to pre-Old English **[auj]* can have run its course and then undone itself. Their derivations give Proto-Germanic **hawja* through **hauj-* to *hīeġ* ‘hay’, and Proto-Germanic **strawjaną* through **straujan* to Anglian *strēgan* ‘strew’ (Ringe and Taylor 2014, 173).

The strongest comparative argument for the doubled stage comes from paradigms in which some cells had **j* in the ending and others had **i*. Only the first group could double, and the two outcomes then sat side by side. Old High German has *hewi* ‘hay’ beside *houwi*, and Old English itself preserves both in one word, *glīg* ‘mirth’ from the undoubled nominative beside *glīowes* ‘of mirth’ in the genitive (Ringe and Taylor 2014, 53, §3.1.3). The continental forms *houwi* and *gistrouwen* ‘bestrew’, and Old Saxon *hoi* ‘hay’, point to the same West Germanic stage, which English alone went on to resolve (Ringe and Taylor 2014, 173).

The resolution did not treat every doubled **w* alike, and the difference is what allows the doubling to be seen apart from it. Campbell sets the two vowel types side by side: *auj* becomes *auuj* and then *auj*, while *iuj* becomes *iuuj* and then *iuj*. Thereafter they part company, since the *u* of *auuj* is generally lost while the *j* of *iuuj* is lost (Campbell 1959, 46, §120.2). Old English *hīeġ* ‘hay’ and *hīew* ‘form, hue’ begin from shapes that differ in a single vowel and end with different survivors. *Hīeġ* keeps its **j*, written *ġ*, while *hīew* keeps its **w*. Campbell’s other examples of the second type are *nīowe* ‘new’, *nīewe* ‘new’ and *glīow* ‘mirth’, *glīw* ‘mirth’.

One qualification belongs in the record. Fulk holds that **w* was never consonantal here, so that Proto-Germanic already had **straujaną* with its diphthong in place and there is no gemination to undo (Fulk 2018, 73, §4.10 n. 1). The account followed here is the handbook one, which the comparative material supports: the doubled stage that the continental and paradigm-internal forms point to is precisely the one Fulk denies ever existed. Both accounts agree that **auj* is what enters Old English, and the disagreement concerns whether a discrete change took place.

37.3 SC029. Resolution of **awj* to **auj* (`OEAwwjResolution`)

```
define OEAwwjResolution [
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j}
```

```
];
```

The change is confined to the **a* type. PGmc **xáwją* ‘hay’ is doubled to **xáwwją* by the West Germanic law and then resolved to **xáują*, and PGmc **stráwjaną* ‘strew’ is doubled to **stráwwjaną* and resolved to **stráujaną*, yielding *hīeġ* ‘hay’ and *strīeġan* ‘strew’ once the later diphthong changes and *i*-umlaut have applied. PGmc **xíwją* ‘form, hue’ is doubled by the same law, and there the doubling holds: it passes through this change untouched and surfaces as *hīew* ‘form, hue’.

The resolution has to precede SC030 OEAuBrightening, since the fronting needs a diphthong to work on. Ringe and Taylor state the same dependence when they observe that these new instances of **au* went on to share the ordinary development (Ringe and Taylor 2014, 173).

37.4 Historical discussion of the brightening of **au*

The fronting of **au* is the application to a diphthong of the same process that fronts plain **a*, the process usually called Anglo-Frisian brightening. Campbell arrives at the point while establishing the order of the early vowel changes, remarking that the normal development of Proto-Germanic **au* to Old English *ēa* shows that the change of *a* to *æ* would affect the first element of a diphthong (Campbell 1959, 52, §132). His ordered list accordingly places West Germanic *a* > Old English *æ* and West Germanic **au* > Old English **æu* in one and the same step. Fulk puts it in a single sentence, saying that this fronting of *a* applied also to the diphthong *au* in Old English (Fulk 2018, 73, §4.12).

The intermediate stage is directly attested. Ringe and Taylor describe **au* as first tensed and fronted to **æu*, a spelling still found occasionally in eighth-century documents, with the offglide unrounded and lowered only later (Ringe and Taylor 2014, 172). Early spellings such as *Eadbald* with an initial *aeo* preserve the rounded offglide, and the rounding survives in late Northumbrian (Fulk 2018, 73, §4.12). The outcome of the whole sequence is the *ēa* of *dēaþ* ‘death’, *ēage* ‘eye’ and *lēaf* ‘leaf’ (Campbell 1959, 53, §135).

The geographical reach of the diphthongal fronting is narrower than that of the plain one. Old Frisian shows no such fronting and has *ā*, so that Old English *ēac* ‘also’, *ēage* ‘eye’ and *bēam* ‘tree’ stand against Old Frisian *āk* ‘also’, *āge* ‘eye’ and *bām* ‘tree’ (Fulk 2018, 73, §4.12; Ringe and Taylor 2014, 172). A later change supplies independent confirmation. Old English *gēac* ‘cuckoo’ has a palatalized initial, which requires a front vowel to have followed it, while Old Frisian *gāk* ‘cuckoo’ has none (Fulk 2018, 73, §4.12). Brightening of plain **a* is shared with Frisian; brightening of the first element of **au* is English.

37.5 SC030. Brightening of **au* to **æu* (`OEAuBrightening`)

```
define OEAuBrightening [
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
];
```

Most of the words that pass through the change carry **au* inherited straight from Proto-Germanic. PGmc **láubą* ‘leaf’ gives *lēaf* ‘leaf’, PGmc **stráumaz* ‘stream’ gives *strēam* ‘stream’, PGmc **bráudą* ‘bread’ gives *brēad* ‘bread’, and PGmc **dráugmaz* ‘dream’ gives *drēam* ‘dream’. Where a following **j* or **i* survives long enough to cause *i*-umlaut, the *ēa* appears in West Saxon as *īe*, as in *ġelīefan* ‘believe’ from PGmc **galáubijaną* ‘believe’ and *nīed* ‘need’ from PGmc **náudiz* ‘need’ (Campbell 1959, 46, §120.2). The two words supplied by SC029 OEAwwjResolution, *hīeġ* ‘hay’ and *strīeġan* ‘strew’, join this second group.

The proportions matter for what the rule is. This is the general Old English treatment of **au*, to which the resolution of **awj* contributes two further inputs; the history of *hīeġ* ‘hay’ and *strīeġan* ‘strew’ does not define it.

The fronted **æu* has no independent life. It is taken up at once by the SC032 OEDiphthongLeveling, which lowers the offglide and delivers *ēa*. Ringe and Taylor give that order explicitly when

they place the tensing and fronting first and the unrounding and lowering later (Ringe and Taylor 2014, 172).

38 Short pre-w diphthongs and the j-created glide chain

38.1 Historical discussion

Ordinary diphthongization before singleton *w* must be distinguished from long diphthong formation in endingless forms and from the independently created *wwj* sequences. Their similar Old English spellings do not establish one sound law assigning long quantity before every retained vowel ending.

For *cneowe* ‘knee (dat.sg.)’, the regular pre-ending stem has short *eo*. Hogg explicitly identifies the dative as a short-diphthong example, and Campbell and Luick distinguish its genitive and dative from the long endingless form (Hogg 2011, 21–22, 86, §§2.33, 5.22; Campbell 1959, 232–33, §584; Luick 1914--1940, 139, §134). Ringe and Taylor likewise distinguish short *cneow-* before syllabic endings from long endingless *cnēo* ‘knee’. The final *w* of *cnēow* ‘knee’ is generalized from the obliques, not retained by an unrestricted prevocalic lengthening law (Ringe and Taylor 2014, 187–88, 387, §§6.2.4, 7.2.4).

The selected comparison is therefore the dative *cneowe*, from the constructed PGmc dative **knéwai*, while citation **knéwą* remains the lexeme-level reconstruction. Fulk prefers the **-ai* analysis for the West Germanic dative line but discusses alternatives; the complete selected input is a paradigm construction, not a quotation of a whole-word reconstruction (Fulk 2018, 147, §7.8).

Its short diphthong is supplied by SC044 OEBreaking, with the source’s exclusion before a following high front vocalic. Long quantity could later spread from endingless forms into obliques, but that leveling is not the selected short dative’s sound-law history. Campbell describes the extension; Ringe and Taylor qualify the knee quantity inference because decisive verse evidence is elusive (Campbell 1959, 233; Ringe and Taylor 2014, 387).

Earlier inherited-glide reanalysis is independently represented by SC031 OEWWSimplification. Chew, dew, four, hew and you complete its products through SC032 OEDiphthongLeveling. The operation below neither repeats that inherited event nor implements the regular singleton history (Ringe and Taylor 2014, 41–42, 65–66, 171–75).

38.2 SC033. J-created palatalized-glide reanalysis (`OEEwLongDiphthong`)

```
define OEEwLongDiphthong [
    {*i} {*w} {*w} -> {*iu} {*w} || _ {*j},
    {*í} {*w} {*w} -> {*íu} {*w} || _ {*j}
];
```

Hīew ‘hue’ enters with inherited PGmc **xíwją*, following Ringe and Taylor’s explicit **hiwją*. Their palatalized geminate develops through **iuw* before the separately realized long diphthong. The conditioner is palatalized *w*, carried here as *ww+j* and subsequently *w+j*; this does not assert a second independent phoneme where the authors leave the phonemic analysis unclear (Ringe and Taylor 2014, 53, 250).

Orel’s **xewjan* and Kroonen’s **heuja-* remain competing e reconstructions, not earlier stages invented to reconcile the authors. Their derivational explanations differ. The realized-i choice uses the inherited raising account, not a later hue-specific law (Orel 2003, 171–72; Kroonen 2013, 224; Ringe 2017, 151–53; Fulk 2018, 59). Campbell supports the contrasting *iwj* outcome, but Fulk questions traditional geminate dismantling and its consonantal premises. The detailed lexical account preserves those objections (Campbell 1959, 46; Fulk 2018, 71–72, 126).

The stable component now supplies offglide reanalysis, not long quantity. Singleton *iwj*, nonpalatal *iww*, unraised *ewwj* and the separate low-vowel *awj* history are excluded. Knee remains a short singleton-w comparison. No exact historical date or independent confidence is fabricated for this support encoding.

38.3 SC106. Retained j-created glide compatibility (OEJWWSimplification)

```
define OEJWWSimplification [
    {*w} {*w} -> {*w} || _ {*j}
];
```

The unchanged technical operation remains composed for compatibility. Hue's reanalysis already leaves a singleton glide, so it no longer feeds this simplification. Absence of a current application does not establish universal redundancy or a newly dated historical event. It neither restores nominative *w* in knee nor duplicates earlier inherited short-Vww reanalysis.

38.4 SC110. Palatalized-glide long-io realization (OEJGlideIO)

```
define OEJGlideIO [
    [{*iu}|{*íu}] -> {*īo} || _ {*w} {*j}
];
```

The palatalized-glide product realizes long *īo*, not the ordinary long *ēo* supplied by SC032 OEDiphthongLeveling. Ringe and Taylor explicitly supply pre-OE long *īow* and early West Saxon long *īew*; the retained semivowel still conditions mutation (Ringe and Taylor 2014, 250). Existing SC055 OEIUmlaut supplies the latter change. Ordinary inherited *iu* without this palatal conditioner remains outside the component, as does short *io*. This distinct realization does not adjudicate all the inherited atomic and split-symbol clauses in SC032 OEDiphthongLeveling.

39 Quantity-distinct English diphthong realization

39.1 Historical discussion

The English histories of inherited diphthongs and their secondary equivalents must be distinguished from both smoothing and paradigm leveling. Ringe and Taylor identify a common tensing or lengthening of the syllabic nuclei, followed by further developments of the offglides; they place that tensing before breaking. This does not make all the eventual English spellings the products of a single simultaneous change (Ringe and Taylor 2014, 170–75).

The fronted continuation of **au* gives long *ēa*, including in *brēad* 'bread', *dēaw* 'dew' and *hēawan* 'hew'. First-element fronting is distinct from subsequent offglide unrounding and lowering. The latter was not universal: some Northumbrian varieties retained a rounded offglide. Thus this outcome is not exclusively West Saxon, but neither is it an exceptionless description of every English dialect (Ringe and Taylor 2014, 172–73; Campbell 1959, 53–54, 116–18).

The long *ēo* outcomes in *ċēowan* 'chew' and *fēower* 'four' belong to the **eu* history. Earlier inherited or assimilation-created geminate-glide reanalysis supplies the diphthong while retaining the following consonantal glide. Dew and hew similarly acquire **au* through SC031 OEWWSimplification, then reach the fronted-au path. Their early reanalysis is not the later English realization (Ringe and Taylor 2014, 41–42, 65–66, 172–74).

The **iu* history is different again. Early *īo* is not simply another spelling for the initial **eu* development. Subsequent io/eo convergence has dialect-specific histories: Northumbrian retains io, whereas West Saxon and parts of Mercian show convergence. The long *ēo* of *frēond* 'friend' and *ēow* 'you' is the selected endpoint here, not evidence that the merger preceded every intervening English change. Friend's contraction-created input and you's earlier glide reanalysis are also distinct origins (Ringe and Taylor 2014, 41–42, 174–75; Campbell 1959, 124–26).

39.2 SC032. Quantity-distinct diphthong realization (OEDiphthongLeveling)

```
define OEDiphthongLeveling [
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
];
```

The retained definition is a compound realization proxy. Its atomic **aeu*, **eu* and **iu* symbols belong to the heavy, long-output tier; an acute mark does not itself indicate length. Atomic **aeu* records a fronted-au intermediate, not a claim that a three-vowel phonetic sequence is demonstrated. The direct **iu* to *ēo* mapping packages the selected later endpoint. The split-symbol e+u, é+u and i+u clauses instead produce short *eo*, retaining stress only in the é+u clause. Campbell distinguishes short breaking and back-umlaut diphthongs from their long equivalents, but his account does not specify these computational token boundaries (Campbell 1959, 116–18).

All 32 current applications use atomic long-tier inputs. The unaccented atomic alternatives and all three split-symbol clauses remain formal compatibility paths without a selected-word application. Their retention does not establish an independent palatal-glide origin, and no additional stressed-i clause is inferred.

The short dative *cneowe* 'knee' remains outside this long-output operation; its pre-w breaking is supplied by SC044 OEBreaking. The separately palatalized-glide history of *hīew* 'hue' reaches long *īo* through SC110 OEJGlideIO before the ordinary iu matcher. It then receives long *īe* through

mutation, not ordinary *ēo* here. These contrasts preserve quantity and conditioning within the adopted selected cells (Campbell 1959, 232–33; Ringe and Taylor 2014, 187–88, 250, 387).

Fronting must supply its product before its offglide continuation: SC030 OEAuBrightening feeds this component, not every clause in the definition. Ringe and Taylor explicitly distinguish the two developments (Ringe and Taylor 2014, 172). Earlier displacement tests that left **aeu* unrealized produced +*?*, a computational rejection with no attested linguistic status. They therefore cannot independently establish the historical interval.

The input of *hēafod* 'head' must be realized into a recognized vowel before SC040 OEMedUnstressedULowering can lower its remaining medial vowel in the model. That internal recognition requirement is not proof that historical offglide change conditioned the lowering. Campbell records both *hēafud* and *hēafod*, with varying preservation of protected unstressed u (Campbell 1959, 155–56). No historical order for the entire nine-clause bundle follows from that one computational dependency.

40 Endingless *aw* and the retained show proxy

40.1 Historical discussion

The attested long nucleus does not establish one general law changing singleton *aw* to *ēaw* before every following vowel. For straw, the inherited nominative/accusative **stráwą* loses its ending and vocalizes final *w*, giving **strau*; the ordinary English development then gives reconstructed **strēa*. Attested *strēaw* ‘straw’ combines that long diphthong with *w* restored from inflected forms. Conversely, the long diphthong spreads into those forms. The regular free simplex and short obliques are explicitly unattested, although the endingless pattern survives in the compound *strēaberige* ‘strawberry’ (Campbell 1959, 46–47, 232–33; Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 172–73, 387).

The model retains attested *strēaw* ‘straw’ as its comparison target and produces reconstructed **strēa* through the regular cascade. This is an explicit mismatch, not an attested reconstructed endpoint or a sound law for analogical restoration. SC111 PWGmcEndinglessAw supplies the narrow endingless fragment; it does not imply that general final-vowel loss was conditioned by an adjacent *w*.

The inherited geminates of *dēaw* ‘dew’ and *hēawan* ‘hew’ instead undergo early SC031 OEWWSimplification, producing **au* plus retained **w*, followed by English fronting and SC032 OEDiphthongLeveling. Campbell distinguishes the glide histories; Ringe and Taylor distinguish earlier reanalysis from later English realization (Campbell 1959, 45–47, 53–54; Ringe and Taylor 2014, 65–66, 171–75). These retained-glide histories differ from endingless straw.

For *sċēawian* ‘show’, Orel and Kroonen print singleton dictionary inputs, while Campbell and Brunner include the verb among inherited geminate examples. The class-II suffix places a vowel between root *w* and *j*; this excludes direct root *aw+j* gemination but does not resolve the different root histories (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49). SC034 OEAwLongDiphthong therefore remains a disclosed support proxy for the three selected show cells, not an established general singleton sound law or a newly dated West Saxon event.

40.2 SC034. Retained show long-diphthong proxy (`OEAwLongDiphthong`)

```
define OEAwLongDiphthong [
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    {*á} {*w} -> {*éa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

The current operation changes the show infinitive, imperative singular and present third singular. Straw already has its regular long nucleus without *w* and is unchanged here. Dew and hew no longer demonstrate a local simplification-then-lengthening chain here: their earlier reanalysis has already removed that input shape. The older displacement results for those geminates describe the superseded representation, not independent dates for the adopted early event. The two proxy clauses remain unchanged; their successful show finals do not decide the unresolved singleton-versus-geminate source premise. The old straw displacement observation measured dependence on a following vowel in the superseded proxy, not independent historical order (Campbell 1959, 45–47; Ringe and Taylor 2014, 171–75).

41 Prefix and compound adjustments

41.1 Historical discussion of prefixal **a*-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* 'believe' supplies the comparative witness for the same development (Campbell 1959, 147, §354; Ringe and Taylor 2014, 245; 2014, 267).

The rule has a narrow historical range and gives prefixed forms the weak vowel inherited by later vocalic changes.

41.2 SC035. Reduction of prefixal **a* (`OEPrefixAReduction`)

```
define OEPrefixAReduction [
    {*a} -> {*ĕ}
        || .#. {*g} _
           [EnglishStarConsonant | EnglishPalatalConsonant]
           EnglishStarVocalic
];
```

The prefix of *ġelīefan* 'believe' supplies the upper boundary for **ga-* > **ge-*. If SC035 OEPrefixAReduction follows SC043 EAFBrightening, PGmc **galáubijaną* 'believe' yields †*ġealīefan* rather than expected OE *ġelīefan* 'believe'. Earlier placement changes no output, so the witness dates prefix reduction before brightening without locating its beginning.

41.3 Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell's discussion of *weorold* 'world' / *weoruld* 'world' and Ringe and Taylor's derivation of **weraldu* 'world' > **weruldu* 'world' > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

41.4 SC036. Raising of medial **a* between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising [
    {*a} -> {*u}
        || PGmcStarVowel EnglishStarConsonant* _
           [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
    {*à} -> {*u}
];
```

The two boundaries have unequal force. Before SC019 PNWGmcFinalLongORaising, PGmc **sáiwalō* 'soul' yields †*sāwel* rather than expected OE *sāwol* 'soul'; after SC040 OEMedUnstressedU-Lowering, it yields †*sāwul* rather than *sāwol*, while PGmc **wír-àldu* 'world' yields †*weoruld* rather than *weorold* 'world'. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In handbook terms, medial **a* > **u* belongs to the *world-* and *soul*-type low-stress vocalism that followed the earlier final-vowel changes.

41.5 Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as *reġnboga* ‘rainbow’. The only boundary the lexical evidence supplies is the immediately following technical stress-stripping stage, which is not a sound change.

41.6 SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope [
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _
           OEAnyConsonant+ PGmcStarGraveVowel
];
```

The *reġnboga* ‘rainbow’ test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc **régna-bùgô* ‘rainbow’ yields †*reġnefoga* rather than expected OE *reġnboga* ‘rainbow’, because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

42 Medial unstressed vowel changes

42.1 Historical discussion

The history of *wuduwe* ‘widow’ orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* ‘world’ / *weoruld* ‘world’ alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu* ‘world’, and **jugunþi* ‘youth’ (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

42.2 SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

```
define OEWICombinativeUUmlaut [
    {*í} -> {*ú}
        || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

The *wuduwe* ‘widow’ derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut follows SC040 OEMedUnstressedULowering, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe*; earlier placement changes no output. The witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

42.3 SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering [
    {*u} -> {*o}
        || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _
           [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

The two witnesses date medial unstressed **u* > **o* at very different scales. Before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe* ‘widow’; after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* ‘youth’ yields †*ġeogoþ* rather than expected *ġeoguþ* ‘youth’. The local *weorold* ‘world’ and widow evidence places lowering after combinative u-umlaut, while the youth form supplies only the distant requirement that lowering precede unstressed long-vowel shortening.

43 Final bare-*a* loss

43.1 Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

43.2 SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

```
define PWGmcFinalBareALoss [
    {*a} -> 0 || _ .#.
];
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final *z*-deletion, the change gives the wrong outputs: PGmc **bárdaz* ‘beard’ yields †*bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* ‘comb’ yields †*camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* ‘craft’ yields †*craft* rather than expected OE *cræft* ‘craft’, and PGmc **dágaz* ‘day’ yields †*dag* rather than expected *dæġ* ‘day’. The distant lower limit follows final *z*-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

44 Surviving bimoric **ō* unrounding

44.1 Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* ‘rest’ nevertheless undergoes unrounding before SC043 EAFBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992b, 102, 105; Ringe and Taylor 2014, 157–58, 189–90).

The sole witness establishes a local relation to brightening but supports no broader generalization.

44.2 SC042. Unrounding of the surviving bimoric **ō* (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding [
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ .#.
];
```

The single *ræste* ‘rest’ derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 EAFFinalZDeletion or after SC043 EAFBrightening, PGmc **rástōz* ‘rest’ yields †*rasta* rather than expected OE *ræste*. Unrounding must therefore follow final *z*-loss and precede brightening, although only the relation to brightening is local.

45 Ordinary English fronting

45.1 Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (Campbell 1959, 52, §131; Hogg 1992b, 102, 105; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

The traditional name does not itself establish one inherited event. Under Campbell's premise that the daughter contractions precede ordinary fronting, comparable English and Frisian outcomes require separate daughter frontings in the strict tree. Chapter 3 develops that conditional argument without removing the ancestral node (Campbell 1939, 90–91). The model adopts the completed ordinary English stressed component on the English daughter under this conventional working account. This is not a claim that every author proves contraction before ordinary fronting: Ringe and Taylor explicitly question whether diphthong nuclei must behave like the plain short vowel (Ringe and Taylor 2014, 170–75). Earlier restricted ancestral fronting remains possible.

The separately retained unstressed and final-vowel components are not dated by that argument. The first supplies an earlier nonnasal contribution; SC070 OEUnstressedFrontingEarly separately implements the broader unstressed domain, including the contrast between coda and heterosyllabic nasals (Campbell 1959, 140–41, §§333–334). The second carries the model's preserved final-vowel quantity into later shortening and merger. Its long intermediate is a representation choice, not independent evidence for a historical long-vowel fronting (Ringe and Taylor 2014, 58–59, 299–300).

45.2 SC043. Fronting of low **a* outside nasal environments (EAFBrightening)

```
define EAFBrightening [
    EAFBrighteningStressed
];
```

45.2.1 Stressed component

```
define EAFBrighteningStressed [
    {*á} -> {*æ} || _ [EnglishStarConsonant - EnglishStarNasal | .#.]
];
```

Slay requires breaking to receive the fronted root vowel: delaying the stressed member until after SC044 OEBreaking gives *sleaan* | *slēaan*, rather than OE *slēan* 'slay'. Rest involves two different changes, root fronting and the separately retained final-vowel representation. Its final-vowel dependency must not be used to date this stressed rule.

45.2.2 Retained unstressed contribution

```
define EAFBrighteningUnstressed [
    {*a} -> {*æ} || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

This earlier contribution is preserved without identifying it with the entire unstressed history. The later syllabic rule also fronts surviving unstressed vowels before heterosyllabic nasals; an exception before every nasal would therefore be too broad (Campbell 1959, 140–41).

45.2.3 Retained final-vowel representation

```
define EAFBrighteningLongFinal [
    {*ā} -> {*ǣ} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
];
```

In *ræste* ‘rest’, this helper receives the length-preserved outcome of SC042 PWGmcSurvivingBimoricOUnrounding. The preceding-nucleus guard excludes stressed monosyllabic *hwā* ‘who’. Later shortening and merger complete the ending; the carried long vowel does not establish a separate dated long-a law (Ringe and Taylor 2014, 58–59, 299–300; Campbell 1959, 49, §125).

46 Breaking and velar-fricative palatalization

46.1 Historical discussion

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

Breaking has the fuller handbook treatment. The *feoh* ‘cattle’ and *feohtan* ‘fight’ type derivations preserve its velar-fricative trigger; they do not subsequently undergo velar-fricative palatalization.

46.2 SC044. Breaking before *h*, *rC*, *lC*, and conditioned *w* (`OEBreaking`)

```
define OEBreaking OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI;
```

Short *e* and *i* also develop short *eo* and *io* before singleton *w*, except when a high front vowel or *j* follows it. This is the regular pre-ending history of *cneowe* ‘knee (dat.sg.)’, not long-diphthong promotion (Luick 1914–-1940, 139, §134; Hogg 2011, 86, §5.22; Ringe and Taylor 2014, 187–88, §6.2.4). The shared w conditioner makes that exclusion explicit:

```
define EnglishBreakingWContext [
    {*w} [[EnglishStarVocalic | EnglishStarConsonant]
        - [{*i} | {*í} | {*ī} | {*ǐ} | {*j}]] |
    {*w} .#.
];
```

Root stress notation does not determine vowel length. The short knee dative is the selected comparison; long endingless *cnēo* ‘knee’ and the restored *w* of *cnēow* ‘knee’ are separate paradigm histories, discussed with SC033 OEEwLongDiphthong.

Breaking must encounter the vowel created by brightening and must precede the rule that would otherwise palatalize its velar trigger in *feoh* ‘cattle’ and *feohtan* ‘fight’. Before SC043 EAFBrightening, PGmc **sláxaną* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. After SC045 OEVelarFricativePalatalization, PGmc **féxu* ‘cattle’ yields †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* ‘fight’ yields †*fehtan* rather than expected *feohtan*. The fronting relation feeds breaking. The fee/fight relation instead protects breaking’s velar trigger from premature palatalization: those forms are displacement negatives, not live SC045 OEVelarFricativePalatalization applications.

46.3 SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization [
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j}
]
    .o. EnglishStarAlphabet*;
```

The live population concerns voiceless x, as in *hēafod* ‘head’ and *heofon* ‘heaven’, not voiced-fricative merger. In *feoh* ‘cattle’ and *feohtan* ‘fight’, breaking removes the original front-vowel context: they do not change at this rule in the live derivation. Moving the rule before SC044

OEBreaking instead consumes its velar trigger, yielding †*fehu* and †*fehtan*. This is counterbleeding protection, not direct feeding.

The distant *six* ‘six’ displacement test supplies only a broader constraint: after SC060 OEWsPalatalUmlaut, PGmc **séxs* ‘six’ yields †*sihs* rather than expected OE *six*. Neither test dates voiced g. Its articulation and disputed merger are separately treated under SC052 OEVelarPalatalization and SC109 OEPalatalFricativeMerger (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32).

47 A-restoration and nasal changes

47.1 Historical discussion of A-restoration

Campbell's restoration of *a* before following back vowels and Ringe and Taylor's later retraction describe the same post-brightening development (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

47.2 SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration (
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 EAFBrightening, PGmc **bákaną* 'bake' yields †*bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* 'fare' yields †*færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákaną* again yields †*bæcan* instead of *bacan*, while PGmc **wádaną* 'wade' yields †*wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

47.3 Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

47.4 SC047. Heavy-syllable nasal apocope of final **ą* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope [
    {*ą} -> 0 || OEAnyConsonant _ .#.
];
```

The old lower-bound test depended on the superseded straw proxy: moving apocope before SC034 OEAwLongDiphthong made PGmc **stráwą* 'straw' yield †*stræw* rather than attested OE *strēaw* 'straw'. The adopted regular endingless branch now reaches reconstructed **strēa* through SC111 PWGmcEndinglessAw, before this apocope. The attested target retains analogically restored *w* and explicitly mismatches the regular output. The old failure therefore does not independently place nasal apocope after a universal singleton long-diphthong law (Campbell 1959, 46–47, 232–33; Ringe and Taylor 2014, 172–73, 387).

The separate recorded upper-bound test is unchanged: after SC048 OESecondaryNasalization, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan*, and PGmc **bíndaną* 'bind'

yields †*binden* rather than expected *bindan*, alongside a broad *-en* failure set. That secondary-nasalization dependency is not newly adjudicated here.

47.5 SC048. Secondary nasalization before final **n* (OESecondaryNasalization)

```
define OESecondaryNasalization [
    {*a} -> {*ą} || _ {*n} .#.
];
```

The broad *-an*/*-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* ‘bake’ yields †*bacen* rather than expected OE *bacan* ‘bake’, and PGmc **bíndaną* ‘bind’ yields †*binden* rather than expected *bindan* ‘bind’. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* ‘steal’ yields †*steolan* rather than expected OE *stelan* ‘steal’, and PGmc **wébaną* ‘weave’ yields †*weofan* rather than expected *wefan* ‘weave’. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

48 B allophony

48.1 Historical discussion

The positional alternation of Germanic **b* is a Proto-Germanic distributional feature. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992b, 108). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

The distribution is narrow, but later changes presuppose the stop-fricative alternation. CAPR implements the rule at a late cascade position for computational reasons: the alternation must interact with consonant environments shaped by intermediate rule applications. Its historical stage is Proto-Germanic.

48.2 SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reġnboga* 'rainbow' develops as †*reġnfoga* rather than expected OE *reġnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

49 Sievers-law syncope

49.1 Historical discussion

Sievers' Law concerns a prosodic and morphological adjustment in heavy stems. It is a distributional rule distinct from b-allophony (SC049 PGmcBAllophony). Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

49.2 SC050. Sievers-law syncope (`SieversLawSyncope`)

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

The Sievers-law reduction **-CijV-** > **-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* 'stretch' yields †*strecċan* rather than expected OE *streċċan* 'stretch'; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

50 Palatalization of **sk* to **sc*

50.1 Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992b, 106–8, 113). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

50.2 SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

The non-fronted vowels of *flasce* ‘flask’ and *wascan* ‘wash’ fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* ‘flask’ and *wæscan* ‘wash’ rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sċeaft* ‘shaft’, *sċēar* ‘shear’, *sċēaþ* ‘sheath’, *sċēap* ‘sheep’, and *sċield* ‘shield’ before SC056 OEWsPalatalDiphthongization. The **sċea-* ’sea’*/**sċie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

51 Class-distinct velar palatalization and fricative merger

51.1 Historical discussion

Singleton g, including initial g, was a fricative when palatalization occurred; gg and postnasal g were stops. Palatal articulation, productive cutoff, merger and later affrication are therefore separate questions (Fulk 2018, 130–32; Laker 2007, 166–68, 175–84). The dotted spelling in *dæġ* ‘day’ cannot establish an early affricate or identify every palatal consonant with inherited j.

The adopted working account keeps a palatal fricative ȷ distinct from j during mutation, then merges it with j. Ringe and Taylor explicitly propose this chronology (Ringe and Taylor 2014, 203–4). It is not unanimous: Hogg objects to the phonetic plausibility of delayed merger and leaves the key/day contradiction unresolved (Hogg 1979, 102–11). We prefer the regular class-distinct account because it explains the contrasting mutation triggers without lexical or grammatical conditions. The objection remains substantive, not silently superseded by a matching final spelling.

Eligibility also requires attribution. Ringe and Taylor’s medial-g table is narrower than Fulk’s before-front-vowel description (Ringe and Taylor 2014, 203–4; Fulk 2018, 130–32). The present fricative implementation retains its broader domain; this merger decision is not a covert adoption of the narrower table.

51.2 SC052. Singleton-fricative articulation (`OEGFricativePalatal`)

```
define OEGFricativePalatal [
    [{*g}|{*ɣ}] -> {*ȷ} ||
        [.#.|[EnglishStarAlphabet - [{*g}|{*n}|{*ŋ}]]]
        _ EnglishStarFrontVowel,
    [{*g}|{*ɣ}] -> {*ȷ} || EnglishStarFrontVowel _ .#.,
    [{*g}|{*ɣ}] -> {*ȷ} || EnglishStarFrontVowel
        _ [EnglishStarConsonant - [{*j}|{*g}]]
] .o. [
    [{*g}|{*ɣ}] -> {*ȷ} ||
        [.#.|[EnglishStarAlphabet - [{*g}|{*n}|{*ŋ}]]] _ {*j}
];
```

The guards keep gg/ng in their stop paths. The value ȷ is a real phonetic class, not a tag recording a consonant’s ancestry. It is transparent to a following mutation trigger, but is not itself j at that date. The source comparison normalizes older fricative notation to modern ɣ/ȷ, never to the digit 3 (Hogg 1979, 105; Ringe and Taylor 2014, 204).

51.3 SC052. K articulation and eventual reflex (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʠ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʠ} || _ [{*i} | {*ī}],
    {*k} -> {*ʠ} || _ {*í},
    {*k} -> {*ʠ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʠ} || {*í} _ EnglishStarFrontVowel,
    {*k} -> {*ʠ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʠ} || {*í} _ .#.
] .o. [
    {*k} {*k} -> {*ʠ} {*ʠ} || _ {*j}
] .o. [
    {*k} -> {*ʠ} || _ {*j}
];
```

Here ʧ is a telescoped eventual-reflex proxy, not a claim that early palatal stops were already affricates. Ringe and Taylor explicitly separate the initial palatal stop from subsequent affrication and its syncope dependencies (Ringe and Taylor 2014, 203–4). The *weccan* ‘wake’, *licgan* ‘lie’ and *lecgan* ‘lay’ examples concern inherited j-clusters, not independent proof of every plain-velar conditioner (Ringe and Taylor 2014, 213–14).

The deterministic kk-before-j path protects *streċċan* ‘stretch’. The original-front versus secondary-front distinction is more directly tested by unrounded key than by rounded *cȳ* ‘cow’; rounding alone can give only a cutoff before later unrounding (Hogg 1979, 100–105; Laker 2007, 167–68).

51.4 SC052. Retained stop paths (`OEVelarPalatalizationStops`)

```
define OEVelarPalatalizationStops [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ȝ} || _ EnglishStarFrontVowel,
    {*g} -> {*ȝ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ȝ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ȝ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ȝ} {*ȝ} || _ {*j}
] .o. [
    {*g} -> {*ȝ} || _ {*j}
];
```

Singleton fricatives have already left this domain. The remaining gg/ng paths use ȝ as an eventual-reflex proxy; the initial singleton fricative is no longer represented as an early affricate. *Wicg* ‘horse’ is a geminate-class illustration, *senġan* ‘singe’ a postnasal control, and *lungen* ‘lung’ the unchanged back-vowel negative (Ringe and Taylor 2014, 203–4, 213–14; Fulk 2018, 131–32).

51.5 SC052. Combined articulation (`OEVelarPalatalization`)

```
define OEVelarPalatalization [
    OEGFricativePalatal .o. OEVelarPalatalizationStops
];
```

The combined operation does not give every class one sharply dated palatalization/affrication/merger event. It supplies the class distinctions needed by ordinary palatal diphthongization and mutation. The common-stem onset remains a separate question (Luick 1914--1940, 835–41; Fulk 2018, 130–32).

51.5.1 Key and day at the mutation checkpoint

Hogg’s oblique comparison starts here from pre-palatal, pre-OE **kājæ* and **dæɣæ*; neither is silently fed through a PGmc prefix (Hogg 1979, 105). The source target is *cǣġe* ‘key’, with velar initial k, versus *dæġe* ‘day’.

Checkpoint	Key	Day under the adopted account
Before palatalization	**kājæ*	**dæɣæ*
Before mutation	Inherited j remains	Palatal ɟ is distinct from j
After mutation	**kǣjæ*, without new initial palatalization	**dæɟæ*, without æ-raising
After merger and native realization	*cǣġe*	*dæġe*

Premature merger instead predicts pre-OE † under these premises. The experiment tests that source-based contrast; the existing day nominative is not the oblique discriminator (Hogg 1979, 105–10; Ringe and Taylor 2014, 204).

The complete key suffix additionally requires SC082 OEIntervocalicJVocalization to retain j after its non-high long front vowel. A correct mutation checkpoint followed by model-only †*cǣie* would not constitute a correct native derivation (Hogg 1979, 105).

51.6 SC109. Postmutation merger serialization (OEPalatalFricativeMerger)

```
define OEPalatalFricativeMerger [
    {*ɟ} -> {*j}
];
```

The historical commitment is merger after mutation (Ringe and Taylor 2014, 204). Its execution after SC089 OELateUnstressedAgSuffix is a computational holding zone, not proof of an exact historical date. It keeps the fricative out of inherited-j normalization while preserving late palatal raising, syllable weight, reduction and native rendering. SC057 OEJClusterCoalescence remains a different cluster process. Stop affrication dates and the precise disputed eligibility domain are not settled by this merger.

52 Post-velar **w*-loss and loss of **w* before final **i*

52.1 Historical discussion

The first rule is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor derive PGmc **singwan* ‘sing’ to Old English *singan* ‘sing’ (Ringe and Taylor 2014, 214, §6.4.2). This comparative evidence establishes the change, although no lexical evidence fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* ‘sea’ from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The first rule is restricted to the **ngw* sequence; the second has a specific lexical witness and defined earlier and later limits.

52.2 SC053. Loss of **w* after velars (`OEPostVelarWLoss`)

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

The comparative development `*singwan > singan` establishes narrow post-velar **w*-loss in the **ngw* sequence, yielding *singan* ‘sing’. Moving SC053 OEPostVelarWLoss earlier or later leaves every output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

52.3 SC054. Loss of **w* before final **i* (`OEWLossBeforeI`)

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

The history of *sǣ* ‘sea’ explains why non-initial **w* disappeared before final unstressed **i*. Campbell describes the loss, Ringe and Taylor derive the form from **saiwi-*/**sawi-*, and Luick gives the parallel trajectory (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 EAFFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields †*sǣw* rather than expected OE *sǣ* ‘sea’. The loss must therefore follow final *z*-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

53 The Old English i-umlaut and West Saxon palatal diphthongization

53.1 Historical discussion

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* 'guest', *giefan* 'give', *hierde* 'shepherd', and *ieldra* 'older' (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992b, 113). His examples, such as *bryd* 'bride', *trymman* 'strengthen', *bedd* 'bed', *ciest* 'chest', and *wiersa* 'worse', likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992b, 113–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13). Ringe and Taylor's examples such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

Two further distinctions are essential to their relative chronology. First, the raising of inherited Germanic **e* to **i* before a following high front vocoid is much earlier than Old English i-umlaut. Ringe reconstructs PGmc **giftiz* 'gift', with OE plural *ġifta* 'wedding', and discusses that earlier raising separately (Ringe 2017, 135, pp. 151–153). Ringe and Taylor likewise caution that raising of inherited **e* occurred hundreds of years before the Old English changes; a later repetition is rare and doubtful (Ringe and Taylor 2014, 220). The selected gift input now has the earlier-raised vowel. Its lack of ordinary diphthongization therefore does not date the general West Saxon process. Published e-reconstructions remain genuine alternatives (Orel 2003, 130; Kluge 2011, 359); the lexical discussion compares their evidence and explains the attributed working choice.

Second, the ordinary diphthongization of inherited/fronted non-high vowels must be distinguished from the later treatment of some mutation products after *sċ*. Campbell separates that small group from his general front-vowel diphthongization: the group includes *sċēaþ* 'sheath' beside *sċǣþ* 'sheath' (Campbell 1959, 68–69, §§184–185). Ringe and Taylor explicitly place the treatment of the ai-derived mutation product after mutation, giving the same two sheath outcomes (Ringe and Taylor 2014, 235). Their ordinary West Saxon derivations, including *ġiest* 'guest' and *ċytel* 'kettle', instead pass through palatal diphthongization before mutation (Ringe and Taylor 2014, 215–17, 222).

The historically preferred working account therefore has ordinary palatal diphthongization before Old English mutation, with the separately evidenced later *sċ* treatment after it. The adopted cascade separates these operations: ordinary diphthongization precedes mutation, and the unchanged late approximation follows it. The sources' alternate spellings do not by themselves supply an exceptionless conditioning law for the later group; its full conditioner remains unresolved.

53.2 SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ǽ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.],
    {*ǽ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ɟ} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.]
];
```

West Saxon *gieldan* ‘pay’, *scield* ‘shield’, and *scieppan* ‘create’ show diphthongization after an already palatal consonant (Ringe and Taylor 2014, 215–16, §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg’s *giefan* ‘give’ and *sceap* ‘sheep’ belong to the same palatal-consonant environment (Hogg 1992b, 112). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (Fulk 2018, 74, §4.13).

The ordinary live applications comprise *ġiefan* ‘give’, *ġiest* ‘guest’, *sċeaft* ‘shaft’, *sċieran* ‘shear’, *sċēap* ‘sheep’, *sċield* ‘shield’ and *ġēar* ‘year’. Gift is unchanged at this rule because its selected vowel already reflects earlier raising. Sheath changes from the mutation product **ǣ* to **ēa*; it supplies evidence for the later *sċ* layer, not the date of the ordinary treatment represented by the other examples.

The adopted ordinary process precedes mutation and preserves the split- diphthong and high-i negative controls. The reported absence of a later displacement failure supplies no positive historical terminus.

53.3 SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǽ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* ‘follow’, *gylden* ‘golden’, *wyrm* ‘worm’, and *giest* ‘guest’ exemplify the same `i`- or `j`-conditioned fronting across different vowels (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The selected cow and lung inputs are negative controls on the present palatalization rule. Moving the composite umlaut rule before that rule yields †*ċȳ* and †*lungen* instead of *cȳ* ‘cow’ and *lungen* ‘lung’. This constrains the current rule’s productive domain; it does not assign every consonantal

layer the same date. In particular the selected *cȳ* is a dative-singular input, not evidence from an assumed generic plural.

The stored gift and sheath displacement failures constrain the current serialization, not a universal historical upper boundary. The former gift input left an earlier raising to this late rule; sheath represents the later *sċ* treatment of a mutation product. Moving the entire palatal-diphthongization bundle earlier strands sheath and diphthongizes the model's still-unraised gift vowel. Neither failure overturns the independently supported ordinary palatal-diphthongization-before-mutation account.

53.4 SC055. Raising under i-umlaut (`OEIUmlautRaising`)

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted æ to e continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The four displacement controls concern the composite rule. They are not independent tests of this raising component: for example, the cow vowel changes in the fronting component, not in this **œ*-raising clause. Component chronology must follow the source-supported assimilatory event, with ordinary palatal diphthongization and the later *sċ* extension kept distinct.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before m and n in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992b, 113).

53.5 SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon ie forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992b, 113–14). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

Those composite displacement controls do not isolate diphthongal mutation: neither the cow input nor the gift input has a diphthong here. In the conventional history, diphthongs produced by ordinary palatal diphthongization can be mutated, as in the source derivation of *ġiest* 'guest' (Ringe and Taylor 2014, 216). The adopted derivation now follows that ea-to-ie source path. The former simple-vowel route converged on the same final form, showing why final agreement alone could not establish the relative chronology.

53.6 SC055. The composite i-umlaut rule (`OEIUmlaut`)

```
define OEIUmlaut OEIUmlautFronting
```

```
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong;
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The cow/lung displacement results require the present productive palatalization rule not to consume those mutation-created environments. The historical interpretation still depends on the relevant consonantal layer and inherited versus secondary front-vowel conditions.

There is no single source-supported upper boundary at the entire West Saxon bundle. Ordinary palatal diphthongization precedes mutation in the handbook account; the later *sċ* treatment of some mutation products follows it. The former gift input and sheath's later layer explained the old bundled serialization, not a date for every historical palatal process.

53.7 Retained late palatal-diphthongization approximation

```
define OELatePalatalDiphthong [
    {*ǣ} -> {*ēa} || .#. [{*ɡ} | {*ɣ} | {*j̵} | {*ʃ} | {*j}] _
     ↪  [EnglishStarConsonant | EnglishPalatalConsonant | .#.]
];
```

This support operation retains exactly the former postmutation long-vowel clause. Its sole current application is *sċēaþ* 'sheath'. The four initial triggers are an unchanged approximation, not a source-proven complete historical conditioner. Campbell and Ringe–Taylor distinguish the later class (Campbell 1959, 68–69; Ringe and Taylor 2014, 235); the incremental implementation does not claim to have resolved it.

54 J-cluster coalescence

54.1 Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīeġan* ‘bend’ and *sēċan* ‘seek’ without assigning this later cluster adjustment the status of a major sound law (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

54.2 SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ȡ}]
    .o. [{*k} {*j} -> {*ȶ}]
);
```

The forms *bīeġan* ‘bend’ and *sēċan* ‘seek’ determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīeġan* ‘bend’ and *sēċan* ‘seek’ are lost. Related forms such as *fylġan* ‘follow’, *heċġ* ‘hedge’, and *sengan* ‘singe’ fail in the same broader palatalization zone. PGmc **báugijaną* ‘bow’ yields †*bēaġan* rather than expected OE *bīeġan*, and PGmc **sōkijaną* ‘seek’ yields †*sōċan* rather than expected *sēċan*. This constrains this computational cluster consumer. It does not independently date all singleton-fricative articulation or merger: the latter is distinct from this rule and is now explicit under SC109 OEPalatalFricativeMerger (Ringe and Taylor 2014, 204; Laker 2007, 167–68). Nothing in the present lexicon supplies a terminus ante quem for the cluster consumer.

55 Back mutation

55.1 Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* ‘gave’ and *weofan* ‘weave’. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (Ringe and Taylor 2014, 319, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (Campbell 1959, 86, §207), while Hogg draws the instructive comparison with breaking (Hogg 1992b, 113). Fulk accordingly separates back mutation from the earlier umlautal changes (Fulk 2018, 69, §4.8).

55.2 SC059. Back mutation before labials and liquids (`OEBackMutation`)

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébaną* ‘give’ produce *ġeofan* ‘give’; the expected form is *ġiefan* ‘give’. **stélaną* ‘steal’ likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébaną* ‘weave’ yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

56 West Saxon palatal umlaut

56.1 Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

56.2 SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage †*mieht* and †*nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

57 Weak-tail nasal loss

57.1 Historical discussion

The pathway from **dōną* ‘do’ to *dōn* ‘do’ supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (Campbell 1959, 144–45, §§345–349; Hogg 1992b, 121; Fulk 2018, 91, §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

57.2 SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

Final weak-tail **-ną* and **-mą* accordingly yield plain **-n* and **-m*.

Only *dōn* ‘do’ constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* ‘do’. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

58 High-vowel apocope

58.1 Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (Campbell 1959, 144–45, §§345–349; Hogg 1992b, 121; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

This later loss is distinct from the West Germanic weak-word loss of SC098 PWGmcUnstressedWordFinalIApocope. A selected proclitic context does not become word-final merely because its citation spelling ends there. Its temporary boundary annotation blocks the final-position conditions below and is removed only after their application; it is not a segment of the linguistic reconstruction. The vowel-replacement clauses themselves are unchanged (Ringe and Taylor 2014, 57–58, 284–303).

58.2 SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
```

```
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    {*u} -> 0 || {*x} _ .#.,
    {*ų} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
] .o. [{*ᶜ} -> 0];
```

Final **i*, **u*, and **ų* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* 'cow' yields †*cū* rather than expected OE *cȳ* 'cow', and PGmc **brūdiz* 'bride' yields †*brūd* rather than expected OE *brȳd* 'bride'. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* 'fright' yields †*fyrht* rather than expected OE *fyrhte* 'fright'. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

59 Post-apocope **n*-loss and medial syncope

59.1 Historical discussion

Evidence for post-apocope reduction is strikingly uneven. The inherited feminine *in*-stem represented by Gothic *faurhtei*, OE *fyrhtu*, and oblique OE *fyrhte* supplies the relevant evidence (Orel 2003, 120; Ringe and Taylor 2014, 380–81; Campbell 1959, 236, §589.7). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (Hogg 1992b, 121; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

59.2 SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

Only final **n* after long **ī* is at issue, as in the inherited *in*-stem behind OE *fyrhte* ‘fright’.

CAPR models the oblique OE form through the Proto-Germanic genitive singular **fúrxtīnaz* ‘fright’, following the project convention of using an appropriate non-nominative paradigm cell when the nominative does not supply the required derivation. Within this selected genitive derivation, the same input fixes both ordering boundaries. Before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. After SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* again yields †*fyrhten* rather than expected *fyrhte* ‘fright’. I therefore order final bare-a loss, stem-final n-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the selected genitive derivation and depend on one inherited lexeme/paradigm.

59.3 SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (Hogg 1992b, 121; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

```
define OEMedialSyncope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
];
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

60 Late syncope and degemination

60.1 Historical discussion

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *netele* 'nettle' supplies a concrete lexical type (Hogg 1992b, 121; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Brunner 1965, 144–45, §§158–159). Fulk places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

60.2 SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope [
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátilōn* 'nettle' yields †*nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* 'spindle' yields †*spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields †*spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

60.3 SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation [
    {*θ} -> 0 || {*t} _
];
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (Hogg 1992b, 121; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

60.4 SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN;
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* 'spindle' (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* 'spindle' yields †*spinnl* rather than expected OE *spinl* 'spindle'. Syncope must therefore create the cluster before degemination simplifies it.

Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

61 Early o-shortening

61.1 Historical discussion

After the principal palatal and umlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

61.2 SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 PNWGmcNStemNLoss, PGmc **nédrōn* ‘adder’ yields †*nǣdran* rather than expected OE *nǣdre* ‘adder’, PGmc **érθōn* ‘earth’ yields †*eorþan* rather than expected *eorþe* ‘earth’, and PGmc **fláskōn* ‘flask’ yields †*flascan* rather than expected *flasce* ‘flask’. The same earlier shift also disrupts forms such as *heorte* ‘heart’ and *līne* ‘line’. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 PNWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

62 Early unstressed fronting and later o-shortening

62.1 Historical discussion

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no lexical evidence fixes its later boundary.

62.2 SC070. Early fronting of unstressed **a* (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

The rule fronts unstressed **a* to **æ* after the earlier shortening has created a frontable vowel but before the later shortening of unstressed **ō*. It produces endings such as OE *-en* in *lungen* ‘lungs’.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* ‘lungs’ yields †*lungen* rather than expected OE *lungen* ‘lungs’. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* ‘bears’ yields †*boreþ* rather than expected OE *boraþ* ‘bears’, and PGmc **mḗnōθz* ‘month’ yields †*mōneþ* rather than expected *mōnaþ* ‘month’. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

62.3 SC071. Later shortening of unstressed **ō* (OELateOShortening)

The following rule handles the later shortening stage.

```
define OELateOShortening [
    {*ō} -> {*o} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
];
```

The rule shortens the remaining unstressed long **ō* after fronting. The shortened vowel is then resolved by the following medial/final distribution, not directly as *a* (Stausland Johnsen 2015, 28–31).

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* ‘bears’ yield †*boreþ* rather than expected OE *boraþ* ‘bears’, and PGmc **líznōθi* ‘learns’ yield †*liorneþ* rather than expected *liornaþ* ‘learns’. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly.

62.4 SC099. Medial raising of shortened unstressed **o* (OEMedUnstressedORaising)

```
define OEMedUnstressedORaising [
    {*o} -> {*u} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]* EnglishStarVocalic
];
```

After SC071 OELateOShortening, the shortened vowel gives *u* in an unstressed medial syllable. The rule encodes Stausland Johnsen's statistically supported account of West Saxon ō-verb pasts, not a general rule for inherited short **o* or for nominal morphology (Stausland Johnsen 2015, 28–31, 36). His diagnostic derivation is PGmc **wúndōdē* 'wounded' > †*wundode* > OE wundude 'wounded' (Stausland Johnsen 2015, 28–29).

62.5 SC100. Final lowering of shortened unstressed **o* (OEFinalUnstressedOLowering)

```
define OEFinalUnstressedOLowering [
    {*o} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]* .#.
];
```

In a final syllable the same shortened vowel gives *a*. Thus the existing month control continues PGmc **mḗnōθz* 'month' through shortened **o* to OE mōnaþ 'month', while **wúndōdē* 'wounded' takes SC099 OEMedUnstressedORaising instead. The medial/final contrast and its chronology after long-vowel shortening are Stausland Johnsen's analysis (Stausland Johnsen 2015, 28–31).

63 Unstressed long-vowel shortening and ae-merger

63.1 Historical discussion

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **œ* with **e* (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

63.2 SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8;
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nédrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

63.3 SC073. Merger of unstressed **œ* with **e* (OEUnstressedAEMerger)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

The rule merges unstressed **œ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **nédrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The lexical evidence fixes the local order after SC072 OEUnstressedLongVowelShortening and places the merger before the later h-loss and contraction.

64 Medial unstressed-i lowering

64.1 Historical discussion

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (Hogg 1992b, 121; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *sċilling* ‘shilling’ type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

64.2 SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 [
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _
];
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhti* rather than expected OE *fyrhte* ‘fright’. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* ‘shilling’ yields †*sċilleng* rather than expected *sċilling* ‘shilling’. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

64.3 SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering [
    {*e} -> {*i} || _ {*n} {*g}
];
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* ‘shilling’.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skíllingaz* ‘shilling’ yield †*sċilleng* rather than expected OE *sċilling* ‘shilling’. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

65 Prefix i-reduction

65.1 Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk's discussion of prefix vowels accounts for OE **be-* and **ne-* (Fulk 2018, 97, §5.7). Hogg and Ringe and Taylor place such weakening within the broader late history of unstressed vowels (Hogg 1992b, 121; Ringe and Taylor 2014, 298–332, §§6.8.3–6.9.6).

The tested forms do not determine the rule's position relative to a neighboring change.

65.2 SC076. Reduction of prefixal **i* in unstressed position (OEPrefixIReduction)

```
define OEPrefixIReduction [
    {*i} -> {*ĕ} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] EnglishStarVocalic
];
```

The rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. The development accounts for later prefix spellings such as OE **be-* and **ne-*.

If the rule is moved earlier or later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not place SC076 OEPrefixIReduction before or after any specific neighboring change.

The handbooks attest late prefix-vowel weakening, but the precise placement remains approximate. No lexical failure fixes it.

66 Weak-tail reduction

66.1 Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (Campbell 1959, 148, §355; Hogg 1992b, 121; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

66.2 SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc ***bákaną*** ‘bake’ yields †***bacen*** rather than expected OE ***bacan*** ‘bake’, and PGmc ***bíndaną*** ‘bind’ yields †***binden*** rather than expected ***bindan*** ‘bind’, alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc ***fléuxaną*** ‘flee’ yields †***flēoan*** rather than expected OE ***flēon*** ‘flee’, and PGmc ***sláxaną*** ‘slay’ yields †***sleaan*** rather than expected ***slēan*** ‘slay’.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

67 Final-j loss and final geminate simplification

67.1 Historical discussion

After SC079 OEJLossAfterHeavy removes **j* in heavy environments, forms such as *lungen* ‘lungs’ acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final *nn* outcome in the following derivation.

67.2 SC079. Loss of **j* after heavy syllables (`OEJLossAfterHeavy`)

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

The rule removes **j* after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is **j*.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* ‘believe’ yields †*ġelēafan* rather than expected OE *ġelīefan* ‘believe’, PGmc **báugijaną* ‘bow’ yields †*bēaġan* rather than expected *bīeġan* ‘bow’, and PGmc **fúlgijaną* ‘follow’ yields †*fulġan* rather than expected *fylġan* ‘follow’. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* ‘lungs’ yields †*lungenn* rather than expected OE *lungen* ‘lungs’. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

67.3 SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* ‘lungs’ yield †*lungenn* rather than expected OE *lungen* ‘lungs’. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

68 J-strengthening, vocalization, and ei-contraction

68.1 Historical discussion

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then normalizes part of the inherited intervocalic **j* domain, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

68.2 SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ǽa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjanǫ* ‘strew’ yields †*strēaġan* rather than expected OE *strīeġan* ‘strew’. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields †*strīeian* rather than *strīeġan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīeġan* ‘strew’ derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

68.3 SC082. Bounded inherited-j normalization (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} ||
        [EnglishStarVocalic - [{*ǣ}|{*ē}|{*ǽ}]] _ EnglishStarVocalic
];
```

The former unrestricted VjV matcher wrongly gave model-only †*cǣie* instead of *cǣġe* ‘key’. Hogg’s pre-OE **kǣjæ* retains the glide (Hogg 1979, 105). The bounded rule therefore excludes preceding non-high long front monophthongs.

This is not a generic physical vocalization law. Hogg distinguishes high-vowel coalescence from spellings that merely represent a consonantal glide and from unstressed alternations (Hogg 2011, 283–86, §§7.69–7.76). The remaining weak-suffix e+j to e+i to i path telescopes later raising/contraction, rather than asserting that its e+i intermediate is a source reconstruction (Ringe and Taylor 2014, 228). SC083 OEUnstressedEIContraction consumes that modeled sequence. The full residual proxy domain is not claimed to be one independently established historical event.

The separately represented fricative ȷ̌ passes through this zone without being treated as inherited j. Its later merger belongs to SC109 OEPalatalFricativeMerger.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjanǫ* ‘strew’ yield †*strīeian* rather than expected OE *strīeġan* ‘strew’. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjanǫ* ‘bore’ yield †*boreian* rather than expected OE *borian* ‘bore’, PGmc **xándlōjanǫ* ‘handle’ yield †*handleian* rather than expected *handlian* ‘handle’, and PGmc **mákōjanǫ* ‘make’ yield †*maceian* rather than expected *macian* ‘make’. The witness

forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

68.4 SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ {*i}
];
```

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* ‘bore’ and *liccian* ‘lick’.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **búrōjaną* ‘bore’ yield †*boreian* rather than expected OE *borian* ‘bore’, PGmc **líznōjaną* ‘learn’ yield †*liorneian* rather than expected *liornian* ‘learn’, and PGmc **líkkōjaną* ‘lick’ yield †*licceian* rather than expected *liccian* ‘lick’. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

69 H-loss and contraction

69.1 Historical discussion

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (Fulk 2018, 270, §12.21), and Luick describes the corresponding West Germanic contractions (Luick 1914--1940, 165).

69.2 SC085. Loss of intervocalic **h* (`OEHLoss`)

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* ‘flee’ yields †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxaną* ‘slay’ yields †*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yields †*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* yields †*tāe* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

69.3 SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*éo},
    {*éo} {*o} -> {*éo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*é} {*a} -> {*é},
    {*é} {*e} -> {*é},
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*í} {*a} -> {*í},
    {*í} {*e} -> {*í},
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
```

```
    {*ū} {*e} -> {*ū}
];
```

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* ‘flee’, *slēan* ‘slay’, and *tēon* ‘draw’.

Moving contraction before SC085 OEHLoss makes PGmc ***fléuxaną*** ‘flee’ yield †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc ***sláxaną*** ‘slay’ yield †*sleaan* rather than expected *slēan* ‘slay’, PGmc ***téxun*** ‘draw’ yield †*teoon* rather than expected *tēon* ‘draw’, and PGmc ***táixōn*** ‘toe’ yield †*tāe* rather than expected *tā* ‘toe’. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

70 R-metathesis

70.1 Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (Brunner 1965, 159, §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (Luick 1914--1940, 201).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

70.2 SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstaną* ‘burst’ yield †*beorstan* rather than expected OE *berstan* ‘burst’. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters no output.

The lexical evidence fixes the earlier relation but does not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

71 Late unstressed suffix raising

71.1 Historical discussion

The late history of unstressed **-ag-* differs from prehistoric i-mutation. Ringe and Taylor give the sequence through fronting and raising to *-ig-*, explicitly placing its final stage long after mutation. Their examples include *hunigʒ* ‘honey’; a velar intermediate can survive before a back vowel (Ringe and Taylor 2014, 334–35, §6.9.6). These are the printed pages, not the PDF sheet labels.

71.2 SC089. Late suffix consumer (`OELateUnstressedAgSuffix`)

```
define OELateUnstressedAgSuffix (
    [{*a} -> {*e} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ _
         ↪  {*g}]
    .o.
    [{*g} -> {*ȷ} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ {*e}
         ↪  _ .#.]
    .o.
    [{*e} -> {*i} ||
        EnglishStarVocalic [EnglishStarConsonant | EnglishPalatalConsonant]+ _
         ↪  [{*ȷ}|{*ʠ}]]
);
```

The singleton consonant is a palatal fricative, not an early affricate (Fulk 2018, 130–32). The raising consumer therefore recognizes ȷ alongside retained stop-reflex compatibility. This does not make ȷ a prehistoric mutation trigger: later raising and earlier i-mutation have different conditioning histories.

The present late applications are *hunigʒ* ‘honey’ and *wīþigʒ* ‘withy’. The word-final guard remains narrow; this implementation is not an exhaustive model of every inflected suffix form. SC109 OEPalatalFricativeMerger follows here as a disclosed serialization, not proof of a uniquely dated historical merger immediately after this raising (Ringe and Taylor 2014, 204, pp. 334–335).

72 Chapter 5. Old English orthography and the written surface

72.1 Historical interval

This short chapter stands apart from the derivational chapters that precede it. The changes of Chapters 1–3 are sound changes: they altered the spoken form of the language. The material treated here belongs instead to the written surface of Old English — scribal conventions that determined how the results of the completed phonological history were committed to parchment.

In the executable model these conventions apply after every phonological rule, at the very end of the cascade, because that is where they belong historically: a spelling practice can only render forms that the spoken language had already produced.

72.2 Scope

The one rule treated here is the West Saxon palatal-glide spelling (SC016), by which back vowels following word-initial [j] — spelled *g* — came to be written with a preceding front glide letter, as in *geoc* ‘yoke’ for spoken [jok] and *geoguþ* ‘youth’ for a form whose root vowel remained [u]. Ringe and Taylor state the modern assessment directly: the *eo* of *geoc* is a spelling convention, and the word was pronounced [jok] (Ringe and Taylor 2014, 5). Hogg reaches the same verdict for the back-vowel cases generally (Hogg 1992b, 113). The older handbooks — Campbell, Brunner, Bülbring, Luick — analysed the same spellings as rising diphthongs; the section below presents both views (Campbell 1959, 17, § 44; Brunner 1965, 64–65, § 92).

72.3 Sources

Ringe and Taylor provide the modern phonological interpretation (Ringe and Taylor 2014, 5). Campbell (Campbell 1959, 17, 64–67, §§ 44, 170–176), Brunner (Brunner 1965, 64–65, § 92), and Bülbring (Bülbring 1902, 120, §§ 298–299) document the distribution of the spellings; Hogg supplies the critical reassessment (Hogg 1992b, 113).

73 West Saxon palatal-glide spelling before back vowels

73.1 Historical discussion

West Saxon spellings such as *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeoguþ* ‘youth’ write a front glide letter between a word-initial palatal and a following back vowel. Campbell describes the phenomenon as the development of rising diphthongs when “palatal glides developed before back vowels” and cites *ġeoc* directly (Campbell 1959, 17, §44); Brunner separates the *u*-cases (*ġeong*, *ġeoguþ*) from the *o*-cases (*ġioc* ‘yoke’, *ġeoc*) (Brunner 1965, 64–65, §92.1); Bülbring likewise treats *iuguð* ‘youth’ and *iuc* under *ju* but derives *ġioc*, *ġeoc* from West Germanic **jok* (Bülbring 1902, 120, §§298–299); and Luick groups all of these under his “schwebende Diphthonge” after palatal onsets (Luick 1914--1940, 158–59, §169).

The phonological interpretation of these spellings is disputed. The older handbook tradition — Campbell, Brunner, Bülbring, Luick — reads them as genuine rising diphthongs. The modern assessment is orthographic: Ringe and Taylor state flatly that *ġeoc* “is /jok/”, the digraph being a spelling convention that became universal after word-initial /j/ (Ringe and Taylor 2014, 5), and Hogg concludes that the back-vowel cases were “never anything more than an orthographic variation”, judging Campbell’s arguments to the contrary “insubstantial” (Hogg 1992b, 113; Campbell 1959, 66–67, §176). This model follows Ringe and Taylor and Hogg: the rule is a spelling convention applied to the finished phonology, and it therefore stands at the end of the derivation, in the written-surface stage of the cascade.

Its position also settles a relative chronology. The *o* of *ġeoc* ‘yoke’ is itself the product of Northwest Germanic u-lowering (SC017 PNWGmcULowering): Fulk lists *ġeoc* as a regular lowering example beside OIcel *ok* and OHG *joh* (Fulk 2018, 56, §4.3), and Campbell gives *ġeoc* among the regular *u* >

o words (Campbell 1959, 43, §115). The lowering therefore feeds the spelling: first **juk-* became **jok-* in Northwest Germanic, and only much later did West Saxon scribes write the result as *ġeoc*. Where lowering did not apply, as in *ġeoguþ* ‘youth’, whose root *u* was protected by the high vowel of the following syllable, the same convention wrote the retained *u* with the same digraph (Brunner 1965, 64–65, §92.1).

73.2 SC016. West Saxon palatal-glide spelling before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
    {*ó} -> {*éo} || .#. ġ _ ,
    {*ú} -> {*éo} || .#. ġ _ ,
    {*o} -> {*eo} || .#. ġ _ ,
    {*u} -> {*eo} || .#. ġ _
];
```

The rule rewrites a back vowel after word-initial *ġ* as the digraph spelling, covering both the lowered *o*-cases (*ġeoc* ‘yoke’) and the retained *u*-cases (*ġeoguþ* ‘youth’). Because it is a convention of the written language, it applies after every phonological change; in particular it follows SC017 PNWGmcULowering, which supplies the *o* of *ġeoc*. If the spelling rule were placed before u-lowering, the derivation would have to treat an Old English scribal practice as a Northwest Germanic sound change, an ordering that no source supports. The witnesses *ġeoc* and *ġeoguþ* between them fix both faces of the rule: one shows the convention applied to lowered *o*, the other to unlowered *u*. The handbook domain is broader (it also includes *a*/*ā*/*ō* contexts after word-initial palatals), but this executable rule is intentionally complete for the currently selected corpus witnesses rather than a maximal dialectal enumeration.

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